

ELITE IAL MATHEMATICS

WMA11 Pure 1 Expertise With Answers

Questions 6 and above with worked-solution pages.

89

questions

WMA11

Pure 1

**Question
bank with
worked
answers**

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Prepared for Dr Eslam Ahmed - eliteigcse.com

TOPIC

Indices & Surds

Question 6

6.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

(a) Expand and simplify

$$\left(r - \frac{1}{r}\right)^2$$

(2)

(b) Express $\frac{1}{3 + 2\sqrt{2}}$ in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

(c) Use the results of parts (a) and (b), or otherwise, to show that

$$\sqrt{3 + 2\sqrt{2}} - \frac{1}{\sqrt{3 + 2\sqrt{2}}} = 2$$

(3)

(Total for Question 6 is 7 marks)

Worked Solution - Question 6

1. Expand the square

$$\left(r - \frac{1}{r}\right)^2 = r^2 - 2 + \frac{1}{r^2}.$$

2. Rationalise the denominator

$$\frac{1}{3 + 2\sqrt{2}} \times \frac{3 - 2\sqrt{2}}{3 - 2\sqrt{2}} = \frac{3 - 2\sqrt{2}}{9 - 8} = 3 - 2\sqrt{2}.$$

3. Let r be the square root

Let $r = \sqrt{3 + 2\sqrt{2}}$. Then $r^2 = 3 + 2\sqrt{2}$ and $\frac{1}{r^2} = 3 - 2\sqrt{2}$.

4. Use part (a)

$$\left(r - \frac{1}{r}\right)^2 = r^2 - 2 + \frac{1}{r^2} = (3 + 2\sqrt{2}) - 2 + (3 - 2\sqrt{2}) = 4.$$

5. Take the positive square root

Since $r > 1$, $r - \frac{1}{r} > 0$, so $r - \frac{1}{r} = 2$.

Final answer

(a) $r^2 - 2 + \frac{1}{r^2}$.

(b) $3 - 2\sqrt{2}$.

(c) The expression equals 2.

TOPIC

Quadratics

Question 8

8. Solve, using algebra, the equation

$$x - 6x^{\frac{1}{2}} + 4 = 0$$

Fully simplify your answers, writing them in the form $a + b\sqrt{c}$, where a , b and c are integers to be found.

(5)

(Total 5 marks)

Worked Solution - Question 8

1. Use a substitution

Let $u = \sqrt{x}$. Then $x = u^2$, so the equation becomes $u^2 - 6u + 4 = 0$.

2. Solve the quadratic

$$u = \frac{6 \pm \sqrt{36 - 16}}{2} = \frac{6 \pm 2\sqrt{5}}{2} = 3 \pm \sqrt{5}.$$

3. Return to x

Since $u = \sqrt{x}$, $x = u^2 = (3 \pm \sqrt{5})^2 = 9 + 5 \pm 6\sqrt{5} = 14 \pm 6\sqrt{5}$.

Final answer

$$x = 14 + 6\sqrt{5} \text{ or } x = 14 - 6\sqrt{5}.$$

Mark scheme checked - Topic checked - WMA11-01_2019_Oct_Q08

Question 6

6. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

- (a) Given that

$$2xy - 3x^2 = 50$$

and

$$y - x^3 + 6x = 0$$

show that

$$2x^4 - 15x^2 - 50 = 0 \quad (2)$$

- (b) Hence solve the simultaneous equations

$$2xy - 3x^2 = 50$$

$$y - x^3 + 6x = 0$$

Give your answers in fully simplified surd form. (5)

(Total 7 marks)

Worked Solution - Question 6

1. Rearrange the second equation

From $y - x^3 + 6x = 0$, we get $y = x^3 - 6x$.

2. Substitute into the first equation

$2x(x^3 - 6x) - 3x^2 = 50$, so $2x^4 - 15x^2 - 50 = 0$.

3. Solve the quadratic in x squared

Let $u = x^2$. Then $2u^2 - 15u - 50 = 0$, giving $u = 10$ or $u = -\frac{5}{2}$.

4. Reject the negative value

Since x^2 cannot be negative, $x^2 = 10$, so $x = \pm\sqrt{10}$.

5. Find y

$y = x^3 - 6x = x(x^2 - 6) = x(10 - 6) = 4x$. Therefore $y = 4\sqrt{10}$ when $x = \sqrt{10}$ and $y = -4\sqrt{10}$ when $x = -\sqrt{10}$.

Final answer

$(x, y) = (\sqrt{10}, 4\sqrt{10})$ or $(-\sqrt{10}, -4\sqrt{10})$.

Question 8

8.

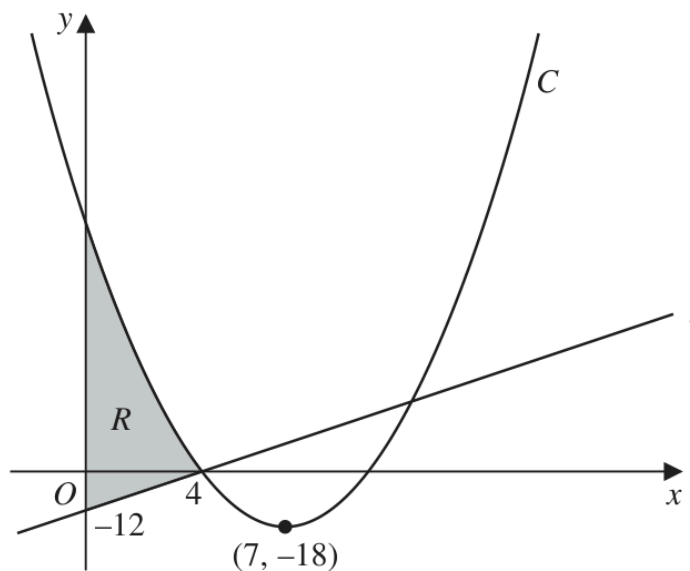


Figure 2

Figure 2 shows a sketch of the straight line l and the curve C .

Given that l cuts the y -axis at -12 and cuts the x -axis at 4 , as shown in Figure 2,

- (a) find an equation for l , writing your answer in the form $y = mx + c$, where m and c are constants to be found.

(2)

Given that C

- has equation $y = f(x)$ where $f(x)$ is a quadratic expression
- has a minimum point at $(7, -18)$
- cuts the x -axis at 4 and at k , where k is a constant

- (b) deduce the value of k ,

(1)

- (c) find $f(x)$.

(3)

The region R is shown shaded in Figure 2.

- (d) Use inequalities to define R .

(2)

(Total for Question 8 is 8 marks)

Worked Solution - Question 8

1. Find the straight line

The line cuts the y-axis at -12 and the x-axis at 4 , so its gradient is $\frac{0 - (-12)}{4 - 0} = 3$.

Hence l is $y = 3x - 12$.

2. Use symmetry of the quadratic

The minimum point has x-coordinate 7 . The roots are 4 and k , so their midpoint is 7 .

3. Find k

$$\frac{4 + k}{2} = 7, \text{ so } k = 10.$$

4. Build the quadratic

Let $f(x) = a(x - 4)(x - 10)$. Since the minimum point is $(7, -18)$,
 $-18 = a(3)(-3) = -9a$, so $a = 2$.

5. Write $f(x)$

$$f(x) = 2(x - 4)(x - 10) = 2x^2 - 28x + 80.$$

6. Identify the shaded region

The region lies between the y-axis and $x = 4$, above the line and below the curve:

$$0 \leq x \leq 4 \text{ and } 3x - 12 \leq y \leq 2x^2 - 28x + 80.$$

Final answer

(a) $y = 3x - 12$.

(b) $k = 10$.

(c) $f(x) = 2(x - 4)(x - 10)$.

(d) $0 \leq x \leq 4, 3x - 12 \leq y \leq 2x^2 - 28x + 80$.

Question 6

6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The equation

$$4(p - 2x) = \frac{12 + 15p}{x + p} \quad x \neq -p$$

where p is a constant, has two distinct real roots.

(a) Show that

$$3p^2 - 10p - 8 > 0 \quad (3)$$

(b) Hence, using algebra, find the range of possible values of p

(3)

(Total for Question 6 is 6 marks)

Worked Solution - Question 6

1. Clear the denominator

$$4(p - 2x) = \frac{12 + 15p}{x + p} \text{ gives } 4(p - 2x)(x + p) = 12 + 15p.$$

2. Form a quadratic in x

Expanding gives $4p^2 - 4px - 8x^2 = 12 + 15p$, so
 $8x^2 + 4px + 12 + 15p - 4p^2 = 0$.

3. Use the discriminant condition

For two distinct real roots, $b^2 - 4ac > 0$.

4. Simplify the inequality

$$(4p)^2 - 4(8)(12 + 15p - 4p^2) > 0, \text{ so } 144p^2 - 480p - 384 > 0.$$

5. Divide and factorise

Dividing by 48 gives $3p^2 - 10p - 8 > 0$, and $3p^2 - 10p - 8 = (3p + 2)(p - 4)$.

6. State the range

The upward-opening quadratic is positive outside its roots, so $p < -\frac{2}{3}$ or $p > 4$.

Final answer

$$p < -\frac{2}{3} \text{ or } p > 4.$$

Question 11

11.

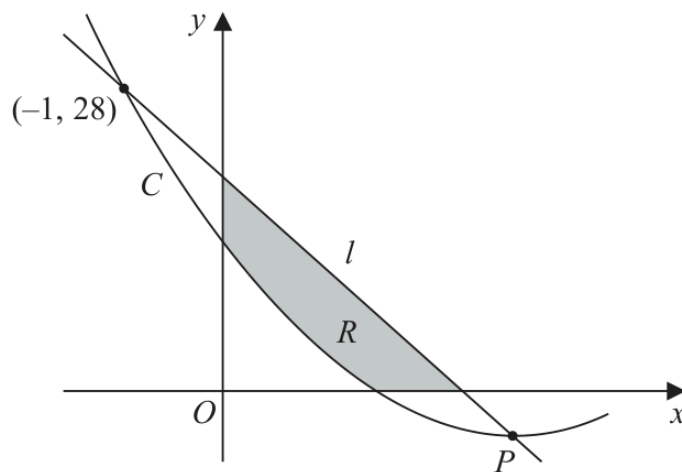


Figure 5

Figure 5 shows part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^2 - 12x + 14$$

- (a) Write $2x^2 - 12x + 14$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

Given that C has a minimum at the point P

- (b) state the coordinates of P

(1)

The line l intersects C at $(-1, 28)$ and at P as shown in Figure 5.

- (c) Find the equation of l giving your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

The finite region R , shown shaded in Figure 5, is bounded by the x -axis, l , the y -axis, and C .

- (d) Use inequalities to define the region R .

(3)

(Total for Question 11 is 10 marks)

Worked Solution - Question 11

1. Complete the square

$$2x^2 - 12x + 14 = 2(x^2 - 6x) + 14 = 2((x - 3)^2 - 9) + 14 = 2(x - 3)^2 - 4.$$

2. State the minimum point

The minimum of $2(x - 3)^2 - 4$ is at $P = (3, -4)$.

3. Find the gradient of l

The line passes through $(-1, 28)$ and $P(3, -4)$, so $m = \frac{-4 - 28}{3 - (-1)} = \frac{-32}{4} = -8$.

4. Find the equation of l

Using $P(3, -4)$, $y + 4 = -8(x - 3)$, so $y = -8x + 20$.

5. Use the axes as boundaries

The region is to the right of the y-axis and above the x-axis, so $x \geq 0$ and $y \geq 0$.

6. Use the line and curve as boundaries

The region is below the line and above the curve, so $y \leq -8x + 20$ and $y \geq 2x^2 - 12x + 14$.

Final answer

(a) $2(x - 3)^2 - 4$.

(b) $P = (3, -4)$.

(c) $y = -8x + 20$.

(d) $x \geq 0, y \geq 0, y \leq -8x + 20, y \geq 2x^2 - 12x + 14$.

Question 9

Quadratics

9. The curve C_1 has equation $y = f(x)$.

Given that

- $f(x)$ is a quadratic expression
- C_1 has a maximum turning point at $(2, 20)$
- C_1 passes through the origin

(a) sketch a graph of C_1 showing the coordinates of any points where C_1 cuts the coordinate axes,

(2)

(b) find an expression for $f(x)$.

(3)

The curve C_2 has equation $y = x(x^2 - 4)$

Curve C_1 and C_2 meet at the origin, and at the points P and Q

Given that the x coordinate of the point P is negative,

(c) using algebra and showing all stages of your working, find the coordinates of P

(5)

(Total for Question 9 is 10 marks)

Worked Solution - Question 9

1. Use symmetry for the sketch

The maximum is at $(2, 20)$ and the curve passes through $(0, 0)$. Since the axis of symmetry is $x = 2$, the other x-intercept is $(4, 0)$.

2. Use completed-square form

Let $f(x) = a(x - 2)^2 + 20$.

3. Use the origin

$0 = a(0 - 2)^2 + 20$, so $4a = -20$ and $a = -5$.

4. Write $f(x)$

$f(x) = -5(x - 2)^2 + 20 = -5x^2 + 20x = -5x(x - 4)$.

5. Equate the two curves

$-5x^2 + 20x = x(x^2 - 4) = x^3 - 4x$.

6. Factorise

$x^3 + 5x^2 - 24x = 0$, so $x(x^2 + 5x - 24) = 0$ and $x(x + 8)(x - 3) = 0$.

7. Use the negative x-coordinate

The negative x-coordinate is $x = -8$.

8. Find y

$y = x(x^2 - 4) = (-8)(64 - 4) = -480$, so $P = (-8, -480)$.

Final answer

(a) downward parabola through $(0, 0)$ and $(4, 0)$ with maximum $(2, 20)$.

(b) $f(x) = -5x(x - 4)$.

(c) $P = (-8, -480)$.

Question 10**10.**

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

$$(k - 1)x^6 + 4x^3 + (k - 4) = 0 \quad \text{where } k \text{ is a constant}$$

- (a) Find the exact solutions to the given equation for $k = 4.5$ **(3)**
- (b) Find the set of possible values of k for which the given equation has no real roots. **(4)**

(Total for Question 10 is 7 marks)

Worked Solution - Question 10

1. Substitute k equals 4.5

The equation becomes $3.5x^6 + 4x^3 + 0.5 = 0$.

2. Clear decimals

Multiplying by 2 gives $7x^6 + 8x^3 + 1 = 0$.

3. Use a quadratic in x cubed

$$7x^6 + 8x^3 + 1 = (x^3 + 1)(7x^3 + 1) = 0.$$

4. Solve for x

$$x^3 = -1 \text{ or } x^3 = -\frac{1}{7}, \text{ so } x = -1 \text{ or } x = -\sqrt[3]{\frac{1}{7}}.$$

5. View the equation as a quadratic

Let $u = x^3$. Then the equation is $(k - 1)u^2 + 4u + (k - 4) = 0$.

6. Use no real roots

Since every real value of u gives a real x , the original equation has no real roots exactly when this quadratic in u has no real roots.

7. Use the discriminant

Require $b^2 - 4ac < 0$: $16 - 4(k - 1)(k - 4) < 0$.

8. Simplify

$$16 - 4(k^2 - 5k + 4) < 0 \text{ gives } 20k - 4k^2 < 0, \text{ so } 4k(5 - k) < 0.$$

9. Solve the inequality

$$4k(5 - k) < 0 \text{ for } k < 0 \text{ or } k > 5.$$

Final answer

(a) $x = -1, -\sqrt[3]{\frac{1}{7}}.$

(b) $k < 0$ or $k > 5.$

Mark scheme checked - Topic checked - WMA11-01_2025_MayJune_Q10

TOPIC

Simultaneous Equations

Question 8

Simultaneous Equations

8. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

The curve C_1 has equation

$$xy = \frac{15}{2} - 5x \quad x \neq 0$$

The curve C_2 has equation

$$y = x^3 - \frac{7}{2}x - 5$$

- (a) Show that C_1 and C_2 meet when

$$2x^4 - 7x^2 - 15 = 0 \quad (2)$$

Given that C_1 and C_2 meet at points P and Q

- (b) find, using algebra, the exact distance PQ (5)

(Total for Question 8 is 7 marks)

Worked Solution - Question 8

1. Write C1 in terms of y

$$xy = \frac{15}{2} - 5x, \text{ so } y = \frac{15}{2x} - 5.$$

2. Equate the two curves

$$\frac{15}{2x} - 5 = x^3 - \frac{7}{2}x - 5.$$

3. Multiply by 2x

$$15 = 2x^4 - 7x^2, \text{ hence } 2x^4 - 7x^2 - 15 = 0.$$

4. Solve for x

$$\text{Let } u = x^2. \text{ Then } 2u^2 - 7u - 15 = 0 = (2u + 3)(u - 5), \text{ so } x^2 = 5 \text{ and } x = \pm\sqrt{5}.$$

5. Find the y-coordinates

$$\text{Using } y = \frac{15}{2x} - 5, \text{ the points are } \left(\sqrt{5}, \frac{3\sqrt{5}}{2} - 5\right) \text{ and } \left(-\sqrt{5}, -\frac{3\sqrt{5}}{2} - 5\right).$$

6. Use the distance formula

$$\Delta x = 2\sqrt{5} \text{ and } \Delta y = 3\sqrt{5}, \text{ so } PQ = \sqrt{(2\sqrt{5})^2 + (3\sqrt{5})^2} = \sqrt{65}.$$

Final answer

$$(a) 2x^4 - 7x^2 - 15 = 0.$$

$$(b) PQ = \sqrt{65}.$$

TOPIC

Inequalities

Question 8

8. The straight line l has equation $y = k(2x - 1)$, where k is a constant.

The curve C has equation $y = x^2 + 2x + 11$

Find the set of values of k for which l does not cross or touch C .

(6)

(Total 6 marks)

Worked Solution - Question 8

1. Equate the line and curve

$$k(2x - 1) = x^2 + 2x + 11, \text{ so } x^2 + (2 - 2k)x + (11 + k) = 0.$$

2. Use the no-intersection condition

For the line not to cross or touch the curve, the quadratic in x must have no real roots, so its discriminant is less than zero.

3. Form the inequality in k

$$(2 - 2k)^2 - 4(1)(11 + k) < 0. \text{ Dividing by 4 gives } (1 - k)^2 - (11 + k) < 0, \text{ so } k^2 - 3k - 10 < 0.$$

4. Solve the quadratic inequality

$$(k - 5)(k + 2) < 0, \text{ so } k \text{ lies between the two roots: } -2$$

Final answer

-2

Mark scheme checked - Topic checked - WMA11-01_2020_Jan_Q08

Question 7

7.

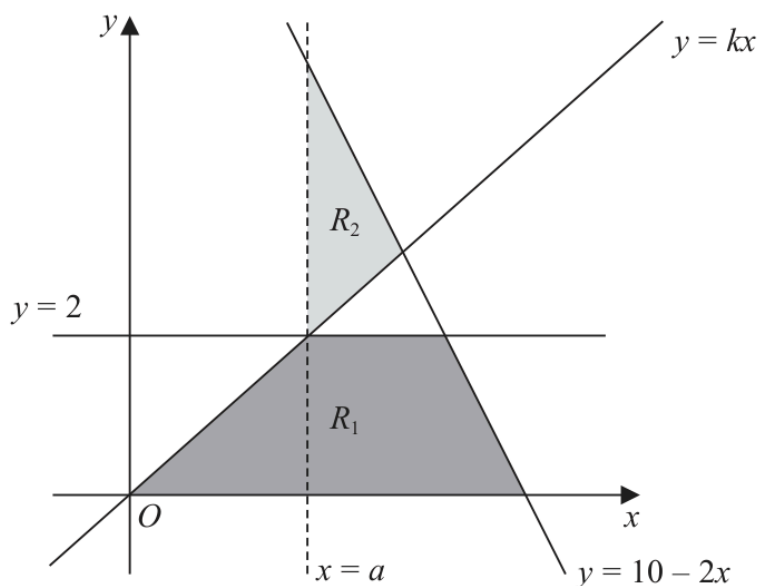


Figure 2

The region R_1 , shown shaded in Figure 2, is defined by the inequalities

$$0 \leq y \leq 2 \quad y \leq 10 - 2x \quad y \leq kx$$

where k is a constant.

The line $x = a$, where a is a constant, passes through the intersection of the lines $y = 2$ and $y = kx$

Given that the area of R_1 is $\frac{27}{4}$ square units,

(a) find

(i) the value of a

(ii) the value of k

(4)

(b) Define the region R_2 , also shown shaded in Figure 2, using inequalities.

(2)

(Total for Question 7 is 6 marks)

Worked Solution - Question 7

1. Find the area of R_1

The bottom horizontal length is 5, from $x = 0$ to the intercept of $y = 10 - 2x$ at $x = 5$.

2. Use the top length

The top horizontal length is $4 - a$, because $y = 2$ meets $y = 10 - 2x$ at $x = 4$.

3. Solve for a

$$\text{Area } R_1 = \frac{1}{2}(5 + 4 - a)(2) = 9 - a = \frac{27}{4}, \text{ so } a = \frac{9}{4}.$$

4. Find k

The line $x = a$ passes through the intersection of $y = 2$ and $y = kx$, so $2 = ka$.

$$\text{Hence } k = \frac{2}{9/4} = \frac{8}{9}.$$

5. Define R_2

Region R_2 is to the right of $x = a$, above $y = kx$, and below $y = 10 - 2x$.

Final answer

$$(a) \ a = \frac{9}{4}, \ k = \frac{8}{9}.$$

$$(b) \ x \geq \frac{9}{4}, \ y \geq \frac{8}{9}x, \ y \leq 10 - 2x.$$

Question 6

6.

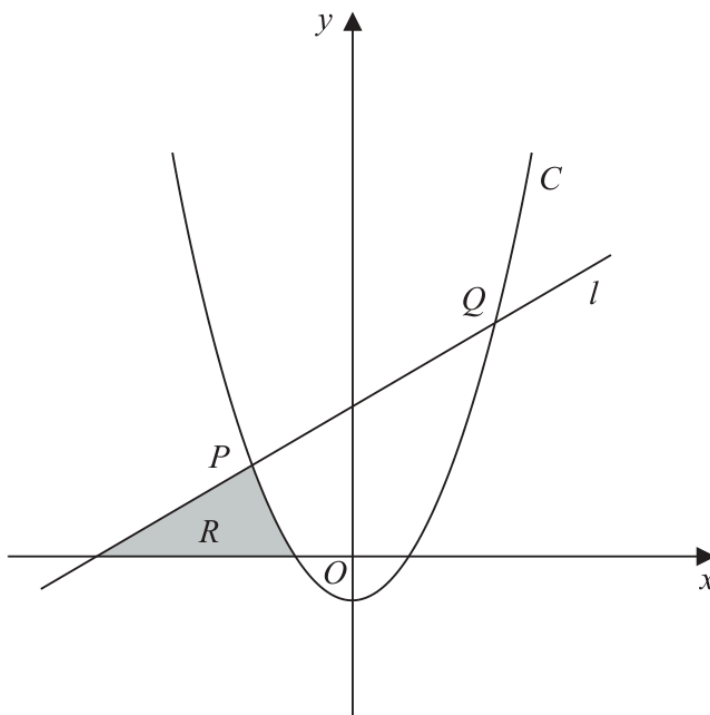


Figure 3

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 3 shows

- the line l with equation $y - 5x = 75$
- the curve C with equation $y = 2x^2 + x - 21$

The line l intersects the curve C at the points P and Q , as shown in Figure 3.

- (a) Find, using algebra, the coordinates of P and the coordinates of Q . (4)

The region R , shown shaded in Figure 3, is bounded by C , l and the x -axis.

- (b) Use inequalities to define the region R . (3)

(Total for Question 6 is 7 marks)

Worked Solution - Question 6

1. Write the line as y equals

$$y - 5x = 75, \text{ so } y = 5x + 75.$$

2. Equate line and curve

$$5x + 75 = 2x^2 + x - 21, \text{ so } 2x^2 - 4x - 96 = 0.$$

3. Solve the quadratic

$$x^2 - 2x - 48 = 0 = (x - 8)(x + 6), \text{ so } x = -6 \text{ or } x = 8.$$

4. Find the y -coordinates

Using $y = 5x + 75$, $x = -6$ gives $y = 45$ and $x = 8$ gives $y = 115$.

5. State P and Q

From the diagram, $P = (-6, 45)$ and $Q = (8, 115)$.

6. Use the x -axis and left branch

The shaded region is above the x -axis and on the left part of the diagram, so $y \geq 0$ and $x \leq 0$.

7. Use the line and curve as upper boundaries

The region is below both the line and the curve, so $y \leq 5x + 75$ and $y \leq 2x^2 + x - 21$.

Final answer

(a) $P = (-6, 45)$, $Q = (8, 115)$.

(b) $y \geq 0$, $y \leq 5x + 75$, $y \leq 2x^2 + x - 21$, $x \leq 0$.

TOPIC

Polynomials

Question 8

Polynomials

8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

- (a) Write $3x^2 + 6x + 9$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

The point P is the minimum point of C_1

- (b) Deduce the coordinates of P .

(1)

A different curve C_2 has equation

$$y = Ax^3 + Bx^2 + Cx + D$$

where A , B , C and D are constants.

Given that C_2

- passes through P
- intersects the x -axis at -4 , -2 and 3

- (c) find, making your method clear, the values of A , B , C and D .

(5)

(Total 9 marks)

Worked Solution - Question 8

1. Complete the square

$$3x^2 + 6x + 9 = 3(x^2 + 2x) + 9 = 3((x + 1)^2 - 1) + 9 = 3(x + 1)^2 + 6.$$

2. Deduce the minimum point

The minimum occurs when $(x + 1)^2 = 0$, so $x = -1$ and $y = 6$. Therefore $P = (-1, 6)$.

3. Use the x-intercepts of C_2

Since C_2 intersects the x-axis at -4 , -2 and 3 , it has form $y = k(x + 4)(x + 2)(x - 3)$.

4. Use point P to find k

Substitute $P = (-1, 6)$: $6 = k(3)(1)(-4) = -12k$, so $k = -\frac{1}{2}$.

5. Expand the cubic

$$(x + 4)(x + 2)(x - 3) = (x^2 + 6x + 8)(x - 3) = x^3 + 3x^2 - 10x - 24.$$

6. Read off A, B, C and D

Thus $y = -\frac{1}{2}x^3 - \frac{3}{2}x^2 + 5x + 12$, so $A = -\frac{1}{2}$, $B = -\frac{3}{2}$, $C = 5$ and $D = 12$.

Final answer

(a) $3(x + 1)^2 + 6$.

(b) $P = (-1, 6)$.

(c) $A = -\frac{1}{2}$, $B = -\frac{3}{2}$, $C = 5$, $D = 12$.

Question 6

6.

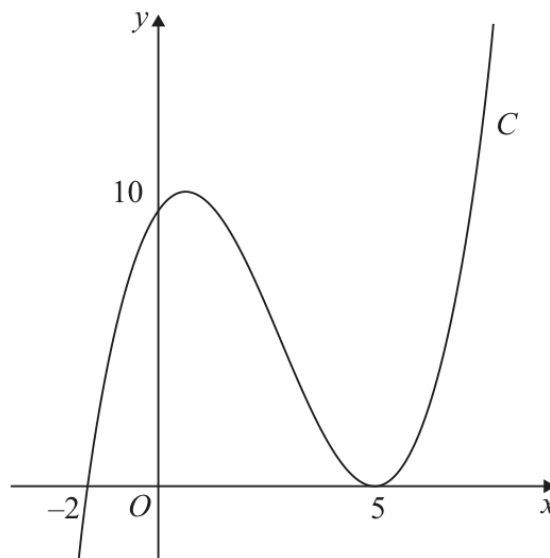
**Figure 3**

Figure 3 shows a sketch of the curve C with equation $y = f(x)$ where $f(x)$ is a cubic function in x .

The curve C

- cuts the x -axis at $(-2, 0)$ and cuts the y -axis at $(0, 10)$
- touches the x -axis at $(5, 0)$

as shown in Figure 3.

(a) Deduce the roots of the equation

(i) $f\left(\frac{1}{3}x\right) = 0$

(ii) $f(x - 3) = 0$

(2)

(b) Find an expression for $f(x)$. You should leave your answer in factorised form.

(3)

The curve C intersects the straight line $y = 10(x + 2)$ at exactly three points.

(c) Use algebra to find the exact x coordinates of the three points of intersection.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

(Total for Question 6 is 9 marks)

Worked Solution - Question 6

1. Roots of f one third x

The roots of $f(x) = 0$ are $x = -2$ and $x = 5$, with $x = 5$ repeated. For $f\left(\frac{1}{3}x\right) = 0$, set $\frac{1}{3}x = -2$ or 5 , giving $x = -6$ or 15 .

2. Roots of f of x minus 3

For $f(x - 3) = 0$, set $x - 3 = -2$ or 5 , giving $x = 1$ or 8 .

3. Use the roots and repeated root

Since the curve cuts at $(-2, 0)$ and touches at $(5, 0)$, write $f(x) = k(x + 2)(x - 5)^2$.

4. Use the y-intercept

The curve passes through $(0, 10)$, so $10 = k(2)(25) = 50k$.

5. State $f(x)$

$k = \frac{1}{5}$, hence $f(x) = \frac{1}{5}(x + 2)(x - 5)^2$.

6. Set equal to the line

$$\frac{1}{5}(x + 2)(x - 5)^2 = 10(x + 2).$$

7. First solution

The factor $x + 2$ gives one solution $x = -2$.

8. Find the other solutions

For $x \neq -2$, divide by $x + 2$: $\frac{1}{5}(x - 5)^2 = 10$, so $(x - 5)^2 = 50$.

9. State all intersections

$x = 5 \pm \sqrt{50} = 5 \pm 5\sqrt{2}$. Therefore the three x-coordinates are -2 , $5 - 5\sqrt{2}$ and $5 + 5\sqrt{2}$.

Final answer

(a)(i) $-6, 15$. (a)(ii) $1, 8$.

(b) $f(x) = \frac{1}{5}(x+2)(x-5)^2$.

(c) $x = -2, 5 \pm 5\sqrt{2}$.

Mark scheme checked - Topic checked - WMA11-01_2026_Jan_Q06

TOPIC

Graphs of Functions

Question 6**Graphs of Functions**

6. The curve C has equation $y = \frac{4}{x} + k$, where k is a positive constant.

- (a) Sketch a graph of C , stating the equation of the horizontal asymptote and the coordinates of the point of intersection with the x -axis.

(3)

The line with equation $y = 10 - 2x$ is a tangent to C .

- (b) Find the possible values for k .

(5)

(Total 8 marks)

Worked Solution - Question 6

1. Read the graph features

For $y = \frac{4}{x} + k$, the horizontal asymptote is $y = k$. Setting $y = 0$ gives $0 = \frac{4}{x} + k$, so the x-intercept is $x = -\frac{4}{k}$.

2. Set the tangent line equal to the curve

A tangent touches once, so solve $10 - 2x = \frac{4}{x} + k$. Multiplying by x gives $2x^2 + (k - 10)x + 4 = 0$.

3. Use the discriminant condition

For one repeated intersection, discriminant = 0: $(k - 10)^2 - 4(2)(4) = 0$. Hence $(k - 10)^2 = 32$.

4. Find k

$k - 10 = \pm 4\sqrt{2}$, so $k = 10 \pm 4\sqrt{2}$. Both are positive, so both satisfy the condition in the question.

Final answer

Asymptote $y = k$, x-intercept $(-\frac{4}{k}, 0)$, and $k = 10 \pm 4\sqrt{2}$.

Question 10

Graphs of Functions

10.

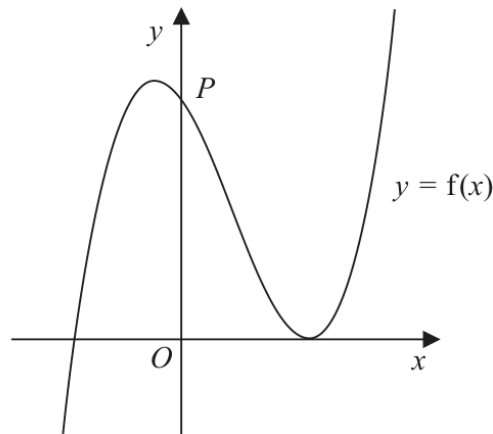


Figure 6

Figure 6 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (2x + 5)(x - 3)^2$$

- (a) Deduce the values of x for which $f(x) \leq 0$ (2)

The curve crosses the y -axis at the point P , as shown.

- (b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a , b , c and d are integers to be found. (3)

- (c) Hence, or otherwise, find

- (i) the coordinates of P ,
- (ii) the gradient of the curve at P . (2)

The curve with equation $y = f(x)$ is translated two units in the positive x direction to a curve with equation $y = g(x)$.

- (d) (i) Find $g(x)$, giving your answer in a simplified factorised form.
- (ii) Hence state the y intercept of the curve with equation $y = g(x)$. (3)

(Total 10 marks)

Worked Solution - Question 10

1. Use the factorised form for the inequality

$f(x) = (2x + 5)(x - 3)^2$. Since $(x - 3)^2 \geq 0$, the sign is controlled by $2x + 5$, with an extra zero at $x = 3$. Hence $f(x) \leq 0$ for $x \leq -\frac{5}{2}$ or $x = 3$.

2. Expand the cubic

$(x - 3)^2 = x^2 - 6x + 9$, so

$$f(x) = (2x + 5)(x^2 - 6x + 9) = 2x^3 - 7x^2 - 12x + 45.$$

3. Find P and the gradient at P

At the y-axis, $x = 0$, so $P = (0, 45)$. Also $f'(x) = 6x^2 - 14x - 12$, so the gradient at P is $f'(0) = -12$.

4. Translate the graph

A translation 2 units in the positive x direction means $g(x) = f(x - 2)$. Therefore $g(x) = (2(x - 2) + 5)((x - 2) - 3)^2 = (2x + 1)(x - 5)^2$.

5. Find the new y-intercept

Set $x = 0$: $g(0) = (1)(25) = 25$, so the y-intercept is 25.

Final answer

$f(x) \leq 0$ for $x \leq -\frac{5}{2}$ or $x = 3$; $f(x) = 2x^3 - 7x^2 - 12x + 45$; $P = (0, 45)$
; gradient = -12 ; $g(x) = (2x + 1)(x - 5)^2$; y-intercept 25.

Question 10

Graphs of Functions

10. The curve C_1 has equation $y = f(x)$, where

$$f(x) = (4x - 3)(x - 5)^2$$

- (a) Sketch C_1 showing the coordinates of any point where the curve touches or crosses the coordinate axes.

(3)

- (b) Hence or otherwise

(i) find the values of x for which $f\left(\frac{1}{4}x\right) = 0$

- (ii) find the value of the constant p such that the curve with equation $y = f(x) + p$ passes through the origin.

(2)

A second curve C_2 has equation $y = g(x)$, where $g(x) = f(x + 1)$

- (c) (i) Find, in simplest form, $g(x)$. You may leave your answer in a factorised form.

- (ii) Hence, or otherwise, find the y intercept of curve C_2

(3)

(Total 8 marks)

Worked Solution - Question 10

1. Use the factorised form for the sketch

$f(x) = (4x - 3)(x - 5)^2$. The curve crosses the x-axis at $x = \frac{3}{4}$ and touches the x-axis at $x = 5$. The y-intercept is $f(0) = (-3)(25) = -75$.

2. Solve $f(x/4)=0$

For $f\left(\frac{1}{4}x\right) = 0$, either $4\left(\frac{x}{4}\right) - 3 = 0$ or $\frac{x}{4} - 5 = 0$. Therefore $x = 3$ or $x = 20$.

3. Find p

The curve $y = f(x) + p$ passes through the origin, so $0 = f(0) + p = -75 + p$.
Hence $p = 75$.

4. Find g(x)

$$g(x) = f(x+1) = (4(x+1) - 3)((x+1) - 5)^2 = (4x+1)(x-4)^2.$$

5. Find the y-intercept of C2

At $x = 0$, $g(0) = (1)(16) = 16$. So the y-intercept is 16.

Final answer

Sketch: crosses at $\left(\frac{3}{4}, 0\right)$, touches at $(5, 0)$, crosses y-axis at $(0, -75)$. Also $x = 3, 20, p = 75, g(x) = (4x+1)(x-4)^2$, y-intercept 16.

Question 6

Graphs of Functions

6. (a) Sketch the curve with equation

$$y = -\frac{k}{x} \quad k > 0 \quad x \neq 0 \quad (2)$$

- (b) On a separate diagram, sketch the curve with equation

$$y = -\frac{k}{x} + k \quad k > 0 \quad x \neq 0$$

stating the coordinates of the point of intersection with the x -axis and, in terms of k , the equation of the horizontal asymptote.

(3)

- (c) Find the range of possible values of k for which the curve with equation

$$y = -\frac{k}{x} + k \quad k > 0 \quad x \neq 0$$

does not touch or intersect the line with equation $y = 3x + 4$

(5)

(Total 10 marks)

Worked Solution - Question 6

1. Sketch $y = -k/x$

Because $k > 0$, $y = -\frac{k}{x}$ has one branch in quadrant II and one branch in quadrant IV, with asymptotes $x = 0$ and $y = 0$.

2. Translate the graph upward

$y = -\frac{k}{x} + k$ is $y = -\frac{k}{x}$ translated up by k units. The vertical asymptote is still $x = 0$, and the horizontal asymptote is $y = k$.

3. Find the x-intercept

Set $y = 0$: $0 = -\frac{k}{x} + k$. Since $k > 0$, $\frac{k}{x} = k$, so $x = 1$. The x-intercept is $(1, 0)$.

4. Set up the intersection equation

For intersections with $y = 3x + 4$, solve $-\frac{k}{x} + k = 3x + 4$. Multiplying by x gives $-k + kx = 3x^2 + 4x$.

5. Use the discriminant

This rearranges to $3x^2 + (4 - k)x + k = 0$. For no touch or intersection, the discriminant must be negative: $(4 - k)^2 - 12k < 0$.

6. Solve the quadratic inequality in k

$k^2 - 20k + 16 < 0$. The roots are $k = 10 \pm 2\sqrt{21}$, so the required range is $10 - 2\sqrt{21} < k < 10 + 2\sqrt{21}$.

Final answer

(b) $(1, 0)$, $y = k$.

$\frac{10-2\sqrt{21}}{c}$

Mark scheme checked - Topic checked - WMA11-01_2021_Jan_Q06

Question 8

Graphs of Functions

8.

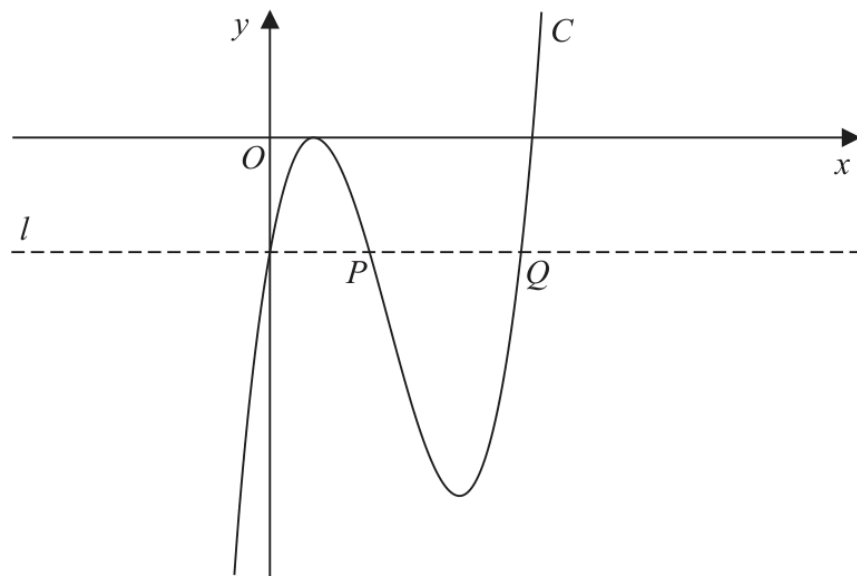


Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x - 2)^2(x - 4)$$

- (a) Deduce the values of x for which $f(x) > 0$

(1)

- (b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a , b , c and d are integers to be found.

(3)

The line l , also shown in Figure 4, passes through the y intercept of C and is parallel to the x -axis.

The line l cuts C again at points P and Q , also shown in Figure 4.

- (c) Using algebra and showing your working, find the length of line PQ . Write your answer in the form $k\sqrt{3}$, where k is a constant to be found.

(Solutions relying entirely on calculator technology are not acceptable.)

(5)

(Total 9 marks)

Worked Solution - Question 8

1. Use the factorised form

$f(x) = (3x - 2)^2(x - 4)$. The squared factor is never negative, and it is zero at $x = \frac{2}{3}$. Therefore the sign of $f(x)$ away from the repeated root is controlled by $x - 4$.

2. Find where $f(x)$ is positive

For $f(x) > 0$, we need $x - 4 > 0$, so $x > 4$.

3. Expand $f(x)$

$(3x - 2)^2 = 9x^2 - 12x + 4$. Therefore

$$f(x) = (9x^2 - 12x + 4)(x - 4) = 9x^3 - 48x^2 + 52x - 16.$$

4. Find the line l

The y-intercept of C is $f(0) = -16$, so the horizontal line l is $y = -16$.

5. Find the other intersections with l

Solve $f(x) = -16$: $9x^3 - 48x^2 + 52x - 16 = -16$, so $x(9x^2 - 48x + 52) = 0$.

The root $x = 0$ is the y-intercept, and the two other roots give P and Q .

6. Solve the quadratic

$9x^2 - 48x + 52 = 0$. Hence

$$x = \frac{48 \pm \sqrt{48^2 - 4(9)(52)}}{18} = \frac{48 \pm 12\sqrt{3}}{18} = \frac{8 \pm 2\sqrt{3}}{3}.$$

7. Find PQ

Because P and Q lie on the same horizontal line, PQ is the difference in their x-

$$\text{coordinates: } PQ = \frac{8 + 2\sqrt{3}}{3} - \frac{8 - 2\sqrt{3}}{3} = \frac{4\sqrt{3}}{3}.$$

Final answer

(a) $x > 4$.

(b) $f(x) = 9x^3 - 48x^2 + 52x - 16$.

(c) $PQ = \frac{4\sqrt{3}}{3}$.

Mark scheme checked - Topic checked - WMA11-01_2021_Jan_Q08

Question 6

Graphs of Functions

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = f(x)$ where

$$f(x) = 2(x + 1)(x - 3)^2$$

(a) Sketch a graph of C .

Show on your graph the coordinates of the points where C cuts or meets the coordinate axes.

(3)

(b) Write $f(x)$ in the form $ax^3 + bx^2 + cx + d$, where a , b , c and d are constants to be found.

(3)

(c) Hence, find the equation of the tangent to C at the point where $x = \frac{1}{3}$

(4)

(Total 10 marks)

Worked Solution - Question 6

1. Use the factorised form for the sketch

$f(x) = 2(x + 1)(x - 3)^2$. The curve cuts the x-axis at $(-1, 0)$ and meets the x-axis at $(3, 0)$ because $x = 3$ is a repeated root.

2. Find the y-intercept

$f(0) = 2(1)(9) = 18$, so the y-intercept is $(0, 18)$.

3. Describe the cubic shape

The leading term is positive, so the cubic goes down to the left and up to the right. It crosses at $x = -1$ and touches at $x = 3$.

4. Expand $f(x)$

$(x + 1)(x - 3)^2 = (x + 1)(x^2 - 6x + 9) = x^3 - 5x^2 + 3x + 9$, so
 $f(x) = 2x^3 - 10x^2 + 6x + 18$.

5. Differentiate

$f'(x) = 6x^2 - 20x + 6$.

6. Find the gradient at $x = 1/3$

$f'\left(\frac{1}{3}\right) = 6\left(\frac{1}{9}\right) - 20\left(\frac{1}{3}\right) + 6 = 0$, so the tangent is horizontal.

7. Find the y-coordinate

$f\left(\frac{1}{3}\right) = 2\left(\frac{4}{3}\right)\left(-\frac{8}{3}\right)^2 = \frac{512}{27}$.

8. Write the tangent

A horizontal tangent through this point has equation $y = \frac{512}{27}$.

Final answer

(a) x -intercepts $(-1, 0)$, $(3, 0)$, y -intercept $(0, 18)$.

(b) $f(x) = 2x^3 - 10x^2 + 6x + 18$.

(c) $y = \frac{512}{27}$.

Mark scheme checked - Topic checked - WMA11-01_2021_Oct_Q06

Question 10

Graphs of Functions

10. The curve C has equation

$$y = \frac{1}{x^2} - 9$$

(a) Sketch the graph of C .

On your sketch

- show the coordinates of any points of intersection with the coordinate axes
- state clearly the equations of any asymptotes

(4)

The curve D has equation $y = kx^2$ where k is a constant.

Given that C meets D at 4 distinct points,

(b) find the range of possible values for k .

(5)

(Total 9 marks)

Worked Solution - Question 10

1. Find intercepts

For x-intercepts, set $\frac{1}{x^2} - 9 = 0$. Then $x^2 = \frac{1}{9}$, so $x = \pm \frac{1}{3}$. There is no y-intercept because $x = 0$ is not in the domain.

2. State asymptotes

The vertical asymptote is $x = 0$. As $|x|$ becomes large, $\frac{1}{x^2} \rightarrow 0$, so the horizontal asymptote is $y = -9$.

3. Set up intersections with D

For intersections with $y = kx^2$, solve $\frac{1}{x^2} - 9 = kx^2$.

4. Use $u = x$ squared

Let $u = x^2$, where $u > 0$. Multiplying by x^2 gives $1 - 9u = ku^2$, so $ku^2 + 9u - 1 = 0$.

5. Require two positive u-values

Four distinct x-values require two distinct positive values of u , since each positive u gives $x = \pm\sqrt{u}$.

6. Find the range of k

For two positive roots, the product $-\frac{1}{k}$ and sum $-\frac{9}{k}$ must both be positive, so $k < 0$. Also the discriminant must be positive: $81 + 4k > 0$, so $k > -\frac{81}{4}$. Therefore $-\frac{81}{4} < k < 0$.

Final answer

(a) x -intercepts $\left(-\frac{1}{3}, 0\right), \left(\frac{1}{3}, 0\right)$; asymptotes $x = 0, y = -9$.

(b) $-\frac{81}{4}$

Mark scheme checked - Topic checked - WMA11-01_2022_Jan_Q10

Question 6

Graphs of Functions

6. (a) Given that k is a positive constant such that $0 < k < 4$ sketch, on **separate axes**, the graphs of

(i) $y = (2x - k)(x + 4)^2$

(ii) $y = \frac{k}{x^2}$

showing the coordinates of any points where the graphs cross or meet the coordinate axes, leaving coordinates in terms of k , where appropriate.

(5)

- (b) State, with a reason, the number of roots of the equation

$$(2x - k)(x + 4)^2 = \frac{k}{x^2}$$

(1)

(Total for Question 6 is 6 marks)

Worked Solution - Question 6

1. Key points for $y = (2x - k)(x + 4)^2$

The graph has x-intercepts $(-4, 0)$ and $\left(\frac{k}{2}, 0\right)$. The root at $x = -4$ is a repeated root, so the graph touches the x-axis there.

2. Use $0 < k < 4$

Since $0 < k < 4$

3. Key features for $y = k/x^2$

The graph is positive on both sides of the y-axis, with asymptotes $x = 0$ and $y = 0$. It has no intercepts with the coordinate axes.

4. Compare the two graphs

The equation $(2x - k)(x + 4)^2 = \frac{k}{x^2}$ represents intersections between the two graphs.

5. State the number of roots

From the sketches, the positive branch of $\frac{k}{x^2}$ meets the increasing positive branch of $(2x - k)(x + 4)^2$ once only. Therefore there is one root.

Final answer

The graphs intersect once, so the equation has one root.

Question 7

Graphs of Functions

7. (a) Sketch the graph of the curve C with equation

$$y = \frac{4}{x - k}$$

where k is a positive constant.

Show on your sketch

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

Given that the straight line with equation $y = 9 - x$ does not cross or touch C

(b) find the range of values of k .

(5)

(Total for Question 7 is 9 marks)

Worked Solution - Question 7

1. Identify the asymptote

For $y = \frac{4}{x - k}$, the vertical asymptote is where the denominator is zero, so $x = k$.

2. Find axis intercepts

At $x = 0$, $y = -\frac{4}{k}$, so the y-intercept is $(0, -\frac{4}{k})$. The graph has no x-intercept because the numerator is never zero.

3. Set up intersection with the line

If the line $y = 9 - x$ meets the curve, then $9 - x = \frac{4}{x - k}$.

4. Form a quadratic

$(9 - x)(x - k) = 4$, which simplifies to $x^2 - (9 + k)x + 9k + 4 = 0$.

5. Use no real intersection

For the line not to cross or touch the curve, the discriminant must be negative.

6. Solve the inequality

$(9 + k)^2 - 4(9k + 4) < 0$, so $k^2 - 18k + 65 < 0$. Since $k^2 - 18k + 65 = (k - 5)(k - 13)$, the range is \$5

Final answer

(a) y-intercept $(0, -\frac{4}{k})$, no x-intercept, vertical asymptote $x = k$.

\$(b) \setminus 5\$

Question 8

Graphs of Functions

8. The curve C_1 has equation

$$y = x(4 - x^2)$$

- (a) Sketch the graph of C_1 showing the coordinates of any points of intersection with the coordinate axes.

(3)

The curve C_2 has equation $y = \frac{A}{x}$ where A is a constant.

- (b) Show that the x coordinates of the points of intersection of C_1 and C_2 satisfy the equation

$$x^4 - 4x^2 + A = 0$$

(1)

- (c) Hence find the range of possible values of A for which C_1 meets C_2 at 4 distinct points.

(3)

(Total for Question 8 is 7 marks)

Worked Solution - Question 8

1. Find the intercepts of C1

$y = x(4 - x^2)$, so $y = 0$ when $x = 0$ or $x = \pm 2$. The curve crosses the axes at $(-2, 0)$, $(0, 0)$ and $(2, 0)$.

2. Use the cubic shape

$y = 4x - x^3$ has negative leading cubic term, so the left tail rises and the right tail falls.

3. Equate C1 and C2

$$x(4 - x^2) = \frac{A}{x}.$$

4. Multiply by x

$x^2(4 - x^2) = A$, so $4x^2 - x^4 = A$ and therefore $x^4 - 4x^2 + A = 0$.

5. Let u equal x squared

Let $u = x^2$. Then $u^2 - 4u + A = 0$.

6. Require two positive u-values

For 4 distinct x-values, the quadratic in u must have two distinct positive roots.

7. Find the range of A

The discriminant gives $16 - 4A > 0$, so $A < 4$. The product of the two roots is A , so $A > 0$. Therefore $0 < A < 4$.

Final answer

(a) x-intercepts $(-2, 0)$, $(0, 0)$, $(2, 0)$.

(b) $x^4 - 4x^2 + A = 0$.

(c) 0

Mark scheme checked - Topic checked - WMA11-
01_2024_MayJune_Q08

Question 6

Graphs of Functions

6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Sketch the curve C with equation

$$y = \frac{1}{2 - x} \quad x \neq 2$$

State on your sketch

- the equation of the vertical asymptote
- the coordinates of the intersection of C with the y -axis

(3)

The straight line l has equation $y = kx - 4$, where k is a constant.

Given that l cuts C at least once,

(b) (i) show that

$$k^2 - 5k + 4 \geq 0$$

(ii) find the range of possible values for k .

(6)

(Total for Question 6 is 9 marks)

Worked Solution - Question 6

1. Identify the graph features

$y = \frac{1}{2-x}$ has vertical asymptote $x = 2$ and horizontal asymptote $y = 0$.

2. Find the y-intercept

At $x = 0$, $y = \frac{1}{2}$, so the y-intercept is $(0, \frac{1}{2})$.

3. Set up the intersection with the line

If $y = kx - 4$ cuts C , then $kx - 4 = \frac{1}{2-x}$.

4. Form the quadratic

$(kx - 4)(2 - x) = 1$, so $kx^2 - (2k + 4)x + 9 = 0$.

5. Use the discriminant

For at least one intersection, the discriminant is non-negative.

6. Derive the given inequality

$(2k + 4)^2 - 4(k)(9) \geq 0$, so $4(k^2 - 5k + 4) \geq 0$, hence $k^2 - 5k + 4 \geq 0$.

7. Solve the inequality

$k^2 - 5k + 4 = (k - 1)(k - 4)$, so $k \leq 1$ or $k \geq 4$.

Final answer

(a) vertical asymptote $x = 2$, y-intercept $(0, \frac{1}{2})$.

(b) $k \leq 1$ or $k \geq 4$.

Question 7

Graphs of Functions

7. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = \frac{2}{x} - k$$

where k is a **positive** constant.

- (a) Sketch the graph of C .

Show on your sketch

- the coordinates of any points of intersection of C with the coordinate axes
- the equation of the horizontal asymptote to C

stating each in terms of k .

(3)

The line l has equation $y = -kx - 6$

Given that l intersects C at 2 distinct points,

- (b) find the range of possible values of k .

(5)

(Total for Question 7 is 8 marks)

Worked Solution - Question 7

1. Identify the horizontal asymptote

$y = \frac{2}{x} - k$ has horizontal asymptote $y = -k$.

2. Find axis intersections

Setting $y = 0$ gives $\frac{2}{x} = k$, so $x = \frac{2}{k}$. There is no y-intercept because $x = 0$ is not in the domain.

3. Set up line-curve intersections

$$\frac{2}{x} - k = -kx - 6.$$

4. Form a quadratic

Multiplying by x gives $2 - kx = -kx^2 - 6x$, so $kx^2 + (6 - k)x + 2 = 0$.

5. Use two distinct intersections

For two distinct points, the discriminant must be positive.

6. Solve the inequality

$(6 - k)^2 - 8k > 0$, so $k^2 - 20k + 36 > 0$. Since $k^2 - 20k + 36 = (k - 2)(k - 18)$, $k < 2$ or $k > 18$.

7. Use k positive

As $k > 0$, the range is \$018\$.

Final answer

(a) x-intercept $(\frac{2}{k}, 0)$, no y-intercept, horizontal asymptote $y = -k$.

\$(b)\ 018\$.

Question 7

Graphs of Functions

7.

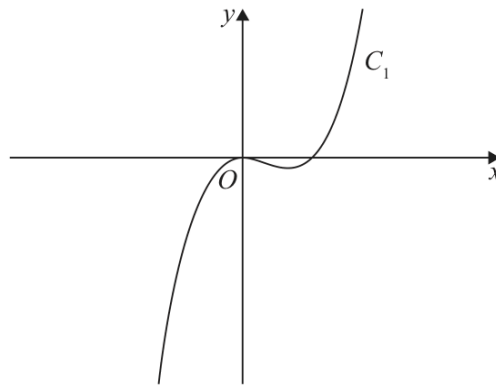


Figure 3

Figure 3 shows a sketch of part of the curve C_1

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a cubic function
- touches the x -axis at the origin and cuts the x -axis at $x = 4$
- passes through the point $(10, 120)$

(a) find $f(x)$

(3)

The curve C_2 has equation $y = 1.2x(8 - x)$

On the following page there is a copy of Figure 3 called Diagram 1.

(b) On Diagram 1 sketch a graph of the curve C_2

(2)

(c) Use algebra to find the coordinates of the points where C_1 and C_2 intersect.
Show each stage of your working.

(5)

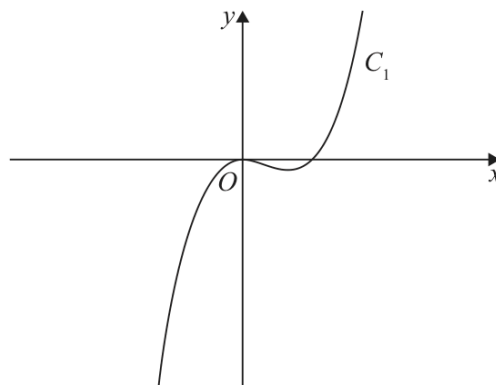


Diagram 1

(Total for Question 7 is 10 marks)

Worked Solution - Question 7

1. Use the repeated root at the origin

Since the cubic touches the x -axis at the origin and cuts it at $x = 4$, write

$$f(x) = \lambda x^2(x - 4).$$

2. Use the point (10,120)

$$120 = \lambda(10)^2(10 - 4) = 600\lambda, \text{ so } \lambda = 0.2.$$

3. State $f(x)$

$$f(x) = 0.2x^2(x - 4).$$

4. Sketch C_2

$C_2 : y = 1.2x(8 - x)$ is a downward-opening parabola through $x = 0$ and $x = 8$, with its maximum between them.

5. Set intersections equal

$$1.2x(8 - x) = 0.2x^2(x - 4).$$

6. Clear decimals

Divide by 0.2 to get $6x(8 - x) = x^2(x - 4)$.

7. Factor

$$48x - 6x^2 = x^3 - 4x^2, \text{ so } x^3 + 2x^2 - 48x = 0 \text{ and } x(x^2 + 2x - 48) = 0.$$

8. Solve for x

$$x = 0, \text{ or } x^2 + 2x - 48 = 0 = (x + 8)(x - 6), \text{ so } x = -8 \text{ or } x = 6.$$

9. Find y -values

Using $y = 1.2x(8 - x)$ gives $(-8, -153.6)$, $(0, 0)$ and $(6, 14.4)$.

Final answer

(a) $f(x) = 0.2x^2(x - 4)$.

(c) $(-8, -153.6)$, $(0, 0)$, $(6, 14.4)$.

Mark scheme checked - Topic checked - WMA11-01_2025_MayJune_Q07

Question 6

Graphs of Functions

6. (a) Sketch the graph of the curve C with equation

$$y = \frac{4k}{x - 2k}$$

where k is a positive constant.

On your sketch show

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

The straight line l has equation

$$y = 6 - 2x$$

Given that there is at least one point of intersection between l and C ,

(b) find the range of possible values of k .

(5)

(Total for Question 6 is 9 marks)

Worked Solution - Question 6

1. Find coordinate-axis intersections

$y = \frac{4k}{x - 2k}$ has no x-intercept because the numerator is non-zero. At $x = 0$,
 $y = \frac{4k}{-2k} = -2$, so the y-intercept is $(0, -2)$.

2. Find the vertical asymptote

The denominator is zero when $x = 2k$, so the vertical asymptote is $x = 2k$.

3. Sketch shape

The curve has one branch in quadrant I to the right of $x = 2k$, and one branch passing through $(0, -2)$ and approaching the x-axis from below as $x \rightarrow -\infty$.

4. Set line and curve equal

$$\frac{4k}{x - 2k} = 6 - 2x.$$

5. Form a quadratic in x

$$4k = (6 - 2x)(x - 2k) = 6x - 12k - 2x^2 + 4kx, \text{ so } 2x^2 - (4k + 6)x + 16k = 0.$$

6. Use at least one intersection

For at least one real intersection, the discriminant must satisfy $b^2 - 4ac \geq 0$.

7. Find the discriminant

$$(4k + 6)^2 - 4(2)(16k) \geq 0, \text{ so } 16k^2 - 80k + 36 \geq 0.$$

8. Solve in k

$$4k^2 - 20k + 9 \geq 0, \text{ so } (2k - 1)(2k - 9) \geq 0.$$

9. Use k positive

Therefore $k \leq \frac{1}{2}$ or $k \geq \frac{9}{2}$. Since k is positive, \$0

Final answer

(a) y-intercept $(0, -2)$, vertical asymptote $x = 2k$.

\$(b) \setminus 0

Mark scheme checked - Topic checked - WMA11-01_2025_Oct_Q06

TOPIC

Transformations

Question 9

Transformations

9. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

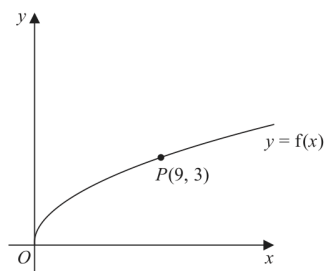


Figure 5

Figure 5 shows a sketch of the curve with equation $y = f(x)$ where

$$f(x) = \sqrt{x} \quad x > 0$$

The point $P(9, 3)$ lies on the curve and is shown in Figure 5.

On the next page there is a copy of Figure 5 called Diagram 1.

- (a) On Diagram 1, sketch and clearly label the graphs of

$$y = f(2x) \quad \text{and} \quad y = f(x) + 3$$

Show on each graph the coordinates of the point to which P is transformed.

(3)

The graph of $y = f(2x)$ meets the graph of $y = f(x) + 3$ at the point Q .

- (b) Show that the x coordinate of Q is the solution of

$$\sqrt{x} = 3(\sqrt{2} + 1)$$

(3)

- (c) Hence find, in simplest form, the coordinates of Q .

(3)

Question 9 continued

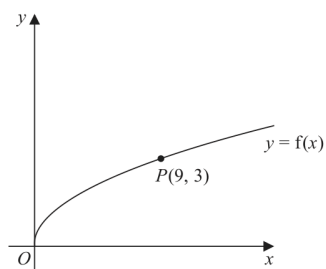
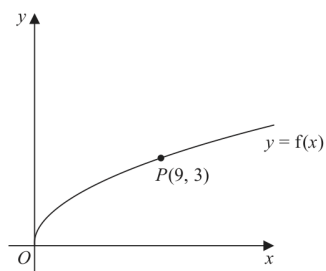


Diagram 1

Turn over for a copy of Diagram 1 if you need to redraw your graphs.

Only use this copy if you need to redraw your graphs.



Copy of Diagram 1

(Total 9 marks)

Worked Solution - Question 9

1. Transform P on $y = f(2x)$

$y = f(2x)$ is a horizontal scale factor of $\frac{1}{2}$, so $P(9, 3)$ maps to $\left(\frac{9}{2}, 3\right)$.

2. Transform P on $y = f(x) + 3$

$y = f(x) + 3$ is a translation 3 units upward, so $P(9, 3)$ maps to $(9, 6)$.

3. Set up the intersection equation

At Q , $f(2x) = f(x) + 3$. Since $f(x) = \sqrt{x}$, this gives $\sqrt{2x} = \sqrt{x} + 3$.

4. Show the required equation

Let $s = \sqrt{x}$. Then $\sqrt{2x} = \sqrt{2}s$, so $\sqrt{2}s = s + 3$. Hence $(\sqrt{2} - 1)s = 3$ and $s = \frac{3}{\sqrt{2} - 1} = 3(\sqrt{2} + 1)$.

5. Find the x-coordinate

$\sqrt{x} = 3(\sqrt{2} + 1)$, so $x = 9(\sqrt{2} + 1)^2 = 9(3 + 2\sqrt{2}) = 27 + 18\sqrt{2}$.

6. Find the y-coordinate

$y = \sqrt{x} + 3 = 3(\sqrt{2} + 1) + 3 = 6 + 3\sqrt{2}$. Therefore $Q = (27 + 18\sqrt{2}, 6 + 3\sqrt{2})$.

Final answer

(a) $P \mapsto \left(\frac{9}{2}, 3\right)$ on $y = f(2x)$, $P \mapsto (9, 6)$ on $y = f(x) + 3$.

(c) $Q = (27 + 18\sqrt{2}, 6 + 3\sqrt{2})$.

Question 7

Transformations

7.

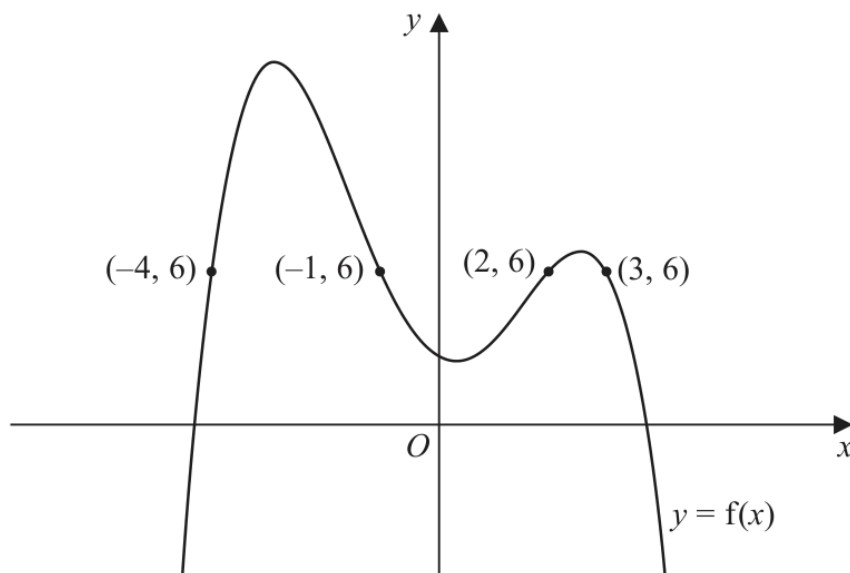


Figure 1

Figure 1 shows the curve with equation $y = f(x)$.

The points $P(-4, 6)$, $Q(-1, 6)$, $R(2, 6)$ and $S(3, 6)$ lie on the curve.

(a) Using Figure 1, find the range of values of x for which

$$f(x) < 6$$

(3)

(b) State the largest solution of the equation

$$f(2x) = 6$$

(1)

(c) (i) Sketch the curve with equation $y = f(-x)$.

On your sketch, state the coordinates of the points to which P , Q , R and S are transformed.

(ii) Hence find the set of values of x for which

$$f(-x) \geq 6 \text{ and } x < 0$$

(4)

(Total for Question 7 is 8 marks)

Worked Solution - Question 7

1. Read $f(x) < 6$ from the graph

The curve is below the line $y = 6$ for $x < -4$, for $\$-13\$$.

2. Solve $f(2x) = 6$

Since $f(u) = 6$ at $u = -4, -1, 2, 3$, we set $2x = -4, -1, 2, 3$. The solutions are $x = -2, -\frac{1}{2}, 1, \frac{3}{2}$.

3. State the largest solution

The largest solution is $x = \frac{3}{2}$.

4. Reflect the points in the y-axis

For $y = f(-x)$, each point (x, y) becomes $(-x, y)$.

5. List the transformed points

$P(-4, 6) \rightarrow (4, 6)$, $Q(-1, 6) \rightarrow (1, 6)$, $R(2, 6) \rightarrow (-2, 6)$, and $S(3, 6) \rightarrow (-3, 6)$.

6. Use the reflected graph

For $x < 0$, the part of the reflected graph on or above $y = 6$ is from $x = -3$ to $x = -2$.

Final answer

(a) $x < -4$, $\$-13\$$

(b) $\frac{3}{2}$

(c)(ii) $-3 \leq x \leq -2$

Question 7

Transformations

7. (a) On Diagram 1, sketch a graph of the curve C with equation

$$y = \frac{6}{x} \quad x \neq 0$$

(2)

The curve C is transformed onto the curve with equation $y = \frac{6}{x-2} \quad x \neq 2$

- (b) Fully describe this transformation.

(2)

The curve with equation

$$y = \frac{6}{x-2} \quad x \neq 2$$

and the line with equation

$$y = kx + 7 \quad \text{where } k \text{ is a constant}$$

intersect at exactly two points, P and Q .

Given that the x coordinate of point P is -4

- (c) find the value of k ,

(2)

- (d) find, using algebra, the coordinates of point Q .

(Solutions relying entirely on calculator technology are not acceptable.)

(4)

Question 7 continued

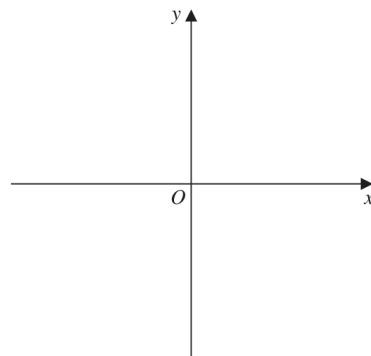
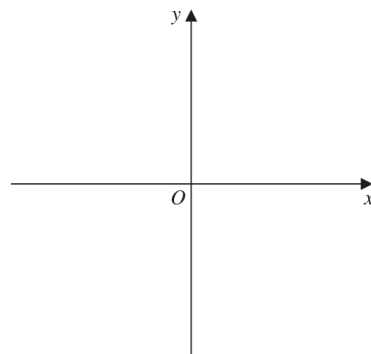


Diagram 1

Only use this copy of Diagram 1 if you need to redraw your graph.



Copy of Diagram 1

(Total for Question 7 is 10 marks)

Worked Solution - Question 7

1. Sketch $y = 6/x$

The graph is a rectangular hyperbola with asymptotes $x = 0$ and $y = 0$, with branches in quadrants I and III.

2. Describe the transformation

$y = \frac{6}{x-2}$ is obtained from $y = \frac{6}{x}$ by a translation 2 units in the positive x -direction.

3. Use P to find k

At $x = -4$ on $y = \frac{6}{x-2}$, $y = \frac{6}{-6} = -1$.

4. Substitute into the line

Since P lies on $y = kx + 7$, $-1 = -4k + 7$, so $k = 2$.

5. Find the second intersection

Solve $\frac{6}{x-2} = 2x + 7$. Multiplying by $x - 2$ gives $6 = (2x + 7)(x - 2)$.

6. Factor and choose Q

$6 = 2x^2 + 3x - 14$, so $2x^2 + 3x - 20 = 0 = (2x - 5)(x + 4)$. The root $x = -4$ is P , so Q has $x = \frac{5}{2}$.

7. Find y for Q

$y = 2\left(\frac{5}{2}\right) + 7 = 12$, so $Q = \left(\frac{5}{2}, 12\right)$.

Final answer

(b) Translation 2 units right.

(c) $k = 2$.

(d) $Q = \left(\frac{5}{2}, 12\right)$.

Mark scheme checked - Topic checked - WMA11-01_2023_Jan_Q07

Question 9

Transformations

9.

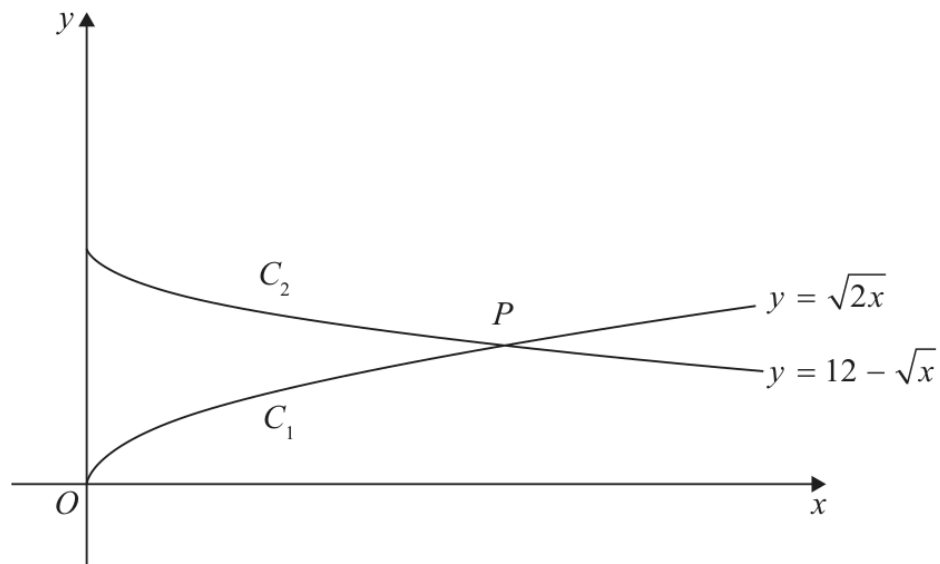


Figure 4

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Figure 4 shows a sketch of

- the graph C_1 with equation $y = \sqrt{2x}$
- the graph C_2 with equation $y = 12 - \sqrt{x}$

(a) Describe fully the single transformation that would transform

- the graph with equation $y = \sqrt{x}$ onto C_1
- the graph with equation $y = -\sqrt{x}$ onto C_2

(4)

The graphs C_1 and C_2 meet at the point P , as shown in Figure 4.

(b) (i) Show that the x coordinate of P is a solution of

$$\sqrt{x} = 12(\sqrt{2} - 1)$$

(ii) Hence find, in simplest form, the exact coordinates of P .

(6)

(Total for Question 9 is 10 marks)

Worked Solution - Question 9

1. Transform y equals root x to C_1

C_1 has equation $y = \sqrt{2x}$. This is obtained from $y = \sqrt{x}$ by a stretch parallel to the x -axis with scale factor $\frac{1}{2}$.

2. Equivalent description

Equivalently, $y = \sqrt{2x} = \sqrt{2}\sqrt{x}$ is a stretch parallel to the y -axis with scale factor $\sqrt{2}$. The mark scheme also accepts the horizontal stretch description.

3. Transform y equals minus root x to C_2

C_2 has equation $y = 12 - \sqrt{x} = -\sqrt{x} + 12$, so it is a translation of $y = -\sqrt{x}$ by $\begin{pmatrix} 0 \\ 12 \end{pmatrix}$.

4. Set the curves equal

At P , $\sqrt{2x} = 12 - \sqrt{x}$.

5. Factor out root x

$\sqrt{2}\sqrt{x} + \sqrt{x} = 12$, so $(\sqrt{2} + 1)\sqrt{x} = 12$.

6. Rationalise

$$\sqrt{x} = \frac{12}{\sqrt{2} + 1} = \frac{12(\sqrt{2} - 1)}{(\sqrt{2} + 1)(\sqrt{2} - 1)} = 12(\sqrt{2} - 1).$$

7. Find x

$$x = [12(\sqrt{2} - 1)]^2 = 144(3 - 2\sqrt{2}).$$

8. Find y

$$y = \sqrt{2x} = \sqrt{2}\sqrt{x} = \sqrt{2} \cdot 12(\sqrt{2} - 1) = 24 - 12\sqrt{2}.$$

Final answer

(a)(i) stretch parallel to the x-axis scale factor $\frac{1}{2}$ (equivalently vertical stretch scale factor $\sqrt{2}$).

(a)(ii) translation by $\begin{pmatrix} 0 \\ 12 \end{pmatrix}$.

(b) $P = (144(3 - 2\sqrt{2}), 24 - 12\sqrt{2})$.

Mark scheme checked - Topic checked - WMA11-01_2025_MayJune_Q09

Question 9

Transformations

9.

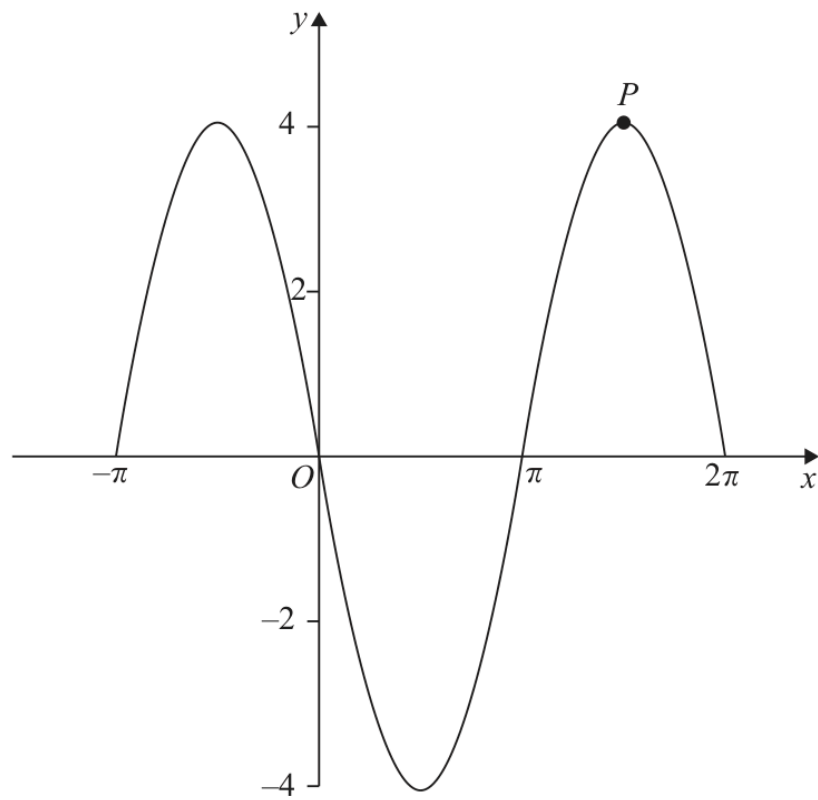


Figure 3

Figure 3 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$

(a) Write down an expression for $f(x)$

(2)

The point P lies on $y = f(x)$ and is shown in Figure 3.

(b) State the coordinates of the point to which P is transformed when the graph of $y = f(x)$ is transformed to the graph with equation

(i) $y = f\left(x - \frac{\pi}{6}\right)$

(2)

(ii) $y = -\frac{1}{2}f(x)$

(2)

(Total for Question 9 is 6 marks)

Worked Solution - Question 9

1. Identify amplitude and direction

The graph has amplitude 4, crosses the origin, and initially moves downwards, so $f(x) = -4 \sin x$.

2. Identify P

The labelled maximum is at $P = \left(\frac{3\pi}{2}, 4\right)$.

3. Transform y equals f of x minus pi over 6

$y = f\left(x - \frac{\pi}{6}\right)$ translates the graph right by $\frac{\pi}{6}$.

4. New coordinates for part b(i)

P becomes $\left(\frac{3\pi}{2} + \frac{\pi}{6}, 4\right) = \left(\frac{5\pi}{3}, 4\right)$.

5. Transform y equals negative half f(x)

$y = -\frac{1}{2}f(x)$ leaves x-coordinates unchanged and multiplies y-coordinates by $-\frac{1}{2}$.

6. New coordinates for part b(ii)

P becomes $\left(\frac{3\pi}{2}, -2\right)$.

Final answer

(a) $f(x) = -4 \sin x$.

(b)(i) $\left(\frac{5\pi}{3}, 4\right)$.

(b)(ii) $\left(\frac{3\pi}{2}, -2\right)$.

TOPIC

Straight Line

Question 6

Straight Line

6. The line l_1 has equation $3x - 4y + 20 = 0$

The line l_2 cuts the x -axis at $R(8,0)$ and is parallel to l_1

- (a) Find the equation of l_2 , writing your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found. (3)

The line l_1 cuts the x -axis at P and the y -axis at Q .

Given that $PQRS$ is a parallelogram, find

- (b) the area of $PQRS$, (3)
- (c) the coordinates of S . (2)

(Total 8 marks)

Worked Solution - Question 6

1. Find the gradient of l_1

$3x - 4y + 20 = 0$ gives $y = \frac{3}{4}x + 5$, so the gradient is $\frac{3}{4}$.

2. Find l_2 through R

Line l_2 is parallel to l_1 and passes through $R(8, 0)$, so $y = \frac{3}{4}(x - 8) = \frac{3}{4}x - 6$.
Therefore $3x - 4y - 24 = 0$.

3. Find P and Q

For l_1 , the x-intercept is $P(-\frac{20}{3}, 0)$ and the y-intercept is $Q = (0, 5)$.

4. Find the area of PQRS

Base $PR = 8 - (-\frac{20}{3}) = \frac{44}{3}$ and height $OQ = 5$. Area = $\frac{44}{3} \cdot 5 = \frac{220}{3}$.

5. Find S using a vector

In parallelogram $PQRS$, $\vec{PS} = \vec{QR} = (8, -5)$. Hence
 $S = P + (8, -5) = (-\frac{20}{3} + 8, -5) = (\frac{4}{3}, -5)$.

Final answer

(a) $3x - 4y - 24 = 0$.

(b) $\frac{220}{3}$.

(c) $S = (\frac{4}{3}, -5)$.

Question 8

Straight Line

8.

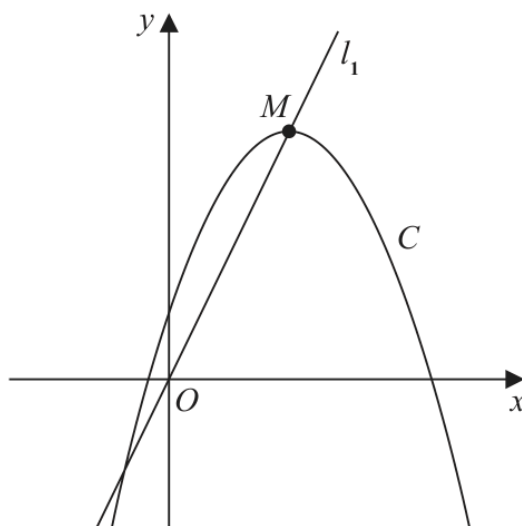


Figure 4

Figure 4 shows a sketch of the curve C with equation

$$y = 4 + 12x - 3x^2$$

The point M is the maximum turning point on C .

- (a) (i) Write $4 + 12x - 3x^2$ in the form

$$a + b(x + c)^2$$

where a , b and c are constants to be found.

- (ii) Hence, or otherwise, state the coordinates of M .

(5)

The line l_1 passes through O and M , as shown in Figure 4.

A line l_2 touches C and is parallel to l_1

- (b) Find an equation for l_2

(5)

(Total 10 marks)

Worked Solution - Question 8

1. Complete the square

$$4 + 12x - 3x^2 = -3(x^2 - 4x) + 4 = -3((x - 2)^2 - 4) + 4 = 16 - 3(x - 2)^2.$$

2. State the maximum point

The maximum occurs when $(x - 2)^2 = 0$, so $M = (2, 16)$.

3. Find the gradient of l_1

Line l_1 passes through $O(0, 0)$ and $M(2, 16)$, so its gradient is $\frac{16}{2} = 8$.

4. Use the parallel tangent gradient

Line l_2 is parallel to l_1 , so its gradient is also 8. For $y = 4 + 12x - 3x^2$,
 $\frac{dy}{dx} = 12 - 6x$.

5. Find the point of contact

Set $12 - 6x = 8$, giving $x = \frac{2}{3}$. Then $y = 4 + 12\left(\frac{2}{3}\right) - 3\left(\frac{4}{9}\right) = \frac{32}{3}$.

6. Write l_2

Using gradient 8 through $\left(\frac{2}{3}, \frac{32}{3}\right)$, $y - \frac{32}{3} = 8\left(x - \frac{2}{3}\right)$. Hence $y = 8x + \frac{16}{3}$.

Final answer

(a)(i) $16 - 3(x - 2)^2$. (a)(ii) $M = (2, 16)$.

(b) $y = 8x + \frac{16}{3}$.

Question 8

Straight Line

8. The line l_1 has equation

$$2x - 5y + 7 = 0$$

- (a) Find the gradient of l_1

(1)

Given that

- the point A has coordinates $(6, -2)$
- the line l_2 passes through A and is perpendicular to l_1

- (b) find the equation of l_2 giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

The lines l_1 and l_2 intersect at the point M .

- (c) Using algebra and showing all your working, find the coordinates of M .

(Solutions relying on calculator technology are not acceptable.)

(3)

Given that the diagonals of a square $ABCD$ meet at M ,

- (d) find the coordinates of the point C .

(2)

(Total 9 marks)

Worked Solution - Question 8

1. Find the gradient of l_1

$2x - 5y + 7 = 0$ gives $y = \frac{2}{5}x + \frac{7}{5}$, so the gradient is $\frac{2}{5}$.

2. Find l_2

The gradient perpendicular to $\frac{2}{5}$ is $-\frac{5}{2}$. Through $A(6, -2)$, $y + 2 = -\frac{5}{2}(x - 6)$, so $y = -\frac{5}{2}x + 13$.

3. Find M

At the intersection, $\frac{2}{5}x + \frac{7}{5} = -\frac{5}{2}x + 13$. Multiplying by 10 gives $4x + 14 = -25x + 130$, so $x = 4$.

4. Find the y-coordinate of M

$y = \frac{2}{5}(4) + \frac{7}{5} = 3$, so $M = (4, 3)$.

5. Use midpoint of the diagonal

In a square, the diagonals bisect each other, so M is the midpoint of AC . Thus $C = 2M - A = (8, 6) - (6, -2) = (2, 8)$.

Final answer

(a) $\frac{2}{5}$.

(b) $y = -\frac{5}{2}x + 13$.

(c) $M = (4, 3)$.

(d) $C = (2, 8)$.

Question 10

Straight Line

10.

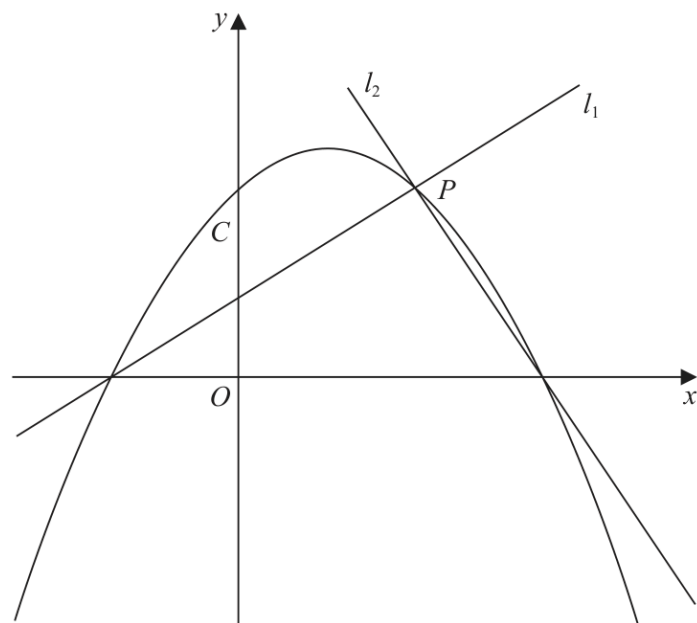


Figure 5

Figure 5 shows a sketch of the quadratic curve C with equation

$$y = -\frac{1}{4}(x+2)(x-b) \quad \text{where } b \text{ is a positive constant}$$

The line l_1 also shown in Figure 5,

- has gradient $\frac{1}{2}$
- intersects C on the negative x -axis and at the point P

(a) (i) Write down an equation for l_1

(1)

(ii) Find, in terms of b , the coordinates of P

(3)

Given that the line l_2 is perpendicular to l_1 and intersects C on the positive x -axis,

(b) find, in terms of b , an equation for l_2

(2)

Given also that l_2 intersects C at the point P

(c) show that another equation for l_2 is

$$y = -2x + \frac{5b}{2} - 4$$

(2)

(d) Hence, or otherwise, find the value of b

(Total for Question 10 is 10 marks)

Worked Solution - Question 10

1. Find l_1

Line l_1 has gradient $\frac{1}{2}$ and passes through the negative x-axis root $(-2, 0)$, so

$$y = \frac{1}{2}(x + 2) = \frac{1}{2}x + 1.$$

2. Find P

Set $-\frac{1}{4}(x + 2)(x - b) = \frac{1}{2}(x + 2)$.

3. Solve the intersection

Multiplying by -4 gives $(x + 2)(x - b) = -2(x + 2)$, so $(x + 2)(x - b + 2) = 0$.

4. Choose the second point

The root $x = -2$ is the x-axis point, so P has $x = b - 2$. Then

$$y = \frac{1}{2}(b - 2) + 1 = \frac{b}{2}.$$

5. Find l_2 from the positive root

Since l_2 is perpendicular to l_1 , its gradient is -2 . It passes through the positive x-axis root $(b, 0)$, so $y = -2(x - b) = -2x + 2b$.

6. Use P to write another l_2

Through $P\left(b - 2, \frac{b}{2}\right)$ with gradient -2 : $y - \frac{b}{2} = -2(x - (b - 2))$, so

$$y = -2x + \frac{5b}{2} - 4.$$

7. Find b

Both equations represent l_2 , so $2b = \frac{5b}{2} - 4$. Hence $b = 8$.

Final answer

$$(a)(i) y = \frac{1}{2}x + 1. \quad (a)(ii) P = \left(b - 2, \frac{b}{2}\right).$$

$$(b) y = -2x + 2b.$$

$$(d) b = 8.$$

Mark scheme checked - Topic checked - WMA11-01_2023_MayJune_Q10

Question 9

Straight Line

9. Given that

- the point A has coordinates $(4, 2)$
- the point B has coordinates $(15, 7)$
- the line l_1 passes through A and B

(a) find an equation for l_1 , giving your answer in the form $px + qy + r = 0$ where p , q and r are integers to be found.

(3)

The line l_2 passes through A and is parallel to the x -axis.

The point C lies on l_2 so that the length of BC is $5\sqrt{5}$

(b) Find both possible pairs of coordinates of the point C .

(4)

(c) Hence find the minimum possible area of triangle ABC .

(2)

(Total for Question 9 is 9 marks)

Worked Solution - Question 9

1. Find the gradient of l_1

$$m = \frac{7 - 2}{15 - 4} = \frac{5}{11}.$$

2. Use point A

$$y - 2 = \frac{5}{11}(x - 4).$$

3. Rearrange

$$11y - 22 = 5x - 20, \text{ so } 5x - 11y + 2 = 0.$$

4. Use the horizontal line l_2

Since l_2 passes through $A(4, 2)$ and is parallel to the x-axis, point C has coordinates $(c, 2)$.

5. Use BC equals 5 root 5

$$(c - 15)^2 + (2 - 7)^2 = (5\sqrt{5})^2.$$

6. Solve for c

$$(c - 15)^2 + 25 = 125, \text{ so } (c - 15)^2 = 100 \text{ and } c = 5 \text{ or } 25.$$

7. Find the smaller area

The height from B to $y = 2$ is 5. For $C = (5, 2)$, $AC = 1$, so the minimum area is $\frac{1}{2}(1)(5) = \frac{5}{2}$.

Final answer

(a) $5x - 11y + 2 = 0$.

(b) $C = (5, 2)$ or $(25, 2)$.

(c) $\frac{5}{2}$.

TOPIC

Radians

Question 7

7.

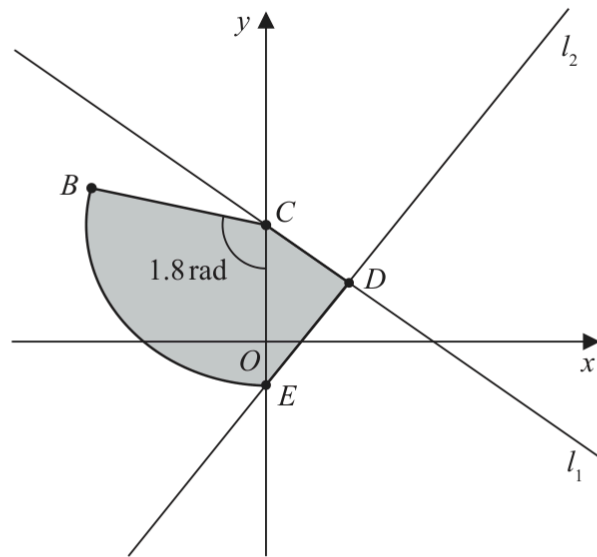


Figure 3

The line l_1 has equation $4y + 3x = 48$

The line l_1 cuts the y -axis at the point C , as shown in Figure 3.

(a) State the y coordinate of C .

(1)

The point $D(8, 6)$ lies on l_1

The line l_2 passes through D and is perpendicular to l_1

The line l_2 cuts the y -axis at the point E as shown in Figure 3.

(b) Show that the y coordinate of E is $-\frac{14}{3}$

(3)

A sector BCE of a circle with centre C is also shown in Figure 3.

Given that angle BCE is 1.8 radians,

(c) find the length of arc BE .

(3)

The region $CBED$, shown shaded in Figure 3, consists of the sector BCE joined to the triangle CDE .

(d) Calculate the exact area of the region $CBED$.

(3)

(Total 10 marks)

Worked Solution - Question 7

1. Find C

On the y-axis, $x = 0$. From $4y + 3x = 48$, $4y = 48$, so the y-coordinate of C is 12.

2. Find the gradient of l_1

$4y + 3x = 48$ gives $y = -\frac{3}{4}x + 12$, so the gradient of l_1 is $-\frac{3}{4}$.

3. Use perpendicular gradients for l_2

Since l_2 is perpendicular to l_1 , its gradient is $\frac{4}{3}$. Through $D(8, 6)$, $y - 6 = \frac{4}{3}(x - 8)$.

4. Show the y-coordinate of E

$y = \frac{4}{3}x - \frac{32}{3} + 6 = \frac{4}{3}x - \frac{14}{3}$. Therefore when $x = 0$, $E_y = -\frac{14}{3}$.

5. Find the radius of the sector

$C = (0, 12)$ and $E = \left(0, -\frac{14}{3}\right)$, so $CE = 12 + \frac{14}{3} = \frac{50}{3}$.

6. Find arc BE

For sector BCE , $r = \frac{50}{3}$ and $\theta = 1.8 = \frac{9}{5}$. Hence arc $BE = r\theta = \frac{50}{3} \cdot \frac{9}{5} = 30$.

7. Find the sector area

Sector area $= \frac{1}{2}r^2\theta = \frac{1}{2}\left(\frac{50}{3}\right)^2\left(\frac{9}{5}\right) = 250$.

8. Find the triangle area and total area

Triangle CDE has vertical base $CE = \frac{50}{3}$ and horizontal height 8, so its area is $\frac{1}{2} \cdot \frac{50}{3} \cdot 8 = \frac{200}{3}$. Total area $= 250 + \frac{200}{3} = \frac{950}{3}$.

Final answer

(a) 12.

(b) $E_y = -\frac{14}{3}$.

(c) 30.

(d) $\frac{950}{3}$.

Mark scheme checked - Topic checked - WMA11-01_2021_MayJune_Q07

Question 7

7.

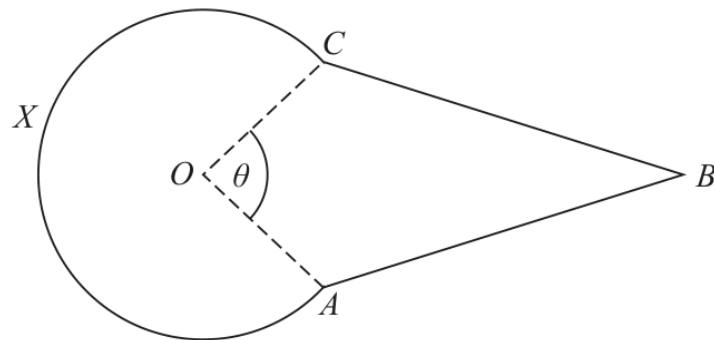


Figure 3

Figure 3 shows the design for a sign at a bird sanctuary.

The design consists of a kite $OABC$ joined to a sector $OCXA$ of a circle centre O .

In the design

- $OA = OC = 0.6\text{ m}$
- $AB = CB = 1.4\text{ m}$
- Angle $OAB = \text{Angle } OCB = 2\text{ radians}$
- Angle $AOC = \theta\text{ radians}$, as shown in Figure 3

Making your method clear,

- show that $\theta = 1.64\text{ radians}$ to 3 significant figures, (4)
- find the perimeter of the sign, in metres to 2 significant figures, (2)
- find the area of the sign, in m^2 to 2 significant figures. (4)

(Total 10 marks)

Worked Solution - Question 7

1. Work in triangle OAB

In triangle OAB , $OA = 0.6$, $AB = 1.4$, and $\angle OAB = 2$ radians.

2. Find OB

By the cosine rule, $OB^2 = 0.6^2 + 1.4^2 - 2(0.6)(1.4) \cos 2$, so $OB = 1.73756 \dots$

3. Find half of theta

Using the cosine rule again, $\cos \angle AOB = \frac{0.6^2 + OB^2 - 1.4^2}{2(0.6)(OB)}$, giving
 $\angle AOB = 0.822199 \dots$ radians.

4. Use symmetry of the kite

The kite is symmetric, so $\theta = 2\angle AOB = 1.644398 \dots$, hence $\theta = 1.64$ radians to 3 significant figures.

5. Find the outside arc length

The sector $OCXA$ uses the major angle $2\pi - \theta$. Its arc length is
 $0.6(2\pi - 1.644398 \dots) = 2.78327 \dots$

6. Find the perimeter

Perimeter = $AB + CB + \text{major arc} = 1.4 + 1.4 + 2.78327 \dots = 5.58327 \dots$, so the perimeter is **5.6** m.

7. Find the kite area

The kite consists of two congruent triangles like OAB , so area of kite
 $= 2 \left(\frac{1}{2} (0.6)(1.4) \sin 2 \right) = 0.76381 \dots$

8. Find the sector area and total area

Major sector area $= \frac{1}{2}(0.6)^2(2\pi - 1.644398\dots) = 0.83498\dots$. Total area $= 0.76381\dots + 0.83498\dots = 1.59879\dots$, so the area is **1.6 m²**.

Final answer

(a) $\theta = 1.64$ rad.

(b) 5.6 m.

(c) 1.6 m².

Mark scheme checked - Topic checked - WMA11-01_2021_Oct_Q07

Question 8

8.

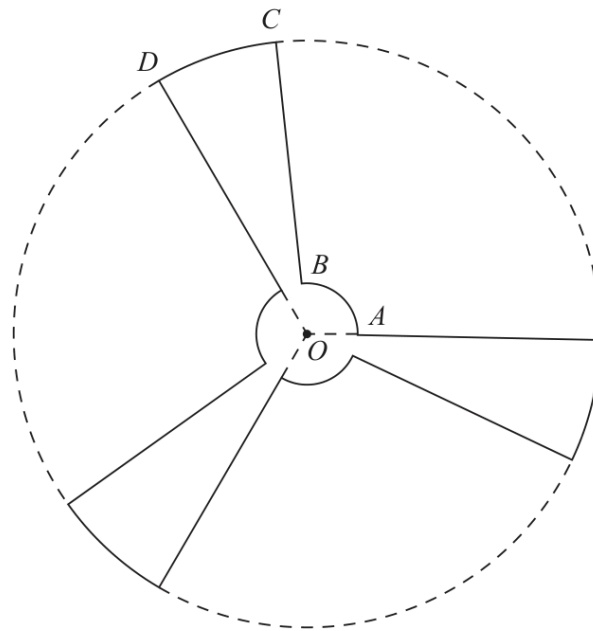


Figure 3

Figure 3 shows a sketch of the outline of the face of a ceiling fan viewed from below.

The fan consists of three identical sections congruent to $OABCD$, shown in Figure 3, where

- $OABO$ is a sector of a circle with centre O and radius 9 cm
- $OBCDO$ is a sector of a circle with centre O and radius 84 cm
- angle $AOD = \frac{2\pi}{3}$ radians

Given that the length of the arc AB is 15 cm,

- (a) show that the length of the arc CD is 35.9 cm to one decimal place.

(3)

The face of the fan is modelled to be a flat surface.

Find, according to the model,

- (b) the perimeter of the face of the fan, giving your answer to the nearest cm,

(2)

- (c) the surface area of the face of the fan.

Give your answer to 3 significant figures and make your units clear.

(5)

(Total 10 marks)

Worked Solution - Question 8

1. Find the small sector angle

For arc AB , $s = r\theta$, so $15 = 9\theta$ and $\theta = \frac{5}{3}$ radians.

2. Find the outer arc angle

Each identical section has total angle $\frac{2\pi}{3}$, so the angle for arc CD is $\frac{2\pi}{3} - \frac{5}{3} = \frac{2\pi - 5}{3}$.

3. Find CD

$$CD = 84 \left(\frac{2\pi - 5}{3} \right) = 28(2\pi - 5) = 35.9 \text{ cm to 3 significant figures.}$$

4. Find the perimeter

There are 3 inner arcs, 3 outer arcs, and 6 straight radial edges of length $84 - 9 = 75$.
Thus $P = 3(15) + 3(35.929\dots) + 6(75) = 602.8\dots \text{ cm}$.

5. Find the area of one section

$$\text{One section has area } \frac{1}{2}(9^2) \left(\frac{5}{3} \right) + \frac{1}{2}(84^2) \left(\frac{2\pi - 5}{3} \right) = 1576.525\dots \text{ cm}^2.$$

6. Find the total area

The total area is $3(1576.525\dots) = 4729.57\dots \text{ cm}^2 = 4.73 \times 10^3 \text{ cm}^2$ to 3 significant figures.

Final answer

(a) $CD = 35.9 \text{ cm}$.

(b) 603 cm .

(c) $4.73 \times 10^3 \text{ cm}^2$.

Question 8

8.

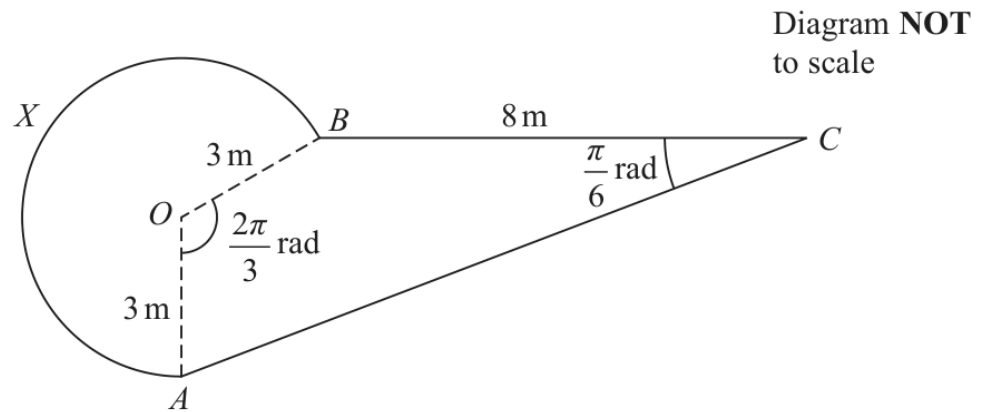


Figure 2

Figure 2 shows the plan view of a design for a pond.

The design consists of a sector $AOBX$ of a circle centre O joined to a quadrilateral $AOBC$.

- $BC = 8$ m
- $OA = OB = 3$ m
- angle AOB is $\frac{2\pi}{3}$ radians
- angle BCA is $\frac{\pi}{6}$ radians

(a) Calculate (i) the exact area of the sector $AOBX$,

(ii) the exact perimeter of the sector $AOBX$.

(5)

(b) Calculate the exact area of the triangle AOB .

(2)

(c) Show that the length AB is $3\sqrt{3}$ m.

(2)

(d) Find the total surface area of the pond. Give your answer in m^2 correct to 2 significant figures.

(5)

(Total for Question 8 is 14 marks)

Worked Solution - Question 8

1. Find the angle of the sector AOBX

The sector AOBX is the major sector, so its angle is $2\pi - \frac{2\pi}{3} = \frac{4\pi}{3}$.

2. Find the sector area

$$\text{Area} = \frac{1}{2}r^2\theta = \frac{1}{2}(3^2)\left(\frac{4\pi}{3}\right) = 6\pi \text{ m}^2.$$

3. Find the sector perimeter

The arc length is $r\theta = 3\left(\frac{4\pi}{3}\right) = 4\pi$. Adding the two radii gives $4\pi + 6 \text{ m}$.

4. Find the area of triangle AOB

$$\text{Area} = \frac{1}{2}(3)(3)\sin\left(\frac{2\pi}{3}\right) = \frac{9}{2} \cdot \frac{\sqrt{3}}{2} = \frac{9\sqrt{3}}{4} \text{ m}^2.$$

5. Show AB

By the cosine rule, $AB^2 = 3^2 + 3^2 - 2(3)(3)\cos\left(\frac{2\pi}{3}\right) = 18 + 9 = 27$, so $AB = 3\sqrt{3}$.

6. Find AC

In triangle ABC, $AB^2 = AC^2 + 8^2 - 2(AC)(8)\cos\left(\frac{\pi}{6}\right)$, so $27 = AC^2 - 8\sqrt{3}AC + 64$.

7. Choose the correct length

Solving gives $AC = 4\sqrt{3} \pm \sqrt{11}$. From the diagram, AC is the longer value, so $AC = 4\sqrt{3} + \sqrt{11}$.

8. Find the total area

Area of triangle ABC is $\frac{1}{2}(8)(AC) \sin\left(\frac{\pi}{6}\right) = 2AC = 8\sqrt{3} + 2\sqrt{11}$. Total area is $6\pi + \frac{9\sqrt{3}}{4} + 8\sqrt{3} + 2\sqrt{11} = 43.236\dots$, so 43 m^2 to 2 significant figures.

Final answer

(a)(i) $6\pi \text{ m}^2$, (ii) $4\pi + 6 \text{ m}$.

(b) $\frac{9\sqrt{3}}{4} \text{ m}^2$.

(c) $AB = 3\sqrt{3} \text{ m}$.

(d) 43 m^2 .

Mark scheme checked - Topic checked - WMA11-01_2022_Oct_Q08

Question 6

6.

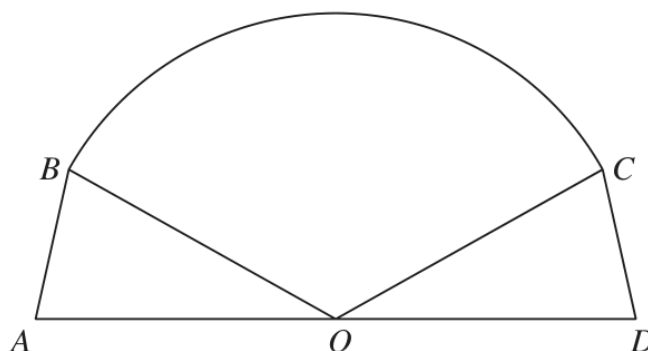
Diagram **NO**
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Figure 1

Figure 1 shows the plan view for the design of a stage.

The design consists of a sector OBC of a circle, with centre O , joined to two congruent triangles OAB and ODC .

Given that

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40 \text{ m}^2$
- AOD is a straight line of length 12.5 m

(a) find the radius of the sector, giving your answer, in m , to 2 decimal places,

(2)

(b) find the size of angle AOB , in radians, to 2 decimal places.

(1)

Hence find

(c) the total area of the stage, giving your answer, in m^2 , to one decimal place,

(3)

(d) the total perimeter of the stage, giving your answer, in m , to one decimal place.

(4)

(Total for Question 6 is 10 marks)

Worked Solution - Question 6

1. Find the radius

Area of sector BOC is $\frac{1}{2}r^2\theta$. So $40 = \frac{1}{2}r^2(2.4) = 1.2r^2$, giving $r = \sqrt{\frac{40}{1.2}} = 5.77$ m.

2. Find angle AOB

The triangles OAB and ODC are congruent, so $\angle AOB = \angle COD$. Since AOD is a straight line, $2\angle AOB + 2.4 = \pi$.

3. Calculate the angle

$$\angle AOB = \frac{\pi - 2.4}{2} = 0.3708\dots, \text{ so } 0.37 \text{ rad.}$$

4. Find the total area

The two congruent triangle areas together are

$$2 \left(\frac{1}{2} (6.25)(5.7735\dots) \sin 0.3708\dots \right) = 13.075\dots$$

5. Add the sector area

Total area = $40 + 13.075\dots = 53.1 \text{ m}^2$ to one decimal place.

6. Find AB

Using the cosine rule, $AB^2 = 6.25^2 + r^2 - 2(6.25)(r) \cos 0.3708\dots$, giving $AB = 2.265\dots$ m.

7. Find the perimeter

Perimeter

$$= AD + AB + CD + \text{arc } BC = 12.5 + 2(2.265\dots) + 5.7735\dots(2.4) = 30.9 \text{ m.}$$

Final answer

(a) $r = 5.77 \text{ m}$.

(b) 0.37 rad .

(c) 53.1 m^2 .

(d) 30.9 m .

Mark scheme checked - Topic checked - WMA11-01_2023_Jan_Q06

Question 9

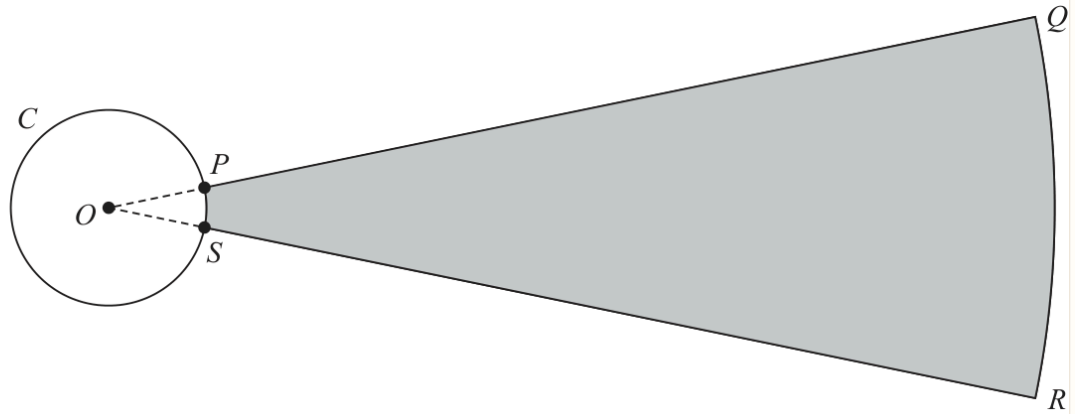
**Figure 3**

Figure 3 shows the plan view of the area being used for a ball-throwing competition.

Competitors must stand within the circle C and throw a ball as far as possible into the target area, $PQRS$, shown shaded in Figure 3.

Given that

- circle C has centre O
- P and S are points on C
- $OPQRSO$ is a sector of a circle with centre O
- the length of arc PS is 0.72 m
- the size of angle POS is 0.6 radians

(a) show that $OP = 1.2$ m

(1)

Given also that

- the target area, $PQRS$, is 90 m^2
- length $PQ = x$ metres

(b) show that

$$5x^2 + 12x - 1500 = 0$$

(3)

(c) Hence calculate the total perimeter of the target area, $PQRS$, giving your answer to the nearest metre.

(3)

(Total for Question 9 is 7 marks)

Worked Solution - Question 9

1. Use arc length

$s = r\theta$, so $0.72 = OP \times 0.6$. Hence $OP = 1.2$ m.

2. Set up the target area

The outer sector has radius $x + 1.2$ and the inner sector has radius 1.2 , with angle 0.6 radians.

3. Use sector area

$$\frac{1}{2}(x + 1.2)^2(0.6) - \frac{1}{2}(1.2)^2(0.6) = 90.$$

4. Simplify to the given quadratic

$0.3((x + 1.2)^2 - 1.44) = 90$, so $0.3x^2 + 0.72x - 90 = 0$. Multiplying by $\frac{50}{3}$ gives $5x^2 + 12x - 1500 = 0$.

5. Find the positive value of x

$$x = \frac{-12 + \sqrt{12^2 + 4(5)(1500)}}{10} = \frac{-6 + 4\sqrt{471}}{5} \approx 16.16.$$

6. Find the perimeter

Perimeter = $PQ + SR + \text{arc } QR + \text{arc } PS = x + x + 0.6(x + 1.2) + 0.72 \approx 43.46$.

7. Round to the nearest metre

The total perimeter is **43 m** to the nearest metre.

Final answer

(a) $OP = 1.2$ m.

(b) $5x^2 + 12x - 1500 = 0$.

(c) 43 m.

Question 8

8.

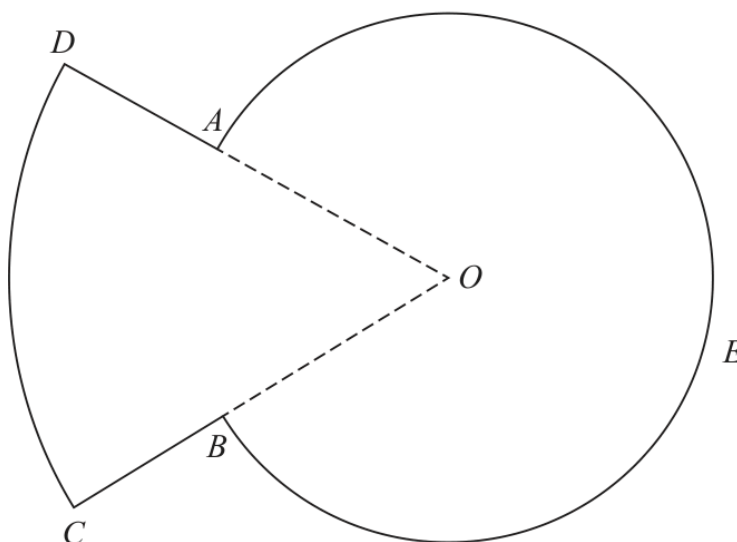


Figure 3

Figure 3 shows a sketch of the plan view of a platform.

The plan view of the platform consists of a sector DOC of a circle centre O joined to a sector $AOBEA$ of a different circle, also with centre O .

Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9$ m
- $DA:AO = 3:5$

(a) show that $AO = 7.03$ m to 3 significant figures.

(3)

(b) Find the perimeter of the platform, in m, to 3 significant figures.

(3)

(c) Find the total area of the platform, giving your answer in m^2 to the nearest whole number.

(3)

(Total for Question 8 is 9 marks)

Worked Solution - Question 8

1. Find the outer radius

For sector DOC , arc length $CD = 9$ and angle 0.8 radians, so

$$OD = \frac{9}{0.8} = 11.25 \text{ m.}$$

2. Use the ratio DA to AO

$DA : AO = 3 : 5$, so $OD = DA + AO$ represents 8 parts. Hence

$$AO = \frac{5}{8}(11.25) = 7.03125 \text{ m} \approx 7.03 \text{ m.}$$

3. Find the straight side lengths

$$DA = CB = OD - AO = 11.25 - 7.03125 = 4.21875 \text{ m.}$$

4. Find the major arc

The major sector $AOBEA$ has angle $2\pi - 0.8$, so arc $AEB = 7.03125(2\pi - 0.8)$.

5. Calculate the perimeter

$$\text{Perimeter} = 9 + 2(4.21875) + 7.03125(2\pi - 0.8) \approx 56.0 \text{ m.}$$

6. Calculate the total area

$$\text{Area} = \frac{1}{2}(11.25)^2(0.8) + \frac{1}{2}(7.03125)^2(2\pi - 0.8) \approx 186 \text{ m}^2 \text{ to the nearest whole number.}$$

Final answer

(a) $AO = 7.03 \text{ m.}$

(b) 56.0 m.

(c) $186 \text{ m}^2.$

Question 8

8.

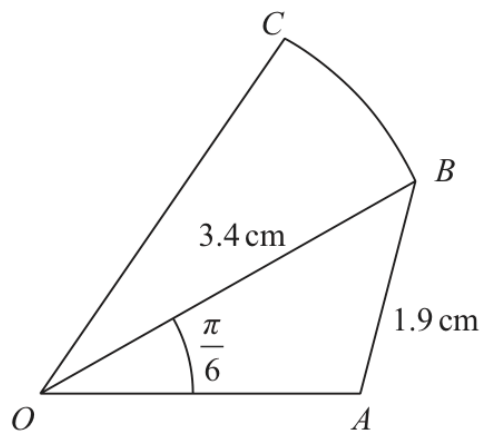


Figure 1

Figure 1 shows a sketch of a design for a badge.

The design consists of a triangle OAB joined to a sector OBC of a circle with centre O
In the design

- $OB = 3.4$ cm
- $AB = 1.9$ cm
- angle $AOB = \frac{\pi}{6}$ radians
- angle $OAB > \frac{\pi}{2}$ radians

Making your method clear,

(a) find the size of angle OAB , giving your answer in radians to 4 significant figures, (3)

(b) find the area of triangle OAB , in cm^2 , giving your answer to 3 significant figures. (2)

Given that the ratio of the area of sector OBC to the area of triangle OAB is 3 : 2

(c) show that angle BOC is 0.462 radians to 3 significant figures. (3)

(d) Hence find the perimeter of the badge, in cm, to the nearest integer. (5)

(Total for Question 8 is 13 marks)

Worked Solution - Question 8

1. Use the sine rule

In triangle OAB , $AB = 1.9$, $OB = 3.4$ and $\angle AOB = \frac{\pi}{6}$.

2. Find sine of angle OAB

$$\frac{\sin \angle OAB}{3.4} = \frac{\sin(\pi/6)}{1.9}, \text{ so } \sin \angle OAB = \frac{17}{19}.$$

3. Choose the obtuse angle

Since $\angle OAB > \frac{\pi}{2}$, $\angle OAB = \pi - \sin^{-1}\left(\frac{17}{19}\right) \approx 2.034$ radians.

4. Find OA

Using the sine rule again, $OA \approx 2.096$ cm.

5. Find the triangle area

$$\text{Area } OAB = \frac{1}{2}(OA)(OB) \sin(\pi/6) \approx 1.78 \text{ cm}^2.$$

6. Use the area ratio

Area sector OBC : Area triangle $OAB = 3 : 2$, so sector area = $\frac{3}{2}(1.78156\dots)$.

7. Find angle BOC

$$\frac{1}{2}(3.4)^2\theta = \frac{3}{2}(1.78156\dots), \text{ so } \theta \approx 0.462 \text{ radians.}$$

8. Find the perimeter

The perimeter is

$$OA + AB + OC + \text{arc } BC = 2.096 + 1.9 + 3.4 + 3.4(0.46234\dots) \approx 8.968 \text{ cm.}$$

9. Round

The perimeter is **9** cm to the nearest integer.

Final answer

(a) $\angle OAB = 2.034$ radians.

(b) 1.78 cm^2 .

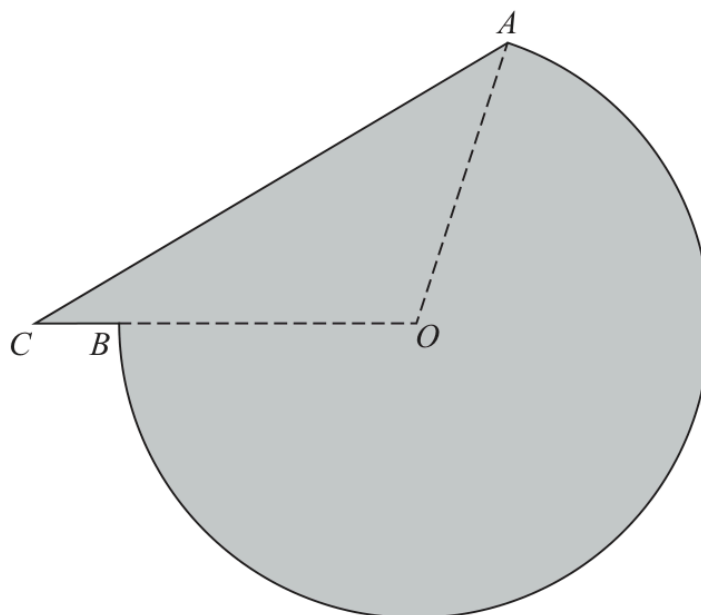
(c) $\angle BOC = 0.462$ radians.

(d) 9 cm .

Mark scheme checked - Topic checked - WMA11-01_2025_Jan_Q08

Question 6

6.



Not to scale

Figure 2

The shaded area in Figure 2 shows the plan view of a helicopter landing pad.

The area consists of the major sector AOB of a circle centre O joined to a triangle AOC .

Given that

- $AO = OB = 15$ m
- $BC = 2$ m
- CBO is a straight line
- angle $ACO = 0.6$ radians

- (a) show that angle COA is 1.847 radians to 3 decimal places. (3)
- (b) Find the total area of the helicopter landing pad.
Give your answer in m^2 to 3 significant figures. (3)
- (c) Find the perimeter of the helicopter landing pad.
Give your answer in metres to 3 significant figures. (3)

(Total for Question 6 is 9 marks)

Worked Solution - Question 6

1. Set up triangle AOC

$AO = 15$, $OC = OB + BC = 17$, and $\angle ACO = 0.6$.

2. Use the sine rule

$$\frac{\sin \angle CAO}{17} = \frac{\sin 0.6}{15}, \text{ so } \angle CAO \approx 0.6944 \text{ radians.}$$

3. Show angle COA

$\angle COA = \pi - 0.6 - 0.6944... = 1.847...$, so $\angle COA = 1.847$ radians to 3 decimal places.

4. Major sector angle

The shaded sector is the major sector AOB , so its angle is $2\pi - 1.847...$

5. Sector area

$$\text{Sector area} = \frac{1}{2}(15)^2(2\pi - 1.847...) = 499.15...$$

6. Triangle area

$$\text{Area } AOC = \frac{1}{2}(15)(17) \sin(1.847...) = 122.73...$$

7. Total area

$499.15... + 122.73... = 621.88...$, so the total area is **622 m²** to 3 significant figures.

8. Find AC

By the cosine rule, $AC^2 = 15^2 + 17^2 - 2(15)(17) \cos(1.847...)$, so $AC \approx 27.57$ m.

9. Major arc length

$$\text{arc } AB = 15(2\pi - 1.847\dots) = 66.54\dots \text{ m.}$$

10. Perimeter

$$66.54\dots + 27.57\dots = 94.11\dots, \text{ so the perimeter is } \mathbf{94.1 \text{ m.}}$$

Final answer

(a) $\angle COA = 1.847 \text{ rad.}$

(b) $622 \text{ m}^2.$

(c) 94.1 m.

Mark scheme checked - Topic checked - WMA11-01_2025_MayJune_Q06

Question 7

7.

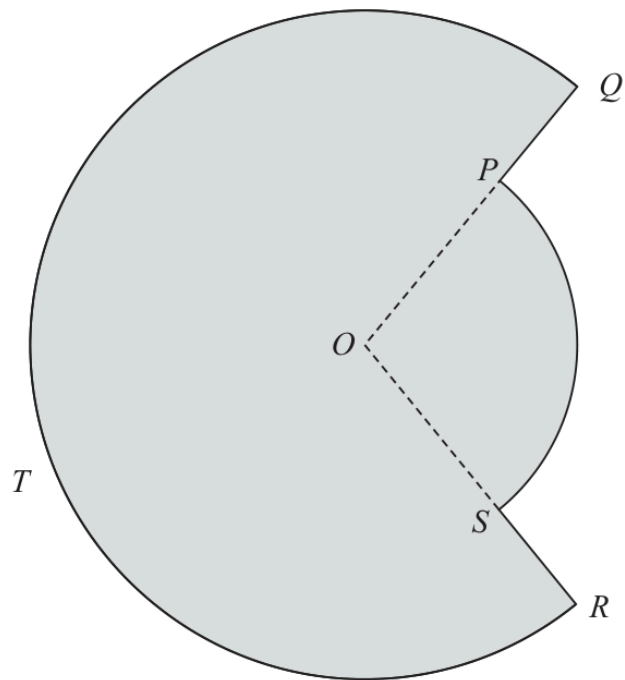


Figure 2

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 2 shows the plan view of a design for a swimming pool.

The design consists of a sector POS of a circle centre O joined to a major sector $QORTQ$ of a different circle, also with centre O .

Given that

- angle POS is 1.65 radians
- the area of sector POS is 30 m^2
- $PQ = 2.8 \text{ m}$

(a) show that, to 3 significant figures, $OQ = 8.83 \text{ m}$

(3)

(b) Find the total surface area of the swimming pool in m^2 to the nearest integer.

(3)

(c) Find the total perimeter of the swimming pool in metres to 2 significant figures.

(3)

(Total for Question 7 is 9 marks)

Worked Solution - Question 7

1. Find OP from the small sector

Let $OP = r$. The area of sector POS is 30, so $\frac{1}{2}r^2(1.65) = 30$.

2. Calculate OP

$$r^2 = \frac{60}{1.65}, \text{ so } r = 6.030\dots \text{ m.}$$

3. Show OQ

$OQ = OP + PQ = 6.030\dots + 2.8 = 8.830\dots$, so $OQ = 8.83$ m to 3 significant figures.

4. Major sector angle

The major sector $QORTQ$ has angle $2\pi - 1.65$ radians.

5. Total area

$$\text{Area} = 30 + \frac{1}{2}(8.830\dots)^2(2\pi - 1.65) = 210.6\dots$$

6. Round area

The total surface area is **211 m²** to the nearest integer.

7. Arc lengths

The inner arc PS has length $6.030\dots(1.65) = 9.95\dots$, and the outer major arc has length $8.830\dots(2\pi - 1.65) = 40.91\dots$

8. Add straight sides

The straight sides PQ and SR each have length **2.8** m.

9. Perimeter

$9.95\dots + 40.91\dots + 2.8 + 2.8 = 56.46\dots$, so the perimeter is **56 m** to 2 significant figures.

Final answer

(a) $OQ = 8.83$ m.

(b) 211 m².

(c) **56 m**.

Mark scheme checked - Topic checked - WMA11-01_2025_Oct_Q07

TOPIC

Trigonometric Functions

Question 9

Trigonometric Functions

9.

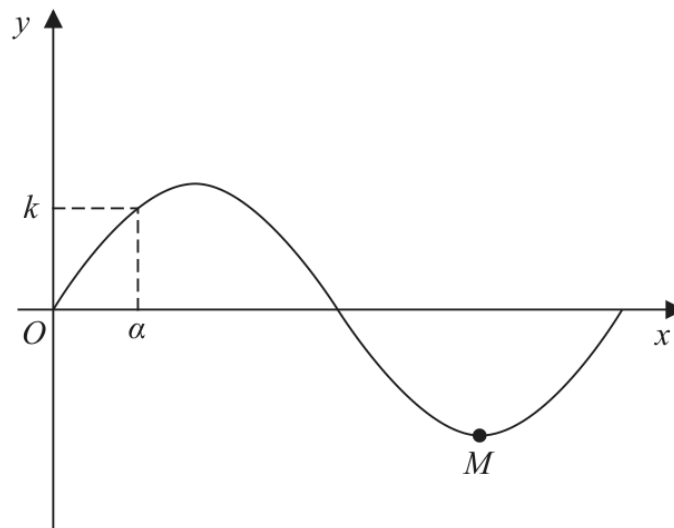


Figure 5

Figure 5 shows a sketch of part of the curve C with equation $y = \sin\left(\frac{x}{12}\right)$, where x is measured in radians. The point M shown in Figure 5 is a minimum point on C .

(a) State the period of C .

(1)

(b) State the coordinates of M .

(1)

The smallest positive solution of the equation $\sin\left(\frac{x}{12}\right) = k$, where k is a constant, is α .

Find, in terms of α ,

(c) (i) the negative solution of the equation $\sin\left(\frac{x}{12}\right) = k$ that is closest to zero,

(ii) the smallest positive solution of the equation $\cos\left(\frac{x}{12}\right) = k$.

(2)

(Total 4 marks)

Worked Solution - Question 9

1. Use the scale factor inside the sine

For $y = \sin\left(\frac{x}{12}\right)$, one full sine cycle needs $\frac{x}{12}$ to increase by 2π , so the period is 24π .

2. Locate the minimum point

The sine function is -1 when its angle is $\frac{3\pi}{2}$. Therefore $\frac{x}{12} = \frac{3\pi}{2}$, so $x = 18\pi$ and $M = (18\pi, -1)$.

3. Use sine symmetry

If the smallest positive solution of $\sin\left(\frac{x}{12}\right) = k$ is α , then the closest negative solution is one half-period before the opposite sine position: $-12\pi - \alpha$.

4. Convert to cosine

Since $\cos \theta = \sin\left(\frac{\pi}{2} - \theta\right)$, the corresponding smallest positive cosine solution is $6\pi - \alpha$.

Final answer

(a) 24π .

(b) $(18\pi, -1)$.

(c)(i) $-12\pi - \alpha$. (ii) $6\pi - \alpha$.

Question 7

Trigonometric Functions

7.

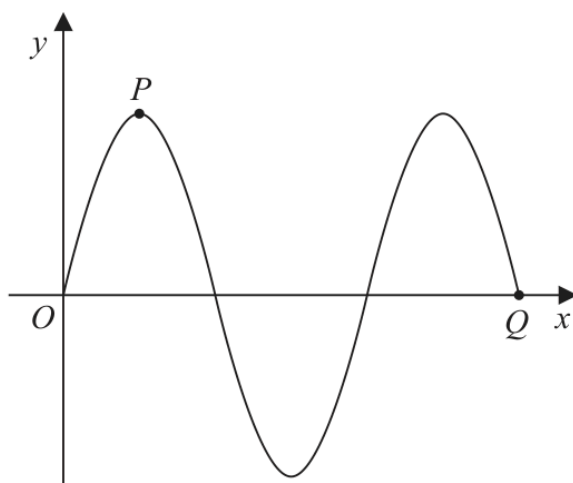


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 3 \sin x$, where x is measured in degrees.

The point P and the point Q lie on C_1 and are shown in Figure 3.

(a) State

- (i) the coordinates of P ,
- (ii) the coordinates of Q .

(3)

A different curve C_2 has equation $y = 3 \sin x + k$, where k is a constant.

The curve C_2 has a maximum y value of 10

The point R is the minimum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)

(Total 5 marks)

Worked Solution - Question 7

1. Read P from $y = 3 \sin x$

The first maximum of $y = 3 \sin x$ occurs at $x = 90^\circ$, with $y = 3$. So $P = (90^\circ, 3)$.

2. Read Q from the next shown zero

After the second maximum, the curve returns to the x-axis at $x = 540^\circ$. So $Q = (540^\circ, 0)$.

3. Find the vertical shift

For C_2 , $y = 3 \sin x + k$. Its maximum is $3 + k = 10$, so $k = 7$.

4. Find the minimum point R

The minimum value is $-3 + 7 = 4$. The smallest positive x-coordinate for a sine minimum is 270° , so $R = (270^\circ, 4)$.

Final answer

(a)(i) $P = (90^\circ, 3)$. (a)(ii) $Q = (540^\circ, 0)$.

(b) $R = (270^\circ, 4)$.

Question 9

Trigonometric Functions

9.

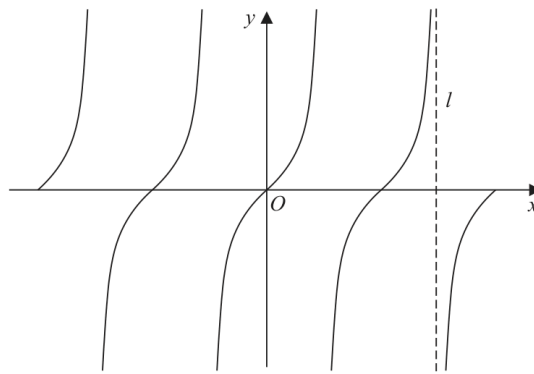


Figure 4

Figure 4 shows a sketch of the curve with equation

$$y = \tan x \quad -2\pi \leq x \leq 2\pi$$

The line l , shown in Figure 4, is an asymptote to $y = \tan x$

(a) State an equation for l .

(1)

A copy of Figure 4, labelled Diagram 1, is shown on the next page.

(b) (i) On Diagram 1, sketch the curve with equation

$$y = \frac{1}{x} + 1 \quad -2\pi \leq x \leq 2\pi$$

stating the equation of the horizontal asymptote of this curve.

(ii) Hence, **giving a reason**, state the number of solutions of the equation

$$\tan x = \frac{1}{x} + 1$$

in the region $-2\pi \leq x \leq 2\pi$

(4)

(c) State the number of solutions of the equation $\tan x = \frac{1}{x} + 1$ in the region

(i) $0 \leq x \leq 40\pi$

(ii) $-10\pi \leq x \leq \frac{5}{2}\pi$

(2)

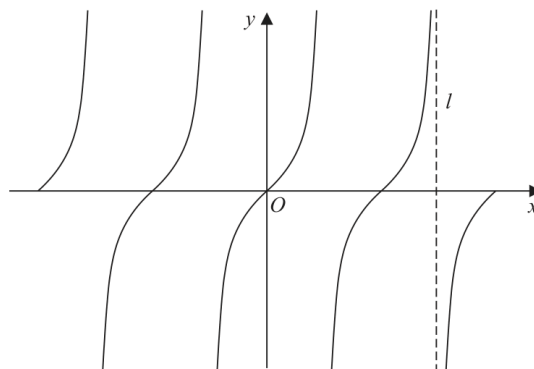


Diagram 1

(Total 7 marks)

Worked Solution - Question 9

1. State the shown tangent asymptote

The shown dashed asymptote is the right-hand positive asymptote before 2π , so l has equation $x = \frac{3\pi}{2}$.

2. Describe the reciprocal graph

For $y = \frac{1}{x} + 1$, the vertical asymptote is $x = 0$ and the horizontal asymptote is $y = 1$. The right branch lies above $y = 1$ and the left branch lies below $y = 1$.

3. Count intersections on -2π to 2π

The number of solutions of $\tan x = \frac{1}{x} + 1$ is the number of intersections of the two graphs. From the sketch on $-2\pi \leq x \leq 2\pi$, there are 5 intersections.

4. Count solutions for $0 \leq x < 40\pi$

From the repeated tangent branches and the reciprocal curve, there is one intersection in each positive interval between consecutive tangent asymptotes. From 0 to 40π , this gives 40 solutions.

5. Count solutions for $-10\pi \leq x \leq 5\pi/2$

Extending the same intersection pattern across the stated interval gives 14 intersections, so there are 14 solutions.

Final answer

(a) $x = \frac{3\pi}{2}$.

(b)(i) $y = 1$ is the horizontal asymptote.

(b)(ii) 5.

(c)(i) 40.

(c)(ii) 14.

Question 9

Trigonometric Functions

9.

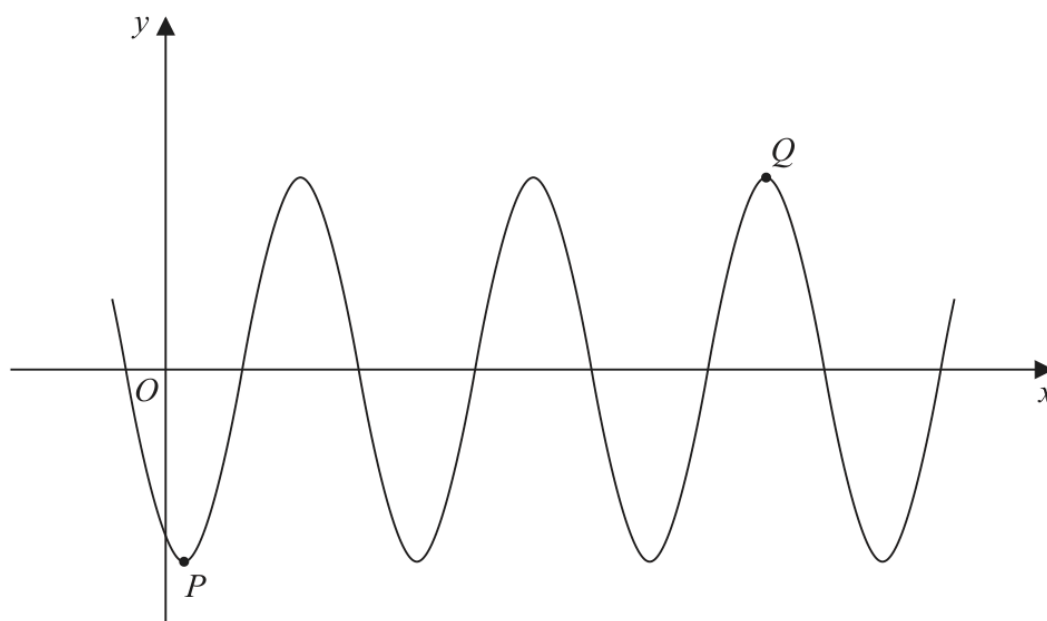


Figure 4

Figure 4 shows part of the curve with equation

$$y = A \cos(x - 30)^\circ$$

where A is a constant.

The point P is a minimum point on the curve and has coordinates $(30, -3)$ as shown in Figure 4.

(a) Write down the value of A .

(1)

The point Q is shown in Figure 4 and is a maximum point.

(b) Find the coordinates of Q .

(3)

(Total 4 marks)

Worked Solution - Question 9

1. Use the minimum point

At $P(30, -3)$, $x - 30 = 0$, so $\cos 0^\circ = 1$. Therefore $A = -3$.

2. Find maximum points

Since $A = -3$, maximum points occur when $\cos(x - 30)^\circ = -1$.

3. Use the shown maximum Q

$x - 30 = 180 + 360n$, so $x = 210 + 360n$. The maximum shown at Q is two full periods after the first maximum, so $x = 210 + 720 = 930$.

4. Write Q

The maximum y-value is 3, hence $Q = (930, 3)$.

Final answer

(a) $A = -3$.

(b) $Q = (930, 3)$.

Question 9

Trigonometric Functions

Figure 4 shows part of the graph of the curve with equation $y = \sin x$

(a) state, in terms of p , the value of

- (i) $2 \sin(180^\circ - \alpha)$
(ii) $\sin(\alpha - 180^\circ)$
(iii) $3 + \sin(180^\circ + \alpha)$

A copy of Figure 4, labelled Diagram 1, is shown on page 27.

On Diagram 1,

- (b) sketch the graph of $y = \sin 2x$

(2)

- (c) Hence find, in terms of α , the x coordinates of any points in the interval $0 < x < 180^\circ$ where

$$\sin 2x = p$$

Diagram 1

(Total 8 marks)

Worked Solution - Question 9

1. Use sine symmetry

$$\sin(180^\circ - \alpha) = \sin \alpha = p, \text{ so } 2 \sin(180^\circ - \alpha) = 2p.$$

2. Use odd symmetry

$$\sin(\alpha - 180^\circ) = -\sin(180^\circ - \alpha) = -p.$$

3. Use the third quadrant identity

$$\sin(180^\circ + \alpha) = -\sin \alpha = -p, \text{ so } 3 + \sin(180^\circ + \alpha) = 3 - p.$$

4. Sketch $y = \sin 2x$

The graph has amplitude 1 and period 180° . It crosses the x-axis at $0^\circ, 90^\circ, 180^\circ$, has a maximum at 45° , and a minimum at 135° .

5. Solve $\sin 2x = p$

Since $\sin \alpha = p$ and $0 < 2x < 360^\circ$, the possible values are $2x = \alpha$ or $2x = 180^\circ - \alpha$.

6. Find x

Therefore $x = \frac{\alpha}{2}$ or $x = 90^\circ - \frac{\alpha}{2}$.

Final answer

(a)(i) $2p$, (ii) $-p$, (iii) $3 - p$.

(c) $x = \frac{\alpha}{2}, 90^\circ - \frac{\alpha}{2}$.

Question 9

Trigonometric Functions

9.

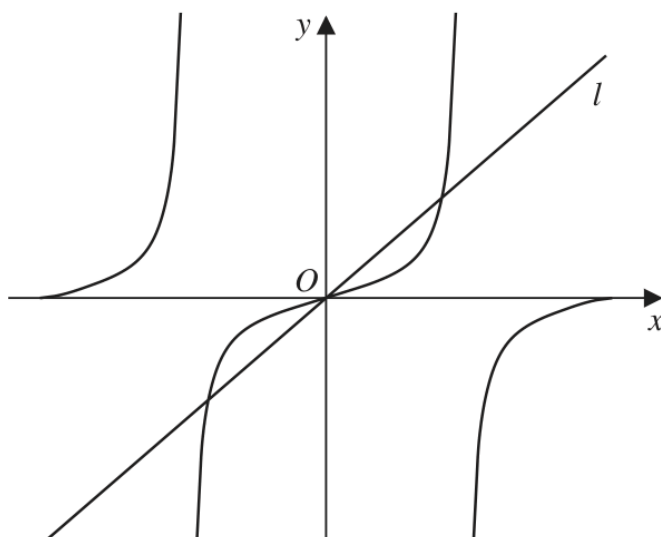


Figure 3

Figure 3 shows a sketch of

- the curve with equation $y = \tan x$
- the straight line l with equation $y = \pi x$

in the interval $-\pi < x < \pi$

(a) State the period of $\tan x$

(1)

(b) Write down the number of roots of the equation

(i) $\tan x = (\pi + 2)x$ in the interval $-\pi < x < \pi$

(1)

(ii) $\tan x = \pi x$ in the interval $-2\pi < x < 2\pi$

(1)

(iii) $\tan x = \pi x$ in the interval $-100\pi < x < 100\pi$

(1)

(Total for Question 9 is 4 marks)

Worked Solution - Question 9

1. State the period

The period of $\tan x$ is π .

2. Count roots for the steeper line

In $[-\pi, \pi]$

3. Extend to $-2\pi < x < 2\pi$

For $y = \pi x$, the interval $[-2\pi, 2\pi]$

4. Extend to $-100\pi < x < 100\pi$

There are 99 positive outer roots, 99 negative outer roots, and 3 central roots, so the total is $99 + 99 + 3 = 201$.

Final answer

(a) π .

(b)(i) 3, (ii) 5, (iii) 201.

Question 9

Trigonometric Functions

9. (i)

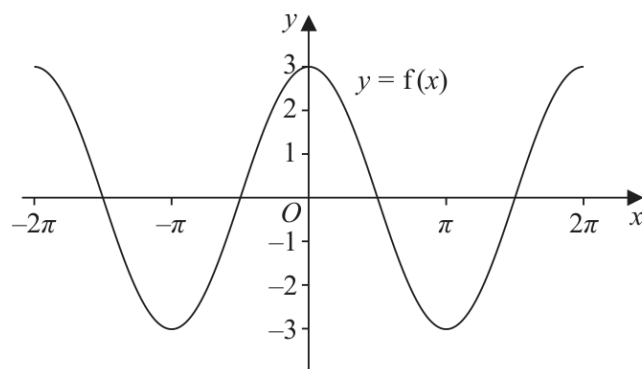


Figure 3

Figure 3 shows part of the graph of the trigonometric function with equation $y = f(x)$

(a) Write down an expression for $f(x)$

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = f\left(x + \frac{\pi}{4}\right)$

Show clearly the coordinates of all the points where the curve intersects the coordinate axes.

(3)

(ii)

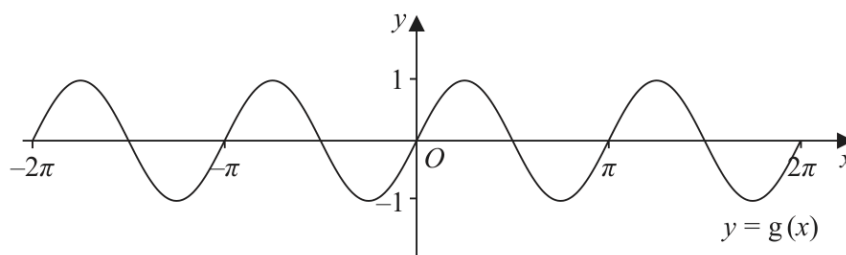


Figure 4

Figure 4 shows part of the graph of the trigonometric function with equation $y = g(x)$

(a) Write down an expression for $g(x)$

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = g(x) - 2$

Show clearly the coordinates of the y intercept.

(3)

(Total for Question 9 is 9 marks)

Worked Solution - Question 9

1. Identify $f(x)$

The first graph has amplitude 3, period 2π , and maximum at $x = 0$, so $f(x) = 3 \cos x$.

2. Transform f

$$y = f\left(x + \frac{\pi}{4}\right) = 3 \cos\left(x + \frac{\pi}{4}\right), \text{ a shift left by } \frac{\pi}{4}.$$

3. Find the x-intercepts

$$\cos\left(x + \frac{\pi}{4}\right) = 0 \text{ gives } x = \frac{\pi}{4} + n\pi, \text{ for } n \in \mathbb{Z}.$$

4. Find the y-intercept

$$\text{At } x = 0, y = 3 \cos \frac{\pi}{4} = \frac{3\sqrt{2}}{2}.$$

5. Identify $g(x)$

The second graph has amplitude 1, period π , and passes through the origin with positive gradient, so $g(x) = \sin 2x$.

6. Transform g

$y = g(x) - 2 = \sin 2x - 2$, so the graph is shifted down by 2. Its y-intercept is $(0, -2)$.

Final answer

(i)(a) $f(x) = 3 \cos x$. (i)(b) x-intercepts $-\frac{7\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}, \frac{5\pi}{4}$ and y-intercept $\frac{3\sqrt{2}}{2}$. (ii)(a) $g(x) = \sin 2x$. (ii)(b) y-intercept $(0, -2)$.

Question 10

Trigonometric Functions

10.

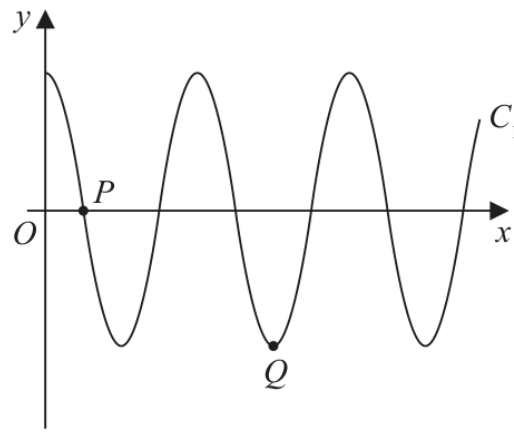


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 3 \cos\left(\frac{x}{n}\right)^\circ \quad x \geq 0$$

where n is a constant.

The curve C_1 cuts the positive x -axis for the first time at point $P(270, 0)$, as shown in Figure 4.

(a) (i) State the value of n

(ii) State the period of C_1

(2)

The point Q , shown in Figure 4, is a minimum point of C_1

(b) State the coordinates of Q .

(2)

The curve C_2 has equation $y = 2 \sin x^\circ + k$, where k is a constant.

The point $R\left(a, \frac{12}{5}\right)$ and the point $S\left(-a, -\frac{3}{5}\right)$, both lie on C_2

Given that a is a constant less than 90

(c) find the value of k .

(2)

(Total for Question 10 is 6 marks)

Worked Solution - Question 10

1. Use the first positive x-intercept

For $y = 3 \cos \left(\frac{x}{n} \right)^\circ$, the first positive zero occurs when $\frac{x}{n} = 90^\circ$.

2. Find n

Since $P = (270, 0)$, $\frac{270}{n} = 90$, so $n = 3$.

3. Find the period

The period is $360n = 360(3) = 1080^\circ$.

4. Read the shown minimum point

A minimum occurs when $\frac{x}{3} = 180^\circ + 360^\circ k$. The point Q shown is the next displayed minimum after one full period, so $x = 1620$ and $y = -3$.

5. Use the two points on C2

For $y = 2 \sin x^\circ + k$, the points give $2 \sin a + k = \frac{12}{5}$ and $-2 \sin a + k = -\frac{3}{5}$.

6. Add the equations

Adding gives $2k = \frac{12}{5} - \frac{3}{5} = \frac{9}{5}$, so $k = \frac{9}{10}$.

Final answer

(a)(i) $n = 3$. (a)(ii) 1080° .

(b) $Q = (1620, -3)$.

(c) $k = \frac{9}{10}$.

Question 6

Trigonometric Functions

6.

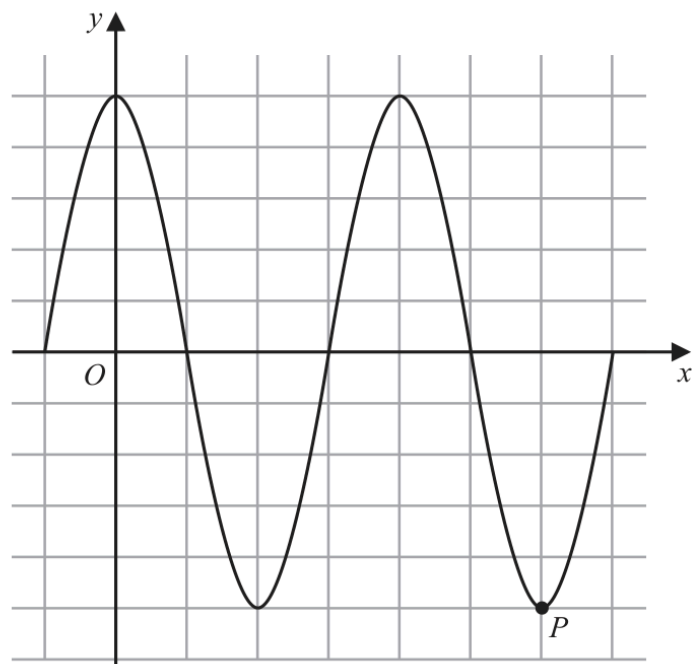


Figure 2

Figure 2 shows a plot of part of the curve C_1 with equation

$$y = 5 \cos x$$

with x being measured in degrees.

The point P , shown in Figure 2, is a minimum point on C_1

(a) State the coordinates of P

(2)

The point Q lies on a different curve C_2

Given that point Q

- is a maximum point on the curve
- is the maximum point with the **smallest** x coordinate, $x > 0$

(b) find the coordinates of Q when

(i) C_2 has equation $y = 5 \cos x - 2$

(ii) C_2 has equation $y = -5 \cos x$

(4)

(Total for Question 6 is 6 marks)

Worked Solution - Question 6

1. Use the cosine graph

For $y = 5 \cos x$, the minimum value is -5 and it occurs at $x = 180^\circ + 360^\circ n$.

2. Read P from the displayed minimum

The plotted point P is the displayed minimum at $x = 540^\circ$, so $P = (540^\circ, -5)$.

3. Translate the graph down

For $y = 5 \cos x - 2$, the maximum value is $5 - 2 = 3$. The smallest positive x -coordinate for a maximum shown is $x = 360^\circ$, so $Q = (360^\circ, 3)$.

4. Reflect in the x-axis

For $y = -5 \cos x$, a maximum occurs where $\cos x = -1$, first at $x = 180^\circ$. The maximum value is 5 , so $Q = (180^\circ, 5)$.

Final answer

(a) $P = (540^\circ, -5)$.

(b)(i) $Q = (360^\circ, 3)$.

(b)(ii) $Q = (180^\circ, 5)$.

Question 11

Trigonometric Functions

11.

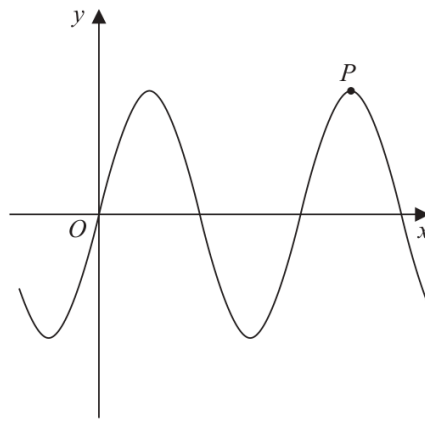


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 12 \sin x$$

where x is measured in radians.

The point P shown in Figure 4 is a maximum point on C_1

(a) Find the coordinates of P .

(2)

The curve C_2 has equation

$$y = 12 \sin x + k$$

where k is a constant.

Given that the **maximum** value of y on C_2 is 3

(b) find the coordinates of the **minimum** point on C_2 which has the **smallest** positive x coordinate.

(2)

The curve C_3 has equation

$$y = 12 \sin(x + B)$$

where B is a positive constant.

Given that $\left(\frac{\pi}{4}, A\right)$, where A is a constant, is the **minimum** point on C_3 which has the **smallest** positive x coordinate,

(c) find

(i) the value of A ,

(ii) the smallest possible value of B .

(2)

(Total for Question 11 is 6 marks)

TOTAL FOR PAPER IS 75 MARKS

Worked Solution - Question 11

1. Use maxima of sine

For $y = 12 \sin x$, maxima occur when $x = \frac{\pi}{2} + 2\pi n$, with value 12.

2. Read point P

The maximum point P shown is the second positive maximum, so $P = (\frac{5\pi}{2}, 12)$.

3. Find k for C2

For $y = 12 \sin x + k$, the maximum value is $12 + k$. Given the maximum is 3, $k = -9$.

4. Find the smallest positive minimum on C2

The minimum value is $-12 - 9 = -21$, and it first occurs at $x = \frac{3\pi}{2}$. So the point is $(\frac{3\pi}{2}, -21)$.

5. Find A for C3

For $y = 12 \sin(x + B)$, any minimum has value -12 , so $A = -12$.

6. Find the smallest positive B

At the minimum $x = \frac{\pi}{4}$, so $\frac{\pi}{4} + B = \frac{3\pi}{2}$. Hence $B = \frac{5\pi}{4}$.

Final answer

(a) $P = (\frac{5\pi}{2}, 12)$.

(b) $(\frac{3\pi}{2}, -21)$.

(c)(i) $A = -12$.

(c)(ii) $B = \frac{5\pi}{4}$.

Question 7

Trigonometric Functions

7.

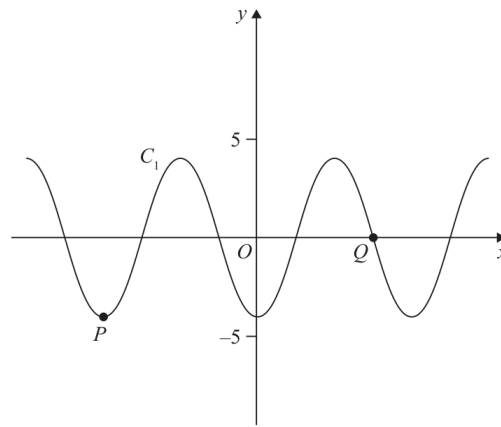


Figure 3

Figure 3 shows a plot of part of the curve C_1 with equation

$$y = -4 \cos x$$

where x is measured in radians.

Points P and Q lie on the curve and are shown in Figure 3.

(a) State

- (i) the coordinates of P
- (ii) the coordinates of Q

(3)

The curve C_2 has equation $y = -4 \cos x + k$ where x is measured in radians and k is a constant.

Given that C_2 has a maximum y value of 11

(b) (i) state the value of k

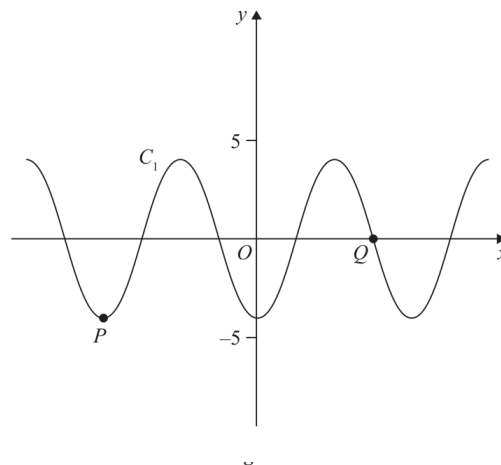
- (ii) state the coordinates of the minimum point on C_2 with the smallest positive x coordinate.

(3)

On the opposite page there is a copy of Figure 3 labelled Diagram 1.

(c) Using Diagram 1, state the number of solutions of the equation

$$-4 \cos x = 5 - \frac{10}{\pi} x$$



(Total for Question 7 is 8 marks)

Worked Solution - Question 7

1. Use y equals negative 4 cos x

For $y = -4 \cos x$, minimum points occur at $x = 2\pi n$ with value -4 .

2. Read P

The displayed minimum to the left of the y -axis is $P = (-2\pi, -4)$.

3. Read Q

The point Q is the positive x -intercept after the maximum at $x = \pi$, so $Q = (\frac{3\pi}{2}, 0)$.

4. Find k for C2

For $y = -4 \cos x + k$, the maximum value is $4 + k$. Given the maximum is 11 , $k = 7$.

5. Find the smallest positive minimum on C2

The minimum value is $-4 + 7 = 3$. The first positive minimum is at $x = 2\pi$, so the point is $(2\pi, 3)$.

6. Interpret the equation graphically

$-4 \cos x = 5 - \frac{10}{\pi}x$ is the intersection of the cosine curve with the straight line
 $y = 5 - \frac{10}{\pi}x$.

7. Count intersections

The line passes through $(0, 5)$ and meets the displayed curve once, at the x -axis point, so there is 1 solution.

Final answer

$$(a)(i) P = (-2\pi, -4). \quad (a)(ii) Q = \left(\frac{3\pi}{2}, 0\right).$$

$$(b)(i) k = 7.$$

$$(b)(ii) (2\pi, 3).$$

(c) 1 solution.

Mark scheme checked - Topic checked - WMA11-01_2024_Oct_Q07

Question 9

Trigonometric Functions

9.

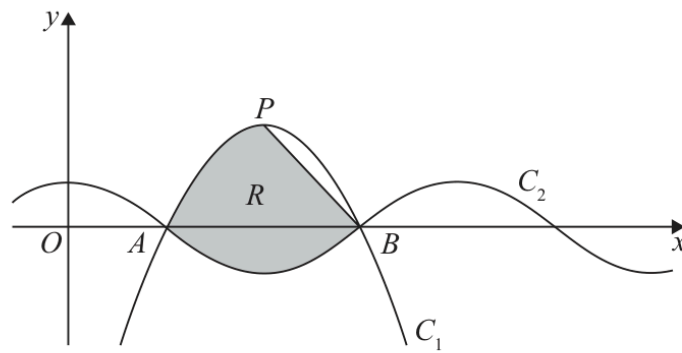


Figure 2

- (a) Express $6x - \frac{27}{4} - x^2$ in the form $a + b(x + c)^2$ where a , b and c are constants to be found.

(3)

Figure 2 shows part of a sketch of curve C_1 with equation

$$y = 6x - \frac{27}{4} - x^2$$

Given that the point P is the maximum point on C_1

- (b) state the coordinates of P

(2)

Figure 2 also shows part of a sketch of curve C_2 with equation

$$y = \cos(kx)$$

where k is a constant and x is measured in radians.

Given that C_1 and C_2 intersect on the x -axis at point A and at point B , as shown in Figure 2,

- (c) (i) state the x coordinate of B
 (ii) state the value of k
 (iii) state the period of C_2

(3)

The line segment L joins P and B .

The region R , shown shaded in Figure 2, is bounded by L , C_1 and C_2

- (d) Use inequalities to define R .

(5)

(Total for Question 9 is 13 marks)

Worked Solution - Question 9

1. Complete the square

$$6x - \frac{27}{4} - x^2 = -(x^2 - 6x) - \frac{27}{4} = -(x - 3)^2 + 9 - \frac{27}{4}.$$

2. State the completed-square form

$$6x - \frac{27}{4} - x^2 = \frac{9}{4} - (x - 3)^2.$$

3. Find the maximum point

The maximum point is $P = (3, \frac{9}{4})$.

4. Find the x-intercepts of C_1

$\frac{9}{4} - (x - 3)^2 = 0$ gives $x = 3 \pm \frac{3}{2}$. Hence A has $x = \frac{3}{2}$ and B has $x = \frac{9}{2}$.

5. Use consecutive zeros of cosine

The distance between consecutive zeros of $\cos(kx)$ is $\frac{\pi}{k}$.

6. Find k and the period

Since $x_B - x_A = \frac{9}{2} - \frac{3}{2} = 3$, $\frac{\pi}{k} = 3$, so $k = \frac{\pi}{3}$. The period is $\frac{2\pi}{k} = 6$.

7. Find the line through P and B

The line through $P(3, \frac{9}{4})$ and $B(\frac{9}{2}, 0)$ has gradient $-\frac{3}{2}$, so L is $y = -\frac{3}{2}x + \frac{27}{4}$.

8. Define the x-range

The region lies between A and B , so $\frac{3}{2} \leq x \leq \frac{9}{2}$.

9. Define the lower boundary

The lower boundary is C_2 , so $y \geq \cos\left(\frac{\pi}{3}x\right)$.

10. Define the upper boundaries

The region is below both C_1 and the line segment L , so $y \leq \frac{9}{4} - (x - 3)^2$ and $y \leq -\frac{3}{2}x + \frac{27}{4}$.

Final answer

(a) $\frac{9}{4} - (x - 3)^2$.

(b) $P = (3, \frac{9}{4})$.

(c)(i) $x_B = \frac{9}{2}$.

(c)(ii) $k = \frac{\pi}{3}$.

(c)(iii) 6.

(d) $\frac{3}{2} \leq x \leq \frac{9}{2}$, $y \geq \cos(\frac{\pi}{3}x)$, $y \leq \frac{9}{4} - (x - 3)^2$, $y \leq -\frac{3}{2}x + \frac{27}{4}$.

Mark scheme checked - Topic checked - WMA11-01_2025_Jan_Q09

Question 9

Trigonometric Functions

9.

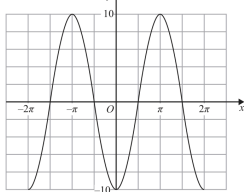


Figure 4

Figure 4 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$.

(a) Write down an expression for $f(x)$. (2)

Copies of Figure 4 (labelled Diagram 1 and Diagram 2) can be found on the following pages.

(b) (i) On Diagram 1 sketch a graph of the curve with equation

$$y = 10 - x^2$$

(ii) Hence find the **number** of solutions of the equation

$$f(x) = 10 - x^2 \quad \text{in the interval } -100\pi \leq x \leq 100\pi$$

(3)

(c) (i) On Diagram 2 sketch a graph of the curve with equation

$$y = \tan x \quad -2\pi \leq x \leq 2\pi$$

(ii) Hence find the **number** of solutions of the equation

$$f(x) = \tan x \quad \text{in the interval } -100\pi \leq x \leq 100\pi$$

giving a reason for your answer.

(4)

(a)

(b)(i)

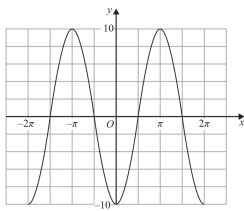
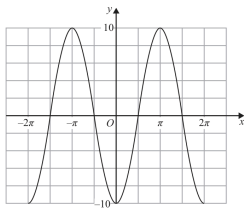


Diagram 1



Copy of Diagram 1

Only use this copy if you need to redraw your graph.

(b)(ii)

(c)(i)

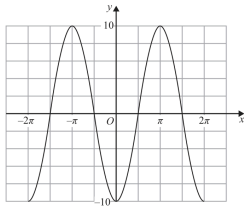
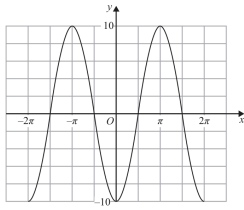


Diagram 2



Copy of Diagram 2

Only use this copy if you need to redraw your graph.

(c)(ii)

(Total for Question 9 is 9 marks)

Worked Solution - Question 9

1. Identify $f(x)$

The graph has amplitude 10, period 2π , and value -10 at $x = 0$, so $f(x) = -10 \cos x$.

2. Sketch y equals 10 minus x squared

The graph $y = 10 - x^2$ is a downward-opening parabola with vertex $(0, 10)$ and x -intercepts at $x = \pm\sqrt{10}$.

3. Count intersections with the parabola

On the given sketch, the parabola meets $f(x)$ twice. Outside the central part the parabola is below the range of $f(x)$, so there are 2 solutions in $-100\pi \leq x \leq 100\pi$.

4. Sketch y equals $\tan x$

The graph $y = \tan x$ has zeros at integer multiples of π and vertical asymptotes at $x = \frac{\pi}{2} + n\pi$.

5. Count in the reference interval

In $-2\pi \leq x \leq 2\pi$, the graph $y = \tan x$ intersects $f(x)$ four times.

6. Scale to the required interval

The interval $-100\pi \leq x \leq 100\pi$ is 50 times as long as $-2\pi \leq x \leq 2\pi$, so the number of solutions is $50 \times 4 = 200$.

Final answer

(a) $f(x) = -10 \cos x$.

(b)(ii) 2.

(c)(ii) 200.

TOPIC

Differentiation

Question 7

Differentiation

7.

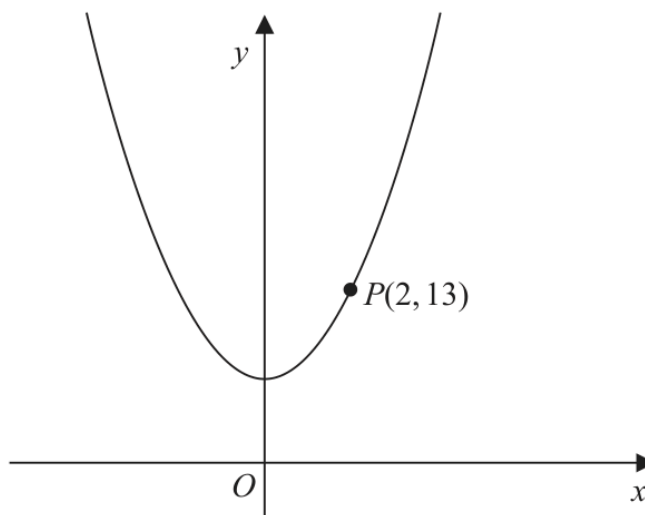


Figure 4

Figure 4 shows part of the curve with equation $y = 2x^2 + 5$

The point $P(2, 13)$ lies on the curve.

(a) Find the gradient of the tangent to the curve at P .

(2)

The point Q with x coordinate $2 + h$ also lies on the curve.

(b) Find, in terms of h , the gradient of the line PQ . Give your answer in simplest form.

(3)

(c) Explain briefly the relationship between the answer to (b) and the answer to (a).

(1)

(Total 6 marks)

Worked Solution - Question 7

1. Differentiate for the tangent gradient

For $y = 2x^2 + 5$, $\frac{dy}{dx} = 4x$. At P , $x = 2$, so the tangent gradient is 8.

2. Find the coordinates of Q

If $x_Q = 2 + h$, then $y_Q = 2(2 + h)^2 + 5 = 13 + 8h + 2h^2$.

3. Find the gradient of PQ

$$\text{gradient } PQ = \frac{(13 + 8h + 2h^2) - 13}{(2 + h) - 2} = \frac{8h + 2h^2}{h} = 8 + 2h.$$

4. Explain the link

As h gets closer to 0, point Q gets closer to P , and the chord gradient $8 + 2h$ tends to the tangent gradient 8.

Final answer

(a) 8.

(b) $8 + 2h$.

(c) As $h \rightarrow 0$, $8 + 2h \rightarrow 8$.

Question 9

Differentiation

9. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$y = \frac{4x^2 + 9}{2\sqrt{x}} \quad x > 0$$

Find the x coordinate of the point on the curve at which $\frac{dy}{dx} = 0$ (6)

(Total 6 marks)

Worked Solution - Question 9

1. Rewrite the function

$$y = \frac{4x^2 + 9}{2\sqrt{x}} = 2x^{3/2} + \frac{9}{2}x^{-1/2}.$$

2. Differentiate

$$\frac{dy}{dx} = 3x^{1/2} - \frac{9}{4}x^{-3/2}.$$

3. Set the derivative equal to zero

$$3x^{1/2} - \frac{9}{4}x^{-3/2} = 0, \text{ so } 3\sqrt{x} = \frac{9}{4x^{3/2}}.$$

4. Solve for x

Multiply by $4x^{3/2}$: $12x^2 = 9$. Hence $x^2 = \frac{3}{4}$. Since $x > 0$, $x = \frac{\sqrt{3}}{2}$.

Final answer

$$x = \frac{\sqrt{3}}{2}.$$

Mark scheme checked - Topic checked - WMA11-01_2020_Jan_Q09

Question 7**Differentiation**

7. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = 2x - 3\sqrt{x} - 5 \quad x > 0$$

(a) Solve the equation

$$f(x) = 9 \quad (4)$$

(b) Solve the equation

$$f''(x) = 6 \quad (5)$$

(Total 9 marks)

Worked Solution - Question 7

1. Solve $f(x) = 9$ without calculator methods

$$2x - 3\sqrt{x} - 5 = 9, \text{ so } 2x - 3\sqrt{x} - 14 = 0.$$

2. Use $u = \text{sqrt}(x)$

$$\text{Let } u = \sqrt{x}, \text{ where } u > 0. \text{ Then } x = u^2, \text{ so } 2u^2 - 3u - 14 = 0.$$

3. Factorise and reject the invalid root

$$2u^2 - 3u - 14 = (2u - 7)(u + 2). \text{ Since } u > 0, u = \frac{7}{2}, \text{ so } x = u^2 = \frac{49}{4}.$$

4. Differentiate twice

$$f(x) = 2x - 3x^{1/2} - 5, \text{ so } f'(x) = 2 - \frac{3}{2}x^{-1/2} \text{ and } f''(x) = \frac{3}{4}x^{-3/2}.$$

5. Solve $f''(x) = 6$

$$\frac{3}{4}x^{-3/2} = 6, \text{ so } x^{-3/2} = 8. \text{ Therefore } x^{3/2} = \frac{1}{8}.$$

6. Find x

$$\text{Since } x^{3/2} = (\sqrt{x})^3, \text{ we get } \sqrt{x} = \frac{1}{2}, \text{ hence } x = \frac{1}{4}.$$

Final answer

$$(a) x = \frac{49}{4}.$$

$$(b) x = \frac{1}{4}.$$

Question 7

Differentiation

7.

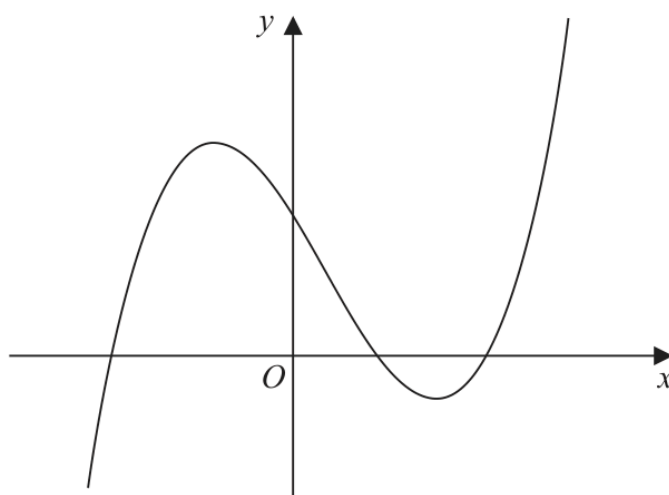


Figure 3

Figure 3 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (x + 4)(x - 2)(2x - 9)$$

Given that the curve with equation $y = f(x) - p$ passes through the point with coordinates $(0, 50)$

(a) find the value of the constant p .

(2)

Given that the curve with equation $y = f(x + q)$ passes through the origin,

(b) write down the possible values of the constant q .

(2)

(c) Find $f'(x)$.

(4)

(d) Hence find the range of values of x for which the gradient of the curve with equation $y = f(x)$ is less than -18

(3)

(Total 11 marks)

Worked Solution - Question 7

1. Find p from the translated curve

$f(0) = (4)(-2)(-9) = 72$. Since $y = f(x) - p$ passes through $(0, 50)$, $72 - p = 50$, so $p = 22$.

2. Use roots for $f(x + q)$

If $y = f(x + q)$ passes through the origin, then $f(q) = 0$. From $f(x) = (x + 4)(x - 2)(2x - 9)$, the possible values are $q = -4, 2, \frac{9}{2}$.

3. Expand $f(x)$

$$(x + 4)(x - 2)(2x - 9) = 2x^3 - 5x^2 - 34x + 72.$$

4. Differentiate

$$f'(x) = 6x^2 - 10x - 34.$$

5. Set the gradient less than -18

$$6x^2 - 10x - 34 < -18, \text{ so } 6x^2 - 10x - 16 < 0, \text{ or } 3x^2 - 5x - 8 < 0.$$

6. Solve the inequality

$3x^2 - 5x - 8 = (3x - 8)(x + 1)$. Since the quadratic opens upward, it is negative between the roots: -1

Final answer

(a) $p = 22$.

(b) $q = -4, 2, \frac{9}{2}$.

(c) $f'(x) = 6x^2 - 10x - 34$.

(d) -1

Mark scheme checked - Topic checked - WMA11-01_2022_Jan_Q07

Question 10

Differentiation

10.

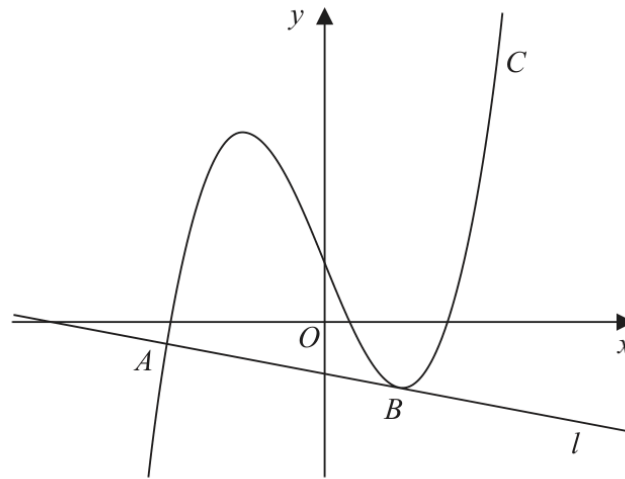


Figure 5

Figure 5 shows a sketch of the curve C with equation

$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where k is a constant.

(a) Find $\frac{dy}{dx}$

(2)

The line l , shown in Figure 5, is the normal to C at the point A with x coordinate $-\frac{7}{2}$

Given that l is also a tangent to C at the point B ,

(b) show that the x coordinate of the point B is a solution of the equation

$$12x^2 + 4x - 33 = 0$$

(4)

(c) Hence find the x coordinate of B , justifying your answer.

(2)

Given that the y intercept of l is -1

(d) find the value of k .

(4)

(Total 12 marks)

Worked Solution - Question 10

1. Differentiate the curve

For $y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$, $\frac{dy}{dx} = \frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}$.

2. Find the tangent gradient at A

At $x = -\frac{7}{2}$, $\frac{dy}{dx} = \frac{6}{7}\left(\frac{49}{4}\right) + \frac{2}{7}\left(-\frac{7}{2}\right) - \frac{5}{2} = \frac{21}{2} - 1 - \frac{5}{2} = 7$.

3. Find the gradient of l

Since l is the normal at A , its gradient is the negative reciprocal of 7, so $m_l = -\frac{1}{7}$.

4. Use that l is tangent at B

At B , the tangent gradient is also $-\frac{1}{7}$, so $\frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2} = -\frac{1}{7}$.

5. Reach the given quadratic

Multiplying by 14 gives $12x^2 + 4x - 35 = -2$, hence $12x^2 + 4x - 33 = 0$.

6. Solve for the x-coordinate of B

$12x^2 + 4x - 33 = 0$ gives $x = \frac{-4 \pm 40}{24}$, so $x = \frac{3}{2}$ or $x = -\frac{11}{6}$. From the diagram, B is the point to the right of the y-axis, so $x_B = \frac{3}{2}$.

7. Use the y-intercept of l

Since the y-intercept is -1 and the gradient is $-\frac{1}{7}$, the line is $y = -\frac{1}{7}x - 1$.

8. Find the y-coordinate of A

At $x = -\frac{7}{2}$ on the line, $y = -\frac{1}{7}\left(-\frac{7}{2}\right) - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$.

9. Substitute A into the curve

$$-\frac{1}{2} = \frac{2}{7} \left(-\frac{7}{2}\right)^3 + \frac{1}{7} \left(-\frac{7}{2}\right)^2 - \frac{5}{2} \left(-\frac{7}{2}\right) + k = -\frac{7}{4} + k.$$

10. Find k

Therefore $k = \frac{5}{4}$.

Final answer

$$(a) \frac{dy}{dx} = \frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}.$$

$$(c) x_B = \frac{3}{2}.$$

$$(d) k = \frac{5}{4}.$$

Mark scheme checked - Topic checked - WMA11-01_2022_MayJune_Q10

Question 9

Differentiation

9.

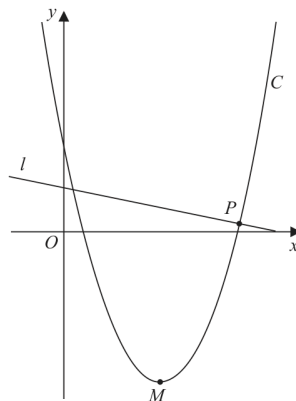


Figure 3

Figure 3 shows a sketch of the curve C with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

(a) Write $\frac{1}{2}x^2 - 10x + 22$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

The point M is the minimum turning point of C , as shown in Figure 3.

(b) Deduce the coordinates of M

(2)

The line l is the normal to C at the point P , as shown in Figure 3.

Given that l has equation $y = k - \frac{1}{8}x$, where k is a constant,

(c) (i) find the coordinates of P

(ii) find the value of k

(6)

Question 9 continues on the next page

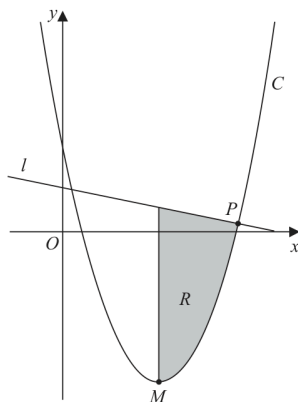


Figure 4

Figure 4 is a copy of Figure 3. The finite region R , shown shaded in Figure 4, is bounded by l , C and the line through M parallel to the y -axis.

(d) Identify the inequalities that define R .

(3)

(Total for Question 9 is 14 marks)

Worked Solution - Question 9

1. Complete the square

$$\frac{1}{2}x^2 - 10x + 22 = \frac{1}{2}(x^2 - 20x) + 22 = \frac{1}{2}((x - 10)^2 - 100) + 22.$$

2. Simplify the completed square form

$$\frac{1}{2}((x - 10)^2 - 100) + 22 = \frac{1}{2}(x - 10)^2 - 28.$$

3. Read the minimum point

The minimum point is $M(10, -28)$.

4. Use the normal gradient

The line l has gradient $-\frac{1}{8}$, so the tangent gradient at P is 8.

5. Differentiate the curve

$$\frac{dy}{dx} = x - 10. \text{ At } P, x - 10 = 8, \text{ so } x = 18.$$

6. Find P

$$\text{When } x = 18, y = \frac{1}{2}(18)^2 - 10(18) + 22 = 4, \text{ so } P = (18, 4).$$

7. Find k

$$\text{Since } P \text{ lies on } y = k - \frac{1}{8}x, 4 = k - \frac{1}{8}(18) = k - \frac{9}{4}, \text{ so } k = \frac{25}{4}.$$

8. Write the upper boundary

$$\text{The line is } y = \frac{25}{4} - \frac{x}{8}.$$

9. Write the lower boundary

$$\text{The curve is } y = \frac{1}{2}x^2 - 10x + 22.$$

10. Identify the region

The region lies to the right of the vertical line through M and between the curve and the normal line: $10 \leq x \leq 18$ and $\frac{1}{2}x^2 - 10x + 22 \leq y \leq \frac{25}{4} - \frac{x}{8}$.

Final answer

(a) $\frac{1}{2}(x - 10)^2 - 28$.

(b) $M(10, -28)$.

(c) $P(18, 4)$, $k = \frac{25}{4}$.

(d) $10 \leq x \leq 18$, $\frac{1}{2}x^2 - 10x + 22 \leq y \leq \frac{25}{4} - \frac{x}{8}$.

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Question 10

Differentiation

10.

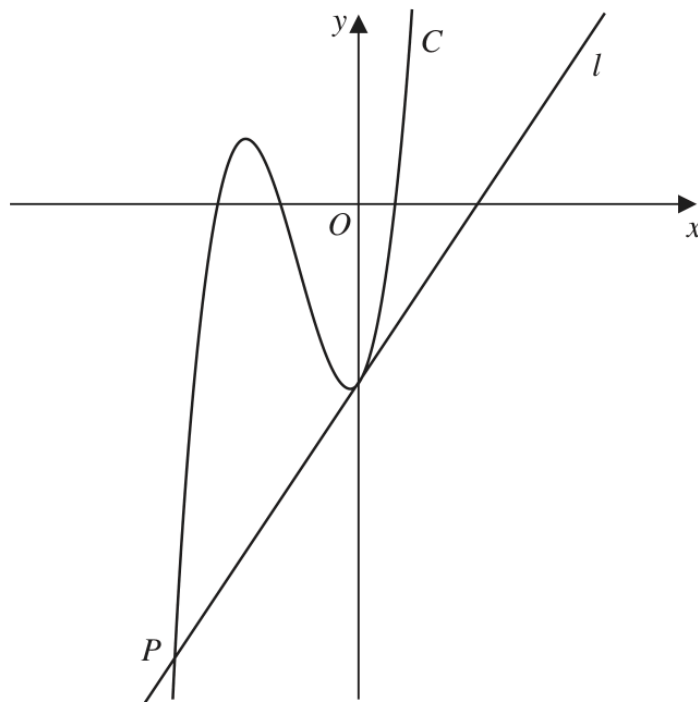
**Figure 4**

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x + 20)(x + 6)(2x - 3)$$

- (a) Use the given information to state the values of x for which

$$f(x) > 0$$

(2)

- (b) Expand $(3x + 20)(x + 6)(2x - 3)$, writing your answer as a polynomial in simplest form.

(3)

The straight line l is the tangent to C at the point where C cuts the y -axis.

Given that l cuts C at the point P , as shown in Figure 4,

- (c) find, using algebra, the x coordinate of P

(Solutions based on calculator technology are not acceptable.)

(5)

(Total for Question 10 is 10 marks)

Worked Solution - Question 10

1. Use the roots

$f(x) = (3x + 20)(x + 6)(2x - 3)$ has roots $x = -\frac{20}{3}$, $x = -6$, and $x = \frac{3}{2}$.

2. State where $f(x)$ is positive

Since the cubic has positive leading coefficient, $f(x) > 0$ for $-\frac{20}{3} < x < -6$ and $x > \frac{3}{2}$.

3. Expand the first two brackets

$$(3x + 20)(x + 6) = 3x^2 + 38x + 120.$$

4. Complete the expansion

$$(3x^2 + 38x + 120)(2x - 3) = 6x^3 + 67x^2 + 126x - 360.$$

5. Find the tangent at the y-axis

$f'(x) = 18x^2 + 134x + 126$, so at $x = 0$ the tangent gradient is 126. Also $f(0) = -360$.

6. Write the tangent equation

The tangent line is $y = 126x - 360$.

7. Find the other intersection

Set $6x^3 + 67x^2 + 126x - 360 = 126x - 360$. This gives $6x^3 + 67x^2 = 0$, so $x^2(6x + 67) = 0$.

8. Choose point P

$x = 0$ is the tangent point on the y-axis, so the other intersection P has $x = -\frac{67}{6}$.

Final answer

$$(a) \quad -\frac{20}{3} \frac{32}{3}$$

$$(b) \quad 6x^3 + 67x^2 + 126x - 360.$$

$$(c) \quad x_P = -\frac{67}{6}.$$

Mark scheme checked - Topic checked - WMA11-01_2023_Jan_Q10

Question 7

Differentiation

7. The curve C has equation $y = f(x)$ where

$$f(x) = 2x^3 - kx^2 + 14x + 24$$

and k is a constant.

(a) Find, in simplest form,

(i) $f'(x)$

(ii) $f''(x)$

(3)

The curve with equation $y = f'(x)$ intersects the curve with equation $y = f''(x)$ at the points A and B .

Given that the x coordinate of A is 5

(b) find the value of k .

(2)

(c) Hence find the coordinates of B .

(3)

(Total for Question 7 is 8 marks)

Worked Solution - Question 7

1. Differentiate once

$$f'(x) = 6x^2 - 2kx + 14.$$

2. Differentiate again

$$f''(x) = 12x - 2k.$$

3. Use the intersection at A

At an intersection of $y = f'(x)$ and $y = f''(x)$, $f'(x) = f''(x)$. Given $x_A = 5$, substitute $x = 5$.

4. Find k

$$6(5)^2 - 2k(5) + 14 = 12(5) - 2k, \text{ so } 164 - 10k = 60 - 2k \text{ and } k = 13.$$

5. Find the other x-coordinate

With $k = 13$, $6x^2 - 26x + 14 = 12x - 26$, so $6x^2 - 38x + 40 = 0$. Hence $(3x - 4)(x - 5) = 0$.

6. Use the point B

The other point has $x = \frac{4}{3}$. Then $y = f''(\frac{4}{3}) = 12(\frac{4}{3}) - 26 = -10$.

Final answer

$$(a)(i) f'(x) = 6x^2 - 2kx + 14. \quad (a)(ii) f''(x) = 12x - 2k.$$

$$(b) k = 13.$$

$$(c) B = (\frac{4}{3}, -10).$$

Question 9

Differentiation

9.

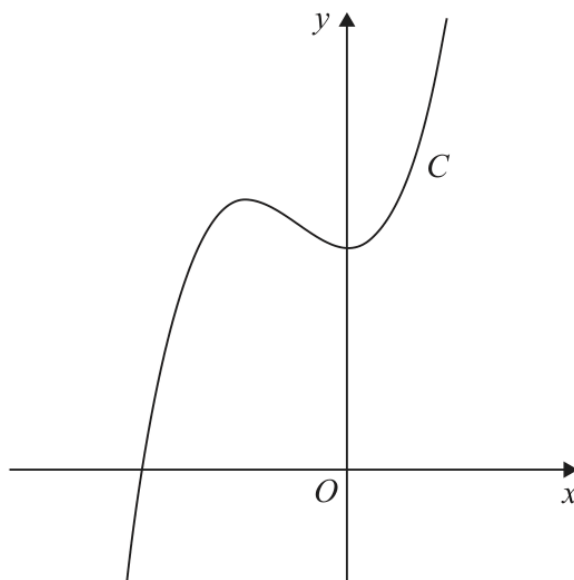


Figure 4

Figure 4 shows a sketch of the curve C with equation $y = f(x)$, where

$$f(x) = (x + 5)(3x^2 - 4x + 20)$$

(a) Deduce the range of values of x for which $f(x) \geq 0$ (1)

(b) Find $f'(x)$ giving your answer in simplest form. (3)

The point $R(-4, 84)$ lies on C .

Given that the tangent to C at the point P is parallel to the tangent to C at the point R

(c) find the x coordinate of P . (4)

(d) Find the point to which R is transformed when the curve with equation $y = f(x)$ is transformed to the curve with equation,

(i) $y = f(x - 3)$

(ii) $y = 4f(x)$ (2)

(Total for Question 9 is 10 marks)

Worked Solution - Question 9

1. Use the positive quadratic factor

$3x^2 - 4x + 20$ has discriminant $(-4)^2 - 4(3)(20) < 0$ and positive leading coefficient, so it is always positive.

2. Deduce the sign of $f(x)$

The sign of $f(x) = (x + 5)(3x^2 - 4x + 20)$ is therefore the sign of $x + 5$. Hence $f(x) \geq 0$ for $x \geq -5$.

3. Expand $f(x)$

$$f(x) = 3x^3 + 11x^2 + 100.$$

4. Differentiate

$$f'(x) = 9x^2 + 22x.$$

5. Find the gradient at R

$$\text{At } R(-4, 84), f'(-4) = 9(16) + 22(-4) = 56.$$

6. Use parallel tangents

For the tangent at P to be parallel, $f'(x) = 56$.

7. Solve for x

$$9x^2 + 22x - 56 = 0, \text{ so } x = \frac{-22 \pm 50}{18}. \text{ The roots are } x = -4 \text{ and } x = \frac{14}{9}.$$

8. Choose the point P

$$x = -4 \text{ is the point } R, \text{ so } x_P = \frac{14}{9}.$$

9. Transform R under y equals f of x minus 3

$y = f(x - 3)$ translates the curve 3 units to the right, so $(-4, 84)$ maps to $(-1, 84)$.

10. Transform R under y equals 4f(x)

$y = 4f(x)$ stretches vertically by scale factor 4, so $(-4, 84)$ maps to $(-4, 336)$.

Final answer

(a) $x \geq -5$.

(b) $f'(x) = 9x^2 + 22x$.

(c) $x_P = \frac{14}{9}$.

(d)(i) $(-1, 84)$.

(d)(ii) $(-4, 336)$.

Mark scheme checked - Topic checked - WMA11-01_2024_Oct_Q09

Question 10

Differentiation

10.

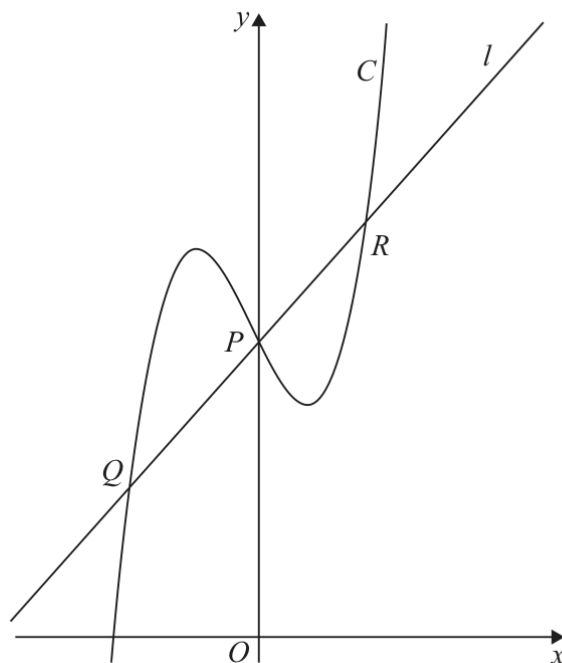


Figure 4

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 4 shows a sketch of the curve C with equation

$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5$$

The line l is the normal to C at the point P where $x = 0$

The line l also intersects C at points Q and R as shown in Figure 4.

(a) Find, using algebra, the x coordinate of point Q .

(6)

The point T lies on C .

Given that

- the tangent to C at T is parallel to l
- the x coordinate of T is positive

(b) find, using algebra, the exact x coordinate of T .

(4)

(Total for Question 10 is 10 marks)

Worked Solution - Question 10

1. Differentiate C

$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5, \text{ so } \frac{dy}{dx} = 6x^2 + x - 2.$$

2. Find the tangent gradient at P

At P , $x = 0$, so the tangent gradient is -2 .

3. Equation of the normal

The normal gradient is $\frac{1}{2}$ and $P = (0, 5)$, so $l : y = \frac{1}{2}x + 5$.

4. Find intersections with the normal

$$\text{Set } 2x^3 + \frac{1}{2}x^2 - 2x + 5 = \frac{1}{2}x + 5.$$

5. Form and factor the cubic

$$2x^3 + \frac{1}{2}x^2 - \frac{5}{2}x = 0. \text{ Multiplying by 2 gives}$$
$$4x^3 + x^2 - 5x = 0 = x(4x^2 + x - 5).$$

6. Solve for x

$$x(4x + 5)(x - 1) = 0, \text{ so the intersections have x-coordinates } 0, -\frac{5}{4} \text{ and } 1.$$

7. Select point Q

$$\text{The point } Q \text{ is the left-hand intersection, so } x_Q = -\frac{5}{4}.$$

8. Parallel tangent condition

$$\text{The tangent at } T \text{ is parallel to } l, \text{ so its gradient is } \frac{1}{2}.$$

9. Set derivative equal to one half

$$6x^2 + x - 2 = \frac{1}{2}, \text{ so } 12x^2 + 2x - 5 = 0.$$

10. Use the positive solution

$$x = \frac{-2 \pm \sqrt{4 + 240}}{24} = \frac{-1 \pm \sqrt{61}}{12}. \text{ Since } x_T \text{ is positive, } x_T = \frac{-1 + \sqrt{61}}{12}.$$

Final answer

$$(a) x_Q = -\frac{5}{4}.$$

$$(b) x_T = \frac{-1 + \sqrt{61}}{12}.$$

Mark scheme checked - Topic checked - WMA11-01_2025_Oct_Q10

Question 7

Differentiation

7. The curve C has equation

$$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3}$$

(a) Show that $\frac{dy}{dx}$ can be written in the form

$$p(x + q)^2 + r$$

where p , q and r are constants to be found.

(5)

(b) Hence state

(i) the minimum value of $\frac{dy}{dx}$

(ii) the value of x at which this minimum value occurs.

(2)

Given that S is the point on C at which the gradient is a minimum,

(c) find the equation of the tangent to C at S , giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

(Total for Question 7 is 10 marks)

Worked Solution - Question 7

1. Differentiate

$$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3} \text{ gives } \frac{dy}{dx} = 2x^2 - 16x + 43.$$

2. Complete the square

$$2x^2 - 16x + 43 = 2(x^2 - 8x) + 43 = 2[(x - 4)^2 - 16] + 43.$$

3. Simplify

$$\frac{dy}{dx} = 2(x - 4)^2 + 11.$$

4. Minimum gradient

Since $2(x - 4)^2 \geq 0$, the minimum value of $\frac{dy}{dx}$ is 11.

5. Where it occurs

The minimum occurs when $x - 4 = 0$, so $x = 4$.

6. Find the point S

$$\text{At } x = 4, y = \frac{2}{3}(4)^3 - 8(4)^2 + 43(4) - \frac{20}{3} = 80.$$

7. Tangent equation

The gradient at S is 11, so $y - 80 = 11(x - 4)$.

8. Simplify

$$y = 11x + 36.$$

Final answer

$$(a) \frac{dy}{dx} = 2(x - 4)^2 + 11.$$

$$(b) \min \frac{dy}{dx} = 11 \text{ at } x = 4.$$

$$(c) y = 11x + 36.$$

Mark scheme checked - Topic checked - WMA11-01_2026_Jan_Q07

TOPIC

Integration

Question 11

11. A curve has equation $y = f(x)$.

The point $P\left(4, \frac{32}{3}\right)$ lies on the curve.

Given that

- $f''(x) = \frac{4}{\sqrt{x}} - 3$
- $f'(x) = 5$ at P

find

(a) the equation of the tangent to the curve at P , writing your answer in the form $y = mx + c$, where m and c are constants to be found,

(2)

(b) $f(x)$.

(8)

(Total 10 marks)

Worked Solution - Question 11

1. Use the given gradient for the tangent

At $P\left(4, \frac{32}{3}\right)$, $f'(4) = 5$, so the tangent has gradient 5.

2. Find the tangent equation

$y - \frac{32}{3} = 5(x - 4)$. Hence $y = 5x - \frac{28}{3}$.

3. Integrate f'' to find f'

$f''(x) = 4x^{-1/2} - 3$, so $f'(x) = 8x^{1/2} - 3x + C$.

4. Use $f'(4)=5$

$5 = 8(2) - 12 + C = 4 + C$, so $C = 1$. Therefore $f'(x) = 8\sqrt{x} - 3x + 1$.

5. Integrate again

$f(x) = \frac{16}{3}x^{3/2} - \frac{3}{2}x^2 + x + D$.

6. Use the point on the curve

Substitute $x = 4$, $f(4) = \frac{32}{3}$: $\frac{32}{3} = \frac{16}{3}(8) - \frac{3}{2}(16) + 4 + D = \frac{68}{3} + D$, so $D = -12$.

Final answer

(a) $y = 5x - \frac{28}{3}$.

(b) $f(x) = \frac{16}{3}x^{3/2} - \frac{3}{2}x^2 + x - 12$.

Question 11

Integration

11. A curve has equation $y = f(x)$, where

$$f''(x) = \frac{6}{\sqrt{x^3}} + x \quad x > 0$$

The point $P(4, -50)$ lies on the curve.

Given that $f'(x) = -4$ at P ,

- (a) find the equation of the normal at P , writing your answer in the form $y = mx + c$, where m and c are constants,

(3)

- (b) find $f(x)$.

(8)

(Total 11 marks)

Worked Solution - Question 11

1. Find the normal gradient

At P , $f'(4) = -4$, so the tangent gradient is -4 . The normal gradient is therefore $\frac{1}{4}$.

2. Find the normal equation

Using $P(4, -50)$: $y + 50 = \frac{1}{4}(x - 4)$, so $y = \frac{1}{4}x - 51$.

3. Integrate f'' to find f'

$f''(x) = 6x^{-3/2} + x$. Therefore $f'(x) = -12x^{-1/2} + \frac{1}{2}x^2 + k$.

4. Use $f'(4) = -4$

$-4 = -\frac{12}{2} + \frac{1}{2}(16) + k = -6 + 8 + k = 2 + k$, so $k = -6$.

5. Integrate again

$f(x) = -24x^{1/2} + \frac{x^3}{6} - 6x + d$.

6. Use the point on the curve

Substitute $P(4, -50)$:

$-50 = -24(2) + \frac{64}{6} - 24 + d = -72 + \frac{32}{3} + d = -\frac{184}{3} + d$. Hence $d = \frac{34}{3}$.

Final answer

(a) $y = \frac{1}{4}x - 51$.

(b) $f(x) = -24\sqrt{x} + \frac{x^3}{6} - 6x + \frac{34}{3}$.

Question 9

Integration

9. (i) Find

$$\int \frac{(3x + 2)^2}{4\sqrt{x}} \, dx \quad x > 0$$

giving your answer in simplest form.

(5)

(ii) A curve C has equation $y = f(x)$.

Given

- $f'(x) = x^2 + ax + b$ where a and b are constants
- the y intercept of C is -8
- the point $P(3, -2)$ lies on C
- the gradient of C at P is 2

find, in simplest form, $f(x)$.

(6)

(Total 11 marks)

Worked Solution - Question 9

1. Expand and simplify the integrand

$$\frac{(3x+2)^2}{4\sqrt{x}} = \frac{9x^2 + 12x + 4}{4x^{1/2}} = \frac{9}{4}x^{3/2} + 3x^{1/2} + x^{-1/2}.$$

2. Integrate term by term

$$\int \left(\frac{9}{4}x^{3/2} + 3x^{1/2} + x^{-1/2} \right) dx = \frac{9}{10}x^{5/2} + 2x^{3/2} + 2x^{1/2} + c.$$

3. Write in simplest form

Factoring out $\sqrt{x}$ gives $\frac{\sqrt{x}(9x^2 + 20x + 20)}{10} + c.$

4. Use the gradient condition at P

Since $f'(x) = x^2 + ax + b$ and the gradient at $P(3, -2)$ is 2, $f'(3) = 2$. Hence $9 + 3a + b = 2$, so $3a + b = -7$.

5. Integrate $f'(x)$

$f(x) = \frac{x^3}{3} + \frac{a}{2}x^2 + bx + c$. The y-intercept is -8 , so $c = -8$.

6. Use $P(3, -2)$

$-2 = \frac{27}{3} + \frac{9a}{2} + 3b - 8$, so $\frac{9a}{2} + 3b = -3$.

7. Solve for a and b

From $3a + b = -7$, $b = -7 - 3a$. Substitute into $\frac{9a}{2} + 3b = -3$ to get $a = -4$, and then $b = 5$.

8. Write $f(x)$

Therefore $f(x) = \frac{x^3}{3} - 2x^2 + 5x - 8$.

Final answer

$$(i) \frac{\sqrt{x}(9x^2 + 20x + 20)}{10} + c. \quad (ii) f(x) = \frac{x^3}{3} - 2x^2 + 5x - 8.$$

Mark scheme checked - Topic checked - WMA11-01_2021_Jan_Q09

Question 6

6. The curve C has equation $y = f(x)$, $x > 0$

Given that

- C passes through the point $P(8, 2)$
- $f'(x) = \frac{32}{3x^2} + 3 - 2\sqrt[3]{x}$

- (a) find the equation of the tangent to C at P . Write your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

- (b) Find, in simplest form, $f(x)$.

(5)

(Total 8 marks)

Worked Solution - Question 6

1. Find the gradient at P

$$f'(x) = \frac{32}{3x^2} + 3 - 2\sqrt[3]{x}. \text{ At } x = 8, f'(8) = \frac{32}{3(64)} + 3 - 2(2) = \frac{1}{6} - 1 = -\frac{5}{6}.$$

2. Write the tangent equation

$$\text{Using } P(8, 2), y - 2 = -\frac{5}{6}(x - 8), \text{ hence } y = -\frac{5}{6}x + \frac{26}{3}.$$

3. Integrate $f'(x)$

$$f'(x) = \frac{32}{3}x^{-2} + 3 - 2x^{1/3}, \text{ so } f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{4/3} + c.$$

4. Use $P(8, 2)$ to find c

$$2 = -\frac{32}{24} + 24 - \frac{3}{2}(16) + c = -\frac{4}{3} + c, \text{ so } c = \frac{10}{3}.$$

5. Write $f(x)$

$$f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{4/3} + \frac{10}{3}.$$

Final answer

$$(a) y = -\frac{5}{6}x + \frac{26}{3}.$$

$$(b) f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{4/3} + \frac{10}{3}.$$

Question 10

10. A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = ax - 12x^{\frac{1}{3}}$, where a is a constant
- $f''(x) = 0$ when $x = 27$
- the curve passes through the point $(1, -8)$

(a) find the value of a .

(3)

(b) Hence find $f(x)$.

(4)

(Total 7 marks)

Worked Solution - Question 10

1. Differentiate $f'(x)$

$$f'(x) = ax - 12x^{1/3}, \text{ so } f''(x) = a - 4x^{-2/3}.$$

2. Use $f''(27) = 0$

$$\text{Since } 27^{2/3} = 9, 0 = a - \frac{4}{9}, \text{ so } a = \frac{4}{9}.$$

3. Integrate $f'(x)$

$$\text{With } a = \frac{4}{9}, f'(x) = \frac{4}{9}x - 12x^{1/3}. \text{ Therefore } f(x) = \frac{2}{9}x^2 - 9x^{4/3} + c.$$

4. Use the point (1, -8)

$$-8 = \frac{2}{9} - 9 + c, \text{ so } c = -8 + 9 - \frac{2}{9} = \frac{7}{9}.$$

5. Write $f(x)$

$$f(x) = \frac{2}{9}x^2 - 9x^{4/3} + \frac{7}{9}.$$

Final answer

$$(a) a = \frac{4}{9}.$$

$$(b) f(x) = \frac{2}{9}x^2 - 9x^{4/3} + \frac{7}{9}.$$

Question 6

Integration

6. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{(x+3)^2}{x\sqrt{x}}$
- the point $P(4, 20)$ lies on C

- (a) (i) find the value of the gradient at P

- (ii) Hence find the equation of the tangent to C at P , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

- (b) Find $f(x)$, simplifying your answer.

(7)

(Total 11 marks)

Worked Solution - Question 6

1. Simplify $f'(x)$

$$f'(x) = \frac{(x+3)^2}{x\sqrt{x}} = \frac{x^2 + 6x + 9}{x^{3/2}} = x^{1/2} + 6x^{-1/2} + 9x^{-3/2}.$$

2. Find the gradient at P

$$\text{At } x = 4, f'(4) = 2 + 3 + \frac{9}{8} = \frac{49}{8}.$$

3. Find the tangent equation

$$\text{Using } P(4, 20), y - 20 = \frac{49}{8}(x - 4). \text{ Multiplying out gives } 49x - 8y - 36 = 0.$$

4. Integrate $f'(x)$

$$f(x) = \int (x^{1/2} + 6x^{-1/2} + 9x^{-3/2}) dx = \frac{2}{3}x^{3/2} + 12x^{1/2} - 18x^{-1/2} + c.$$

5. Use $P(4, 20)$

$$20 = \frac{2}{3}(8) + 12(2) - \frac{18}{2} + c = \frac{61}{3} + c, \text{ so } c = -\frac{1}{3}.$$

6. Write $f(x)$

$$f(x) = \frac{2}{3}x\sqrt{x} + 12\sqrt{x} - \frac{18}{\sqrt{x}} - \frac{1}{3}.$$

Final answer

$$(a)(i) \frac{49}{8}. \quad (a)(ii) 49x - 8y - 36 = 0.$$

$$(b) f(x) = \frac{2}{3}x\sqrt{x} + 12\sqrt{x} - \frac{18}{\sqrt{x}} - \frac{1}{3}.$$

Question 7

Integration

7. The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{2}{\sqrt{x}} + \frac{A}{x^2} + 3$, where A is a constant
- $f''(x) = 0$ when $x = 4$

(a) find the value of A .

(4)

Given also that

- $f(x) = 8\sqrt{3}$, when $x = 12$

(b) find $f(x)$, giving each term in simplest form.

(5)

(Total 9 marks)

Worked Solution - Question 7

1. Differentiate $f'(x)$

$$f'(x) = 2x^{-1/2} + Ax^{-2} + 3, \text{ so } f''(x) = -x^{-3/2} - 2Ax^{-3}.$$

2. Use $f''(4) = 0$

$$-4^{-3/2} - 2A(4^{-3}) = 0, \text{ so } -\frac{1}{8} - \frac{A}{32} = 0 \text{ and } A = -4.$$

3. Substitute A into $f'(x)$

$$f'(x) = \frac{2}{\sqrt{x}} - \frac{4}{x^2} + 3.$$

4. Integrate to find $f(x)$

$$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x + c.$$

5. Use $f(12) = 8\sqrt{3}$

$$f(12) = 8\sqrt{3} + \frac{1}{3} + 36 + c = 8\sqrt{3}, \text{ so } c = -\frac{109}{3}.$$

Final answer

$$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x - \frac{109}{3}.$$

Question 11

Integration

11. A curve C has equation $y = f(x)$, $x > 0$

Given that

- $f''(x) = 4x + \frac{1}{\sqrt{x}}$
- the point P has x coordinate 4 and lies on C
- the tangent to C at P has equation $y = 3x + 4$

(a) find an equation of the normal to C at P

(2)

(b) find $f(x)$, writing your answer in simplest form.

(6)

(Total for Question 11 is 8 marks)

Worked Solution - Question 11

1. Find point P

Since P has x-coordinate 4 and lies on the tangent $y = 3x + 4$, $P = (4, 16)$.

2. Find the normal

The tangent gradient is 3, so the normal gradient is $-\frac{1}{3}$. Hence the normal is $y - 16 = -\frac{1}{3}(x - 4)$.

3. Integrate $f''(x)$

$$f''(x) = 4x + x^{-1/2}, \text{ so } f'(x) = 2x^2 + 2\sqrt{x} + A.$$

4. Use the tangent gradient

At $x = 4$, $f'(4) = 3$. Thus $2(16) + 2(2) + A = 3$, so $A = -33$.

5. Integrate again

$$f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + B.$$

6. Use P(4, 16)

$$16 = \frac{2}{3}(64) + \frac{4}{3}(8) - 33(4) + B = \frac{160}{3} - 132 + B.$$

7. Find B

$$B = \frac{284}{3}, \text{ so } f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + \frac{284}{3}.$$

Final answer

$$(a) y - 16 = -\frac{1}{3}(x - 4).$$

$$(b) f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + \frac{284}{3}.$$

Question 8

8.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

- (a) Find the equation of the tangent to the curve with equation

$$y = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

at the point $P(4, 12)$

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(5)

The curve with equation $y = f(x)$ also passes through the point $P(4, 12)$

Given that

$$f'(x) = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

- (b) find $f(x)$ giving the coefficients in simplest form.

(5)**(Total for Question 8 is 10 marks)**

Worked Solution - Question 8

1. Differentiate the curve

$$\text{For } y = \frac{1}{4}x^3 - 8x^{-1/2}, \frac{dy}{dx} = \frac{3}{4}x^2 + 4x^{-3/2}.$$

2. Find the gradient at P

$$\text{At } x = 4, \frac{dy}{dx} = \frac{3}{4}(16) + 4(4^{-3/2}) = 12 + \frac{1}{2} = \frac{25}{2}.$$

3. Find the tangent

$$\text{Using } P(4, 12), y - 12 = \frac{25}{2}(x - 4).$$

4. Write in integer form

Multiplying by 2 gives $2y - 24 = 25x - 100$, hence $25x - 2y - 76 = 0$.

5. Integrate $f'(x)$

$$f'(x) = \frac{1}{4}x^3 - 8x^{-1/2} \text{ gives } f(x) = \frac{x^4}{16} - 16\sqrt{x} + c.$$

6. Use $P(4, 12)$

$$12 = \frac{4^4}{16} - 16\sqrt{4} + c = 16 - 32 + c, \text{ so } c = 28.$$

Final answer

$$(a) 25x - 2y - 76 = 0.$$

$$(b) f(x) = \frac{x^4}{16} - 16\sqrt{x} + 28.$$

Question 7

7. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{4x^2 + 10 - 7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}$

- the point $P(4, -1)$ lies on C

(a) (i) find the value of the gradient of C at P

(ii) Hence find the equation of the normal to C at P , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

(b) Find $f(x)$.

(6)

(Total for Question 7 is 10 marks)

Worked Solution - Question 7

1. Simplify f' prime

$$f'(x) = \frac{4x^2 + 10 - 7x^{1/2}}{4x^{1/2}} = x^{3/2} + \frac{5}{2}x^{-1/2} - \frac{7}{4}.$$

2. Find the gradient at P

$$\text{At } x = 4, f'(4) = 4^{3/2} + \frac{5}{2}(4^{-1/2}) - \frac{7}{4} = 8 + \frac{5}{4} - \frac{7}{4} = \frac{15}{2}.$$

3. Find the normal gradient

The normal gradient is the negative reciprocal, so $m_n = -\frac{2}{15}$.

4. Use point P

$$\text{Using } P(4, -1), y + 1 = -\frac{2}{15}(x - 4).$$

5. Write the normal in integer form

$$15y + 15 = -2x + 8, \text{ hence } 2x + 15y + 7 = 0.$$

6. Integrate f' prime

$$f(x) = \int \left(x^{3/2} + \frac{5}{2}x^{-1/2} - \frac{7}{4} \right) dx = \frac{2}{5}x^{5/2} + 5x^{1/2} - \frac{7}{4}x + c.$$

7. Use P to find c

$$-1 = \frac{2}{5}(4^{5/2}) + 5(4^{1/2}) - \frac{7}{4}(4) + c = \frac{64}{5} + 10 - 7 + c, \text{ so } c = -\frac{84}{5}.$$

Final answer

$$(a)(i) \frac{15}{2}. \quad (a)(ii) 2x + 15y + 7 = 0.$$

$$(b) f(x) = \frac{2}{5}x^{5/2} + 5x^{1/2} - \frac{7}{4}x - \frac{84}{5}.$$

Question 10

10. In this question you must show all stages of your working.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- the point $P(2, 8\sqrt{2})$ lies on C
- $f'(x) = 4\sqrt{x^3} + \frac{k}{x^2}$ where k is a constant
- $f''(x) = 0$ at P

(a) find the exact value of k ,

(4)

(b) find $f(x)$, giving your answer in simplest form.

(4)

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(Total for Question 10 is 8 marks)

Worked Solution - Question 10

1. Differentiate f prime

$$f'(x) = 4x^{3/2} + kx^{-2}, \text{ so } f''(x) = 6x^{1/2} - 2kx^{-3}.$$

2. Use f double prime equals zero at P

$$\text{At } P, x = 2 \text{ and } f''(2) = 0, \text{ so } 6\sqrt{2} - 2k(2^{-3}) = 0.$$

3. Find k

$$6\sqrt{2} - \frac{k}{4} = 0, \text{ hence } k = 24\sqrt{2}.$$

4. Integrate f prime

$$f(x) = \int (4x^{3/2} + 24\sqrt{2}x^{-2}) dx = \frac{8}{5}x^{5/2} - 24\sqrt{2}x^{-1} + c.$$

5. Use the point P

$$P(2, 8\sqrt{2}) \text{ lies on } C, \text{ so } 8\sqrt{2} = \frac{8}{5}(2^{5/2}) - \frac{24\sqrt{2}}{2} + c.$$

6. Find c

$$8\sqrt{2} = \frac{32\sqrt{2}}{5} - 12\sqrt{2} + c, \text{ so } c = \frac{68\sqrt{2}}{5}.$$

Final answer

$$(a) k = 24\sqrt{2}.$$

$$(b) f(x) = \frac{8}{5}x^{5/2} - \frac{24\sqrt{2}}{x} + \frac{68\sqrt{2}}{5}.$$

Question 10

Integration

10. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = 6x - \frac{(2x-1)(3x+2)}{2\sqrt{x}}$

- the point $P(4, 12)$ lies on C

(a) find the equation of the normal to C at P , giving your answer in the form $y = mx + c$ where m and c are integers to be found,

(4)

(b) find $f(x)$, giving each term in simplest form.

(6)

(Total for Question 10 is 10 marks)

Worked Solution - Question 10

1. Expand the numerator

$$(2x - 1)(3x + 2) = 6x^2 + x - 2.$$

2. Simplify f prime

$$f'(x) = 6x - \frac{6x^2 + x - 2}{2\sqrt{x}} = 6x - 3x^{3/2} - \frac{1}{2}x^{1/2} + x^{-1/2}.$$

3. Find the tangent gradient at P

$$f'(4) = 24 - 3(8) - \frac{1}{2}(2) + \frac{1}{2} = -\frac{1}{2}.$$

4. Use the normal gradient

The normal gradient is the negative reciprocal of $-\frac{1}{2}$, so it is 2.

5. Find the normal equation

Through $P(4, 12)$, $y - 12 = 2(x - 4)$, hence $y = 2x + 4$.

6. Integrate f prime

$$f(x) = 3x^2 - \frac{6}{5}x^{5/2} - \frac{1}{3}x^{3/2} + 2x^{1/2} + c.$$

7. Use P to find c

$$12 = 3(4)^2 - \frac{6}{5}(4^{5/2}) - \frac{1}{3}(4^{3/2}) + 2(4^{1/2}) + c.$$

8. Simplify c

$$12 = 48 - \frac{192}{5} - \frac{8}{3} + 4 + c = \frac{164}{15} + c, \text{ so } c = \frac{16}{15}.$$

Final answer

(a) $y = 2x + 4.$

(b) $f(x) = 3x^2 - \frac{6}{5}x^{5/2} - \frac{1}{3}x^{3/2} + 2x^{1/2} + \frac{16}{15}.$

Mark scheme checked - Topic checked - WMA11-01_2024_MayJune_Q10

Question 8

8. A curve C has equation $y = f(x)$.

The point P with x coordinate 3 lies on C

Given

- $f'(x) = 4x^2 + kx + 3$ where k is a constant
- the normal to C at P has equation $y = -\frac{1}{24}x + 5$

(a) show that $k = -5$

(3)

(b) Hence find $f(x)$.

(4)

(Total for Question 8 is 7 marks)

Worked Solution - Question 8

1. Find the y-coordinate of P

The normal is $y = -\frac{1}{24}x + 5$. At $x = 3$, $y = -\frac{3}{24} + 5 = \frac{39}{8}$.

2. Use the normal gradient

The normal gradient is $-\frac{1}{24}$, so the tangent gradient is 24.

3. Use f prime at x equals 3

$$f'(3) = 4(3)^2 + 3k + 3 = 39 + 3k.$$

4. Find k

$$39 + 3k = 24, \text{ so } k = -5.$$

5. Integrate f prime

With $k = -5$, $f'(x) = 4x^2 - 5x + 3$, so $f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x + c$.

6. Use point P

$$\frac{39}{8} = \frac{4}{3}(27) - \frac{5}{2}(9) + 3(3) + c = \frac{45}{2} + c.$$

7. Find c

$$c = \frac{39}{8} - \frac{45}{2} = -\frac{141}{8}.$$

Final answer

$$k = -5 \text{ and } f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x - \frac{141}{8}.$$

Question 6

6. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- the point $P(4, -5)$ lies on C
- $f'(x) = \frac{2x^2 + ax + b}{4\sqrt{x}}$, where a and b are constants
- the gradient of the tangent to C at P is 7

(a) show that

$$4a + b = 24 \quad (2)$$

Given also that $a + b = -9$

(b) find, in simplest form, $f(x)$ (7)

Curve C is transformed to the curve with equation $y = f(x - 3)$

Given that point P is transformed to the point Q ,

(c) state the coordinates of Q . (1)

(Total for Question 6 is 10 marks)

Worked Solution - Question 6

1. Use the gradient at P

At $x = 4$, $f'(4) = 7$. Since $f'(x) = \frac{2x^2 + ax + b}{4\sqrt{x}}$, we get $\frac{32 + 4a + b}{8} = 7$.

2. Show the required equation

$32 + 4a + b = 56$, so $4a + b = 24$.

3. Solve for a and b

Together with $a + b = -9$, subtracting gives $3a = 33$, so $a = 11$ and $b = -20$.

4. Rewrite f prime

$$f'(x) = \frac{2x^2 + 11x - 20}{4\sqrt{x}} = \frac{1}{2}x^{3/2} + \frac{11}{4}x^{1/2} - 5x^{-1/2}.$$

5. Integrate

$$f(x) = \frac{1}{5}x^{5/2} + \frac{11}{6}x^{3/2} - 10x^{1/2} + c.$$

6. Use P to find c

$$-5 = \frac{1}{5}(4^{5/2}) + \frac{11}{6}(4^{3/2}) - 10(4^{1/2}) + c = \frac{16}{15} + c.$$

7. State f(x)

$$c = -\frac{91}{15}, \text{ so } f(x) = \frac{1}{5}x^{5/2} + \frac{11}{6}x^{3/2} - 10x^{1/2} - \frac{91}{15}.$$

8. Transform P

$y = f(x - 3)$ translates the curve 3 units to the right, so $P(4, -5)$ becomes $Q(7, -5)$.

Final answer

$$(b) f(x) = \frac{1}{5}x^{5/2} + \frac{11}{6}x^{3/2} - 10x^{1/2} - \frac{91}{15}.$$

$$(c) Q = (7, -5).$$

Mark scheme checked - Topic checked - WMA11-01_2025_Jan_Q06

Question 8

8.

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

A curve has equation $y = f(x)$, $x > 0$

The point $P(4, 12)$ lies on the curve.

Given that

- $f'(x) = 3\sqrt{x} + kx^2$ where k is a constant
- the equation of the tangent to the curve at P has equation $y = 10x + c$ where c is a constant

(a) (i) show that $k = \frac{1}{4}$

(ii) find the value of c

(4)

(b) Hence find the value of $f''(x)$ at P .

(3)

(c) Find $f(x)$.

(4)

(Total for Question 8 is 11 marks)

Worked Solution - Question 8

1. Use the tangent gradient

The tangent at P has equation $y = 10x + c$, so the gradient at P is 10.

2. Show k

$$f'(4) = 3\sqrt{4} + k(4)^2 = 6 + 16k = 10, \text{ so } 16k = 4 \text{ and } k = \frac{1}{4}.$$

3. Find c

Substitute $P(4, 12)$ into $y = 10x + c$: $12 = 40 + c$, so $c = -28$.

4. Differentiate f prime

$$\text{With } k = \frac{1}{4}, f'(x) = 3x^{1/2} + \frac{1}{4}x^2.$$

5. Find f double prime

$$f''(x) = \frac{3}{2}x^{-1/2} + \frac{1}{2}x.$$

6. Evaluate at P

$$f''(4) = \frac{3}{2} \cdot \frac{1}{2} + 2 = \frac{3}{4} + 2 = \frac{11}{4}.$$

7. Integrate f prime

$$f(x) = \int \left(3x^{1/2} + \frac{1}{4}x^2 \right) dx = 2x^{3/2} + \frac{1}{12}x^3 + d.$$

8. Use P to find d

$$12 = 2(4^{3/2}) + \frac{1}{12}(4^3) + d = 16 + \frac{16}{3} + d.$$

9. State $f(x)$

$$d = -\frac{28}{3}, \text{ so } f(x) = 2x^{3/2} + \frac{1}{12}x^3 - \frac{28}{3}.$$

Final answer

$$(a)(i) \, k = \frac{1}{4}. \quad (a)(ii) \, c = -28.$$

$$(b) \, f''(P) = \frac{11}{4}.$$

$$(c) \, f(x) = 2x^{3/2} + \frac{1}{12}x^3 - \frac{28}{3}.$$

Mark scheme checked - Topic checked - WMA11-01_2025_MayJune_Q08

Question 8

8. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = 2x + \frac{8}{x^2} + k$, where k is a constant
- the equation of the tangent to the curve at $x = \sqrt{2}$ is $y = 5x - 3\sqrt{2}$

(a) find the exact value of k .

(2)

(b) Find an equation of the normal to the curve at $x = \sqrt{2}$

(2)

(c) Find $f(x)$, writing your answer in simplest form.

(4)

(Total for Question 8 is 8 marks)

Worked Solution - Question 8

1. Use tangent gradient

The tangent has equation $y = 5x - 3\sqrt{2}$, so the gradient at $x = \sqrt{2}$ is 5.

2. Find k

$$f'(\sqrt{2}) = 2\sqrt{2} + \frac{8}{(\sqrt{2})^2} + k = 2\sqrt{2} + 4 + k = 5, \text{ so } k = 1 - 2\sqrt{2}.$$

3. Find the point on the curve

At $x = \sqrt{2}$, the tangent gives $y = 5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2}$.

4. Normal gradient

The tangent gradient is 5, so the normal gradient is $-\frac{1}{5}$.

5. Equation of the normal

$$y - 2\sqrt{2} = -\frac{1}{5}(x - \sqrt{2}), \text{ hence } y = -\frac{1}{5}x + \frac{11\sqrt{2}}{5}.$$

6. Integrate f prime

$$f(x) = \int \left(2x + \frac{8}{x^2} + k \right) dx = x^2 - \frac{8}{x} + kx + c.$$

7. Substitute k

$$f(x) = x^2 - \frac{8}{x} + (1 - 2\sqrt{2})x + c.$$

8. Use the point

$$2\sqrt{2} = 2 - \frac{8}{\sqrt{2}} + (1 - 2\sqrt{2})\sqrt{2} + c = -2 - 3\sqrt{2} + c.$$

9. Find c

$$c = 2 + 5\sqrt{2}, \text{ so } f(x) = x^2 - \frac{8}{x} + (1 - 2\sqrt{2})x + 2 + 5\sqrt{2}.$$

Final answer

$$(a) k = 1 - 2\sqrt{2}.$$

$$(b) y = -\frac{1}{5}x + \frac{11\sqrt{2}}{5}.$$

$$(c) f(x) = x^2 - \frac{8}{x} + (1 - 2\sqrt{2})x + 2 + 5\sqrt{2}.$$

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Question 8

8.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

A curve C has equation $y = f(x)$, $x > 0$

A point P lies on C .

Given that

- $f'(x) = 4x^2 + \frac{6}{x^2} - 4$
 - the equation of the tangent to C at P is $y = 10x - 6\sqrt{3}$
- (a) (i) verify that $\sqrt{3}$ is a possible x coordinate of P ,
(ii) find, using algebra, the other possible x coordinate of P .

(6)

Given that the x coordinate of P is $\sqrt{3}$

- (b) find an equation of the normal to C at P .
- (c) Find $f(x)$, writing your answer in simplest form.
You must show each stage of your working.

(2)

(4)

(Total for Question 8 is 12 marks)

Worked Solution - Question 8

1. Verify x equals root 3

$f'(\sqrt{3}) = 4(\sqrt{3})^2 + \frac{6}{(\sqrt{3})^2} - 4 = 12 + 2 - 4 = 10$, which matches the tangent gradient.

2. Find all possible x -coordinates

Set $4x^2 + \frac{6}{x^2} - 4 = 10$.

3. Form a quadratic in x squared

Multiplying by x^2 gives $4x^4 - 14x^2 + 6 = 0$, so $2x^4 - 7x^2 + 3 = 0$.

4. Factor

$$2x^4 - 7x^2 + 3 = (2x^2 - 1)(x^2 - 3) = 0.$$

5. Use x greater than zero

$x = \sqrt{3}$ or $x = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$. The other possible x -coordinate is $\frac{\sqrt{2}}{2}$.

6. Find the point P

Given $x_P = \sqrt{3}$, the tangent gives $y_P = 10\sqrt{3} - 6\sqrt{3} = 4\sqrt{3}$.

7. Normal equation

The normal gradient is $-\frac{1}{10}$, so $y - 4\sqrt{3} = -\frac{1}{10}(x - \sqrt{3})$.

8. Integrate f prime

$$f(x) = \int \left(4x^2 + \frac{6}{x^2} - 4 \right) dx = \frac{4}{3}x^3 - \frac{6}{x} - 4x + c.$$

9. Use P to find c

$$4\sqrt{3} = \frac{4}{3}(\sqrt{3})^3 - \frac{6}{\sqrt{3}} - 4\sqrt{3} + c = 4\sqrt{3} - 2\sqrt{3} - 4\sqrt{3} + c.$$

10. State f(x)

$$c = 6\sqrt{3}, \text{ so } f(x) = \frac{4}{3}x^3 - \frac{6}{x} - 4x + 6\sqrt{3}.$$

Final answer

(a)(i) verified.

$$(a)(ii) \ x = \frac{\sqrt{2}}{2}.$$

$$(b) \ y - 4\sqrt{3} = -\frac{1}{10}(x - \sqrt{3}).$$

$$(c) \ f(x) = \frac{4}{3}x^3 - \frac{6}{x} - 4x + 6\sqrt{3}.$$

Mark scheme checked - Topic checked - WMA11-01_2026_Jan_Q08